Supplement to "Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects"

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October 19, 2026

This supplement reports extensions and qualifications of the results of Section 3, two of the three empirical applications, the detailed results of the three simulation studies, and proofs of all results stated in the main paper. References of the form Theorem 1, Lemma 2, Section 2.4 and so on refer to the main paper; sections, tables and equations carrying an S prefix are internal to this supplement.

S1 Extensions and Qualifications of the Main Results

This section collects material that elaborates Section 3 of the main paper without being needed to state its results: the comparison of the finite-difference Riesz representation with the classical integration-by-parts one, the reading of the results when the step size is data-dependent, the binary and categorical cases, the form the nuisance conditions take under the unbounded-support assumption, and what the window effect's relation to resampling is. Each subsection is self-contained and referenced from the point in the main text it belongs to.

S1.1 Interpreting Effects on Heavily Trimmed Features

The practical consequence is that effects on heavily trimmed features are not comparable in magnitude with effects on untrimmed ones. A feature retained for only eight percent

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of the sample reports a number roughly a twelfth the size of its conditional counterpart, and reporting the two side by side without the retained share invites exactly the wrong comparison. Both $\mathbb{E}[w(X)]$ and the implied scale factor are recoverable from the trimmed fractions reported for every feature in Table S2 of the supplement, and Section 4 flags the cases where the distinction changes the reading.

S1.2 Relationship to Classical Average Marginal Effects

Average marginal effects are a standard tool in the econometrics of nonlinear models (Greene, 2012; Wooldridge, 2010). For parametric models the estimator $\widehat{\text{AME}}_j$ defined above reduces exactly to the classical result. Consider a generalized linear model with inverse link function μ and linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, so that $f(\mathbf{x}_i) = \mu(\eta_i)$. The marginal effect of variable j at observation i is:

$$\left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{x}_i} = \beta_j \cdot \mu'(\eta_i) \tag{S1.1}$$

and the population average marginal effect is $\text{AME}_j = \beta_j \cdot E[\mu'(\eta)]$. For the identity link $\mu'(\eta) = 1$ and $\text{AME}_j = \beta_j$ — the marginal effect is constant across observations and equals the OLS coefficient. For the logistic link $\mu'(\eta) = \Lambda(\eta)(1 - \Lambda(\eta))$ and the AME averages the logistic density over the sample (Greene, 2012). For the probit link $\mu'(\eta) = \phi(\eta)$, the standard normal density evaluated at the linear predictor (Wooldridge, 2010).

In each case $\widehat{\text{AME}}_j$ recovers the classical result automatically — the finite difference applied to the model output numerically approximates $\beta_j \cdot \mu'(\eta_i)$ without requiring knowledge of the link function. The framework therefore nests the entire GLM family as a special case and reduces to familiar output when the model is parametric. This backward compatibility is what the `marginaleffects` package in R and Python implements (Arel-Bundock et al., 2024); the present framework extends that implementation with formal asymptotic guarantees for machine learning models.

S1.3 Data-Dependent Step Sizes

Remark S1 (Data-dependent step sizes)**.** Definition 1 fixes h, and every result below is a statement about the estimand at that fixed value. The default of Section 2 sets $h_j = \max(10^{-4}, 0.05\,\hat{\sigma}_j)$, which is a function of the data. The two are reconciled by reading all results conditional on the realized $\hat{h}$: the reported quantity is the window effect at the step actually used, which is why h_j is reported alongside every estimate in Table S2 rather than being treated as a tuning parameter. Nothing in the argument requires h to be non-random — the representation of Lemma 1, the orthogonality of Lemma 2 and the expansion of Theorem 1 all hold pointwise in h — and the conditional reading is the natural one, because the estimand is genuinely indexed by the perturbation scale the analyst chose.

It is worth being explicit about what the conditional reading buys, since the alternative is not free. If one insists instead that the target is θ_{h_0} at the deterministic limit $h_0 = 0.05\,\sigma_j$, then whenever $h \mapsto \theta_h(f)$ is differentiable at h_0 the delta method gives $\theta_{\hat{h}} - \theta_{h_0} = \theta'_{h_0}\,(\hat{h} - h_0) + o_p(n^{-1/2})$, and since $\hat{\sigma}_j$ is root-n consistent this is $O_p(n^{-1/2})$ — the same order as the sampling error itself. The substitution is therefore *not* asymptotically negligible: it contributes a term to the limiting variance, and an unconditional treatment would have to add that term to V_h. We do not pursue this, because the conditional statement is the one that matches how the estimand is used and reported. An analyst who wants the unconditional reading can fix h from prior information or from a pilot sample, at which point the two coincide.

S1.4 Comparison with the Classical Average Derivative Representation

The proof is a change of variables in each half of the finite difference and nothing more. This is worth dwelling on, because it is the point at which the present theory parts ways with the classical average derivative literature. For the derivative functional $\mathbb{E}[\partial_j f(X)]$, the analogous representation is obtained by integration by parts, the representer is the

negative score $-\partial_j \log p$ of the covariate density, and a boundary term appears that must be killed by assuming the density vanishes on the boundary — an assumption incompatible with a density bounded away from zero on a compact support, and the reason Powell et al. (1989) pass to density-weighted variants. For the window functional the integration by parts is replaced by exact summation: no boundary term ever arises, because the trimming weight guarantees that every shifted evaluation point stays inside Ω. The representer involves the density only through ratios of its values at points $2h$ apart — no derivative of p, and hence no density derivative estimation, is ever required. The mean-zero property follows from applying (3.5) to $g \equiv 1$, and the uniform bound makes α_h a bounded weight function, in contrast to the unbounded score-based representers of the derivative functional. The cost of the exactness is the factor h^{-1} in the bound: smaller windows mean noisier representers, the familiar bias–variance logic of a bandwidth, here appearing in the variance of the efficient influence function rather than in a smoothing bias.

S1.5 Binary and Categorical Variables

Remark S2 (Binary and Categorical Variables)**.** For a binary $X_j \in \{0, 1\}$ the contrast estimand of Section 2 is $\theta^{\text{bin}} = \mathbb{E}[w(X_{-j})\{f(X_{-j}, 1) - f(X_{-j}, 0)\}]$, and the same change-of-variables argument — with the integral over x_j replaced by a sum over its two support points — yields the representer:

$$\alpha^{\text{bin}}(u) = w(u_{-j}) \left[\frac{\mathbf{1}\{u_j = 1\}}{\pi(u_{-j})} - \frac{\mathbf{1}\{u_j = 0\}}{1 - \pi(u_{-j})} \right], \qquad \pi(x_{-j}) = \mathbb{P}(X_j = 1 \mid X_{-j} = x_{-j}) \tag{S1.2}$$

These are inverse propensity weights, and the resulting orthogonal score is the augmented inverse probability weighting score of Robins et al. (1994) with X_j in the role of the treatment. The overlap condition $\varepsilon \le \pi \le 1 - \varepsilon$ plays exactly the role the trimming weight plays in the continuous case, and every result of Section 3.4 applies verbatim with (α_h, θ_h) replaced by $(\alpha^{\text{bin}}, \theta^{\text{bin}})$. Continuous, binary, and categorical effects are thus instances of one construction rather than a definition and two patches.

S1.6 Nuisance Conditions under Unbounded Support

Under Assumption 2 in place of the density bounds, the three conditions are read as follows. Condition (a) is strengthened to $\left\|\hat{f}^{(-k)} - f\right\|_{P,2+\delta} = o_p(1)$ for a $\delta > 0$ with $(2+\delta)/2$ conjugate to the r of (3.1); condition (c) is replaced by $\left\|\hat{\alpha}^{(-k)}\right\|_{P,q} \leq \bar{A}_q < \infty$ for the q of Assumption 2(c); and condition (b) is unchanged.

That last point is the one worth emphasizing. What changes between the two assumption sets are a consistency requirement and a boundedness requirement; the product condition (b), which governs how fast the nuisances must converge, is identical in both. The reason is the exactness of Lemma 2: the bias is a product of two L^2 errors with no linearization remainder to control, so the step of the argument that carries the rates never touches a density bound. Everything in Section S5 therefore applies verbatim under either assumption set. Had the bias carried the usual remainder term, the higher integrability would have propagated into the rate conditions and the thresholds of Section S5 would have moved.

Where the learner's error is uniformly bounded — as it is in the compactly supported settings in which the rate results of Section S5 are proved, whose estimators are truncated at a known bound on f — the strengthening of (a) is free. Writing $\Delta = \hat{f}^{(-k)} - f$, interpolation gives

$$\|\Delta\|_{P,2+\delta} \leq \|\Delta\|_{\infty}^{\delta/(2+\delta)} \, \|\Delta\|_{P,2}^{2/(2+\delta)} \tag{S1.3}$$

so L^2 consistency implies $L^{2+\delta}$ consistency with no additional assumption. On unbounded support it is a genuine additional requirement, though a mild one, and it holds for the learners used in the Gaussian design of Section 5.

Theorem 1, Theorem 2 and Proposition 2 hold under either assumption set. The proofs in the supplement carry the two cases side by side at the three points where they differ.

S1.7 Resampling

Remark S3 (Resampling-Free and Resampling-Based Inference)**.** Corollary 1 requires no resampling: the coefficient table of Section 2 is produced from a single pass of cross-fitting, with K model fits in place of the B refits of the bootstrap — a reduction of one to two orders of magnitude in computation. When a resampling distribution is nonetheless wanted (for simultaneous bands across coefficients, for example), asymptotic linearity delivers one for free: the multiplier bootstrap that perturbs the estimated influence function values $\hat{\psi}_i$ by i.i.d. mean-zero weights is consistent (van der Vaart and Wellner, 1996, Chapter 3.6) under the conditions of Theorem 1, and costs no additional model fits.

Remark S4 (The Model-Refitting Bootstrap)**.** The refitting bootstrap of Section 2, applied to the uncorrected plug-in average, is not covered by these guarantees, for two reasons that the present framework makes visible. First, it recenters at the fitted model, so it measures variability around $\mathbb{E}[w\Delta_{j,h}\hat{f}]$ rather than around $\theta_h(f)$: the regularization bias $\mathbb{E}[\alpha_h(\hat{f} - f)]$ — first order for slowly converging learners — is invisible to it. This is the mechanism behind the collapse of random forest coverage documented in Section 5. Second, evaluating each refitted model at the original sample holds the empirical covariate distribution fixed and thereby omits the first variance component of (3.15) from the bootstrap distribution. The refitting bootstrap retains a legitimate role as a diagnostic for how far the fitted function moves under resampling, variability which the asymptotic theory conditions away; it should be read as that diagnostic, not as the inferential procedure.

S1.8 The Mirror Direction of the Double Robustness

The bias identity of Lemma 2 is symmetric in its two arguments, and Proposition 2 exploits only one direction of it. The other is worth recording: a learner converging at the parametric rate — a correctly specified generalized linear model, for instance — tolerates an arbitrarily poor representer estimate, since the product $\|\alpha_h - \tilde{\alpha}\|_{P,2} \cdot \left\|\tilde{f} - f\right\|_{P,2}$ is $o_p(n^{-1/2})$ as soon as either factor is. This is how the classical delta-method validity of

parametric average marginal effects is nested in the present framework, as the opposite corner of one identity rather than as a separate theory.

S2 Function Spaces and Well-Behaved Estimators

The gradient results of the next two sections are stated in two function spaces, each the natural habitat of an estimator class. This section fixes them and the regularity conditions they carry.

We work with two function spaces depending on the estimator class. The choice of function space is not arbitrary—each space is the natural habitat of the estimator class in question, in the sense that estimators of that class belong to the space with probability one under mild conditions.

The **Sobolev space** $W^{k,p}(\Omega)$ for $k \geq 1$ and $p \geq 1$ consists of functions whose weak derivatives up to order k are in $L^p(\Omega)$, with norm:

$$\|g\|_{W^{k,p}} = \sum_{|\alpha| \leq k} \|\partial^\alpha g\|_{L^p} \tag{S2.1}$$

The gradient $\nabla g \in L^p(\Omega)^d$ is a vector valued L^p function for any $g \in W^{1,p}(\Omega)$. Sobolev spaces are the natural setting for smooth estimators because they quantify not just how well the function is approximated in L^p but how well its derivatives are approximated simultaneously. This is exactly the property needed to justify gradient based interpretation—L^p consistency of $\hat{f}_n$ alone is insufficient because differentiation is an unbounded operator on L^p, and convergence of a function does not in general imply convergence of its derivatives without additional regularity (Evans, 2010, Chapter 5).

The **space of functions of bounded variation** $BV(\Omega)$ consists of functions $g \in L^1(\Omega)$ whose distributional gradient Dg is a vector valued Radon measure with finite total variation $|Dg|(\Omega) < \infty$. The BV norm is:

$$\|g\|_{BV} = \|g\|_{L^1} + |Dg|(\Omega) \tag{S2.2}$$

The BV framework extends the notion of gradient to functions that are not differentiable in the classical or even weak sense—precisely the situation with tree based estimators, which have jump discontinuities at leaf boundaries. For piecewise constant functions the distributional gradient is a sum of surface measures concentrated on the boundaries between constant regions, capturing the intuition that the "derivative" of a step function is a spike at the jump location. This makes BV the natural space for random forests and boosting with tree base learners. The total variation $|Dg|(\Omega)$ measures the total size of all jumps, weighted by the boundary surface area—a quantity that is finite and well controlled for any piecewise constant function with finitely many pieces. The standard reference for the theory of BV functions is Ambrosio et al. (2000).

The relationship between the two spaces is important. If $f \in W^{1,p}(\Omega)$ for $p \geq 1$, then $f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$—that is, the distributional gradient of f is absolutely continuous with respect to Lebesgue measure with density ∇f. This embedding $W^{1,p} \hookrightarrow BV$ means that assuming $f \in W^{1,p}(\Omega)$ is sufficient for both the Sobolev and BV results below. We make this assumption throughout.

Definition S1 (Well-Behaved Estimator, Sobolev Sense)**.** We say that $\hat{f}_n$ is *well-behaved in the Sobolev sense* if

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = O_p(1) \quad \text{and} \quad \left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0, \tag{S2.3}$$

where the first condition means that the sequence is tight in $W^{2,p}$: for every $\eta > 0$ there is an $M < \infty$ with

$$\sup_n \mathbb{P}\left(\left\|\hat{f}_n\right\|_{W^{2,p}} > M\right) \leq \eta. \tag{S2.4}$$

Definition S2 (Well-Behaved Estimator, BV Sense)**.** We say that $\hat{f}_n$ is *well-behaved in the BV sense* if:

$$\sup_n \mathbb{E}\left[|D\hat{f}_n|(\Omega)\right] < \infty \quad \text{and} \quad \left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0 \tag{S2.5}$$

In both cases the well-behaved condition has the same structure: a boundedness condition in a strong space combined with consistency in a weaker space. The boundedness

condition is the key regularity requirement—it says that the estimator sequence does not become increasingly irregular as n grows, even as it fits the data more closely. This condition is verifiable from properties of the estimator class and is established for each class in the theorems below. The consistency condition is the standard requirement that the estimator converges to the truth and is available from existing results in each case.

The two boundedness conditions are stated differently, and the difference is not cosmetic. Both $\hat{f}_n$ and its norms are random, so a condition of the form $\sup_n \left\|\hat{f}_n\right\| < \infty$ with a single deterministic bound is stronger than any of the estimators below deliver: the boosting bound of Section S25 is proportional to $\|Y\|_{L^2}$, which is $O_p(1)$ but not bounded by a constant. Definition S1 therefore asks only for tightness. Definition S2 asks for a bound in expectation, which is what the Mondrian argument of Section S26 produces directly; by Markov's inequality it implies tightness of $|D\hat{f}_n|(\Omega)$, so the two conditions are of the same type, and either is implied by an almost sure uniform bound. What both exclude is norms that diverge in probability.

S3 What Learner Gradients Estimate

The inference theory of Section 3.4 used no property of the fitted learner beyond an L^2 rate. The question it leaves open is interpretive: the first component of the score averages the finite differences of the fitted model, and one wants to know what that object measures about the model — and when the model's gradient information converges to the population's. This subsection answers both questions in the function spaces the learners naturally inhabit. Four lemmas establish that gradients exist and converge in the Sobolev space of smooth estimators and the BV space of piecewise constant ones; a pairing identity (Theorem 3) then expresses the window effect of any BV estimator exactly as its jump measure integrated against a locally averaged covariate density, which is, to our knowledge, the first characterization of the population quantity that the marginal effect of a tree-based model measures. All proofs are in the supplement.

The parallel structure of Lemmas S1 through S4 reflects the parallel structure of the

Sobolev and BV frameworks and is the formal expression of the analogy between smooth and piecewise constant estimators. None of these results is needed for the validity of Theorem 1; they characterize the regime in which the *uncorrected* plug-in average is already consistent, and hence in which the correction term of (3.9) is a refinement rather than a repair.

Lemma S1 (Derivative Existence in Sobolev Spaces)**.** *Let $\hat{f}_n \in W^{2,p}(\Omega)$ for some $p > 1$. Then $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$ exists as a well defined L^p function and satisfies:*

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S3.1}$$

Lemma S1 is essentially definitional—membership in $W^{2,p}(\Omega)$ implies by definition that the first and second weak derivatives exist and are in L^p. The content of the lemma is the norm bound, which establishes that the gradient operator $\nabla : W^{2,p} \to W^{1,p}$ is bounded and linear. This boundedness is what makes the subsequent consistency argument work—a bounded linear operator preserves convergence, so once we have convergence of $\hat{f}_n$ in the right space, convergence of $\nabla \hat{f}_n$ follows automatically. The reason we require $\hat{f}_n \in W^{2,p}(\Omega)$ rather than just $W^{1,p}(\Omega)$ is that we need one additional degree of differentiability to apply the compactness argument in Lemma S2. The relevant theory of Sobolev spaces and their norms is developed in Evans (2010, Chapter 5).

Lemma S2 (Derivative Consistency in Sobolev Spaces)**.** *Let $f \in W^{1,p}(\Omega)$ and suppose $\hat{f}_n$ is well-behaved in the Sobolev sense. Then:*

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{S3.2}$$

Lemma S2 is the harder result and the one that does the real work in the Sobolev setting. The difficulty is that L^p consistency of $\hat{f}_n$ alone does not imply L^p consistency of $\nabla \hat{f}_n$—differentiation is an unbounded operator on L^p and limits and derivatives cannot be exchanged without additional structure. The uniform $W^{2,p}(\Omega)$ boundedness condition provides that structure. The proof applies the Rellich–Kondrachov compact embedding theorem, which states that a uniformly bounded sequence in $W^{2,p}(\Omega)$ has a subsequence

converging strongly in $W^{1,p}(\Omega)$ (Evans, 2010, Theorem 5, Section 5.7). Combined with the L^p consistency condition identifying the limit as f, this gives strong convergence of $\hat{f}_n$ to f in $W^{1,p}(\Omega)$, from which L^p convergence of the gradients follows immediately from the boundedness of ∇ established in Lemma S1. The key insight is that the uniform boundedness condition ensures the estimator sequence lives in a compact subset of $W^{1,p}(\Omega)$, preventing the gradients from escaping to infinity even as the estimator fits the data more closely.

Lemma S3 (Derivative Existence in BV)**.** *Let $\hat{f}_n \in BV(\Omega)$. Then the distributional gradient $D\hat{f}_n$ exists as a well defined vector valued Radon measure in $\mathcal{M}(\Omega)^d$ with:*

$$|D\hat{f}_n|(\Omega) < \infty \tag{S3.3}$$

For piecewise constant estimators with leaf partition $\{R_j\}$ and leaf means $\{\bar{y}_j\}$, the distributional gradient takes the explicit form:

$$D\hat{f}_n = \sum_{j \sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k} \tag{S3.4}$$

where ν_{jk} is the unit normal between adjacent leaves and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure on leaf boundaries.

Lemma S3 is the BV analog of Lemma S1 and is similarly structural in nature. The first part—that BV membership implies the distributional gradient exists as a Radon measure—follows from the definition of BV as developed in Ambrosio et al. (2000, Chapter 3). The content of the lemma is the explicit formula for piecewise constant estimators, which makes concrete what the distributional gradient actually looks like for a tree based model. The formula says that the gradient is not a function at all but a measure concentrated on the boundaries between leaves, with mass proportional to the jump in response values across the boundary weighted by the boundary surface area. This is the precise sense in which a random forest or boosted tree has a gradient—not a pointwise derivative, which does not exist, but a distributional object that captures where and how much the

estimator changes. This explicit representation is also what makes the total variation $|D\hat{f}_n|(\Omega)$ computable in terms of observable quantities—the leaf means and the partition geometry—which is essential for verifying the well-behaved condition in the theorems below.

Lemma S4 (Derivative Consistency in BV)**.** *Let $f \in W^{1,p}(\Omega)$ for some $p > 1$, so that $f \in BV(\Omega)$ with $Df = \nabla f \cdot \mathcal{L}^d$. Suppose $\hat{f}_n$ is well-behaved in the BV sense. Then:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \tag{S3.5}$$

If additionally $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$, then convergence is strict.

Lemma S4 is the hardest of the four function space lemmas and the BV analog of Lemma S2. The convergence is necessarily weaker than in the Sobolev case—weak-$*$ convergence of measures rather than strong L^p convergence—because the BV space is not reflexive and the stronger norm convergence is not available in general. Weak-$*$ convergence means that for any continuous test function ϕ:

$$\int_\Omega \phi \, d(D\hat{f}_n) \to \int_\Omega \phi \, d(Df) \tag{S3.6}$$

which is the appropriate notion of convergence for the distributional gradients of piecewise constant estimators. In the context of average marginal effects, this means that integrated gradient information—which is exactly what the AME computes— converges to the correct population quantity. The proof applies the BV compactness theorem of Ambrosio et al. (2000, Theorem 3.23), which is the analog of Rellich–Kondrachov in the BV setting: a uniformly bounded sequence in BV has a subsequence converging in L^1, and the L^1 consistency condition identifies the limit as f. The strict convergence upgrade follows from the lower semicontinuity of the total variation functional and is the stronger result one would want for applications requiring convergence of the total size of gradient information rather than just its distribution.

The parallel between Lemmas S2 and S4 is worth noting explicitly. In both cases the argument has the same structure: uniform boundedness in a strong space gives com-

pactness, consistency in a weak space identifies the limit, and the combination gives convergence of the derivative in the appropriate topology. The difference is in the topology of the conclusion—strong L^p convergence in the Sobolev case, weak-$*$ convergence of measures in the BV case—and this difference is fundamental rather than technical. It reflects the fact that smooth estimators carry gradient information in a stronger sense than piecewise constant estimators, which is itself a reflection of the different inductive biases of the estimator classes.

The pairing identity, Theorem 3 of the main paper, is the capstone of the interpretation theory: by Lemma S3 the distributional gradient of a piecewise constant estimator is a measure concentrated on leaf boundaries, and the identity integrates that measure against the smoothed density $K_h \star q$ of (3.16). One consequence of it is worth recording separately.

Corollary S1. *If in addition q extends to a continuous function of compact support (for instance w continuous and p continuous on Ω), then for fixed $g \in BV(\Omega)$:*

$$\lim_{h \downarrow 0} \int_\Omega (K_h \star q)\, dD_j g = \int_\Omega q\, dD_j g \tag{S3.7}$$

and when $g \in W^{1,1}(\Omega)$ the right side equals $\mathbb{E}[w(X)\, \partial_j g(X)]$, consistent with Proposition 1.

Theorem 3 settles a question that the interpretability literature has left informal: what does the average marginal effect of a random forest *measure*? The answer given by (3.18) is exact and geometric: it is the total jump of the fitted surface across leaf boundaries in direction j, with each boundary segment weighted by its share of the covariate mass in an h-window — the places where the model changes its prediction, weighted by how many observations live near those places. The identity reconciles the two readings of a marginal effect that Section 2 could only juxtapose. Read with $g = \hat{f}$, (3.17) is a descriptive statement about the fitted model, available in finite samples with no assumptions at all. Read through Theorem 1, the same plug-in quantity, corrected by the orthogonal score, estimates the population functional $\theta_h(f)$ at the parametric rate. The descriptive and the inferential interpretations are the two sides of (3.9), and the

pairing identity is what makes the first of them precise.

S4 Gradient Convergence of Specific Learners

The theorems of this subsection verify the well-behaved conditions for three estimator classes and conclude, via the lemma pairs above, that the learner's own gradient information converges to the population's. This is the strongest interpretability property a learner can have — the fitted model's local behaviour, not merely its window averages, tracks the truth — and it implies consistency of the uncorrected plug-in average for the small-h estimand. We emphasize the division of labor: these results calibrate how much interpretive weight the fitted gradients themselves can bear, while the inference of Section 3.4 requires none of them. In each case the argument has two components: verifying that the estimator satisfies the well-behaved condition, and citing the appropriate lemma pair. The first component is the estimator-specific contribution; the second is the functional analytic machinery established above.

Theorem S1 (Gradient Consistency for Neural Networks)**.** *Let $\hat{f}_n$ be a neural network estimator with smooth activations $\sigma \in C^2$, bounded weight norms $\prod_\ell \left\|W_\ell^{(n)}\right\| = O_p(1)$, and $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ for $p > d$. Then $\hat{f}_n$ is well-behaved in the Sobolev sense and:*

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{S4.1}$$

The conditions of Theorem S1 are natural and verifiable. Smooth activations — sigmoid, tanh, GELU — are standard in practice and ensure the network function is twice differentiable. The bounded weight norm condition formalizes the effect of L^2 regularization during training, which prevents the network weights from growing without bound. The L^p consistency condition is the standard requirement that the network is learning the true function. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are capable of approximating an arbitrary function and its derivatives to arbitrary accuracy, providing the foundation for the consistency condition and establishing that the function class is rich enough to contain the estimators

we consider. The condition $p > d$ is required by the pointwise differentiability theorem of Evans and Gariepy (2015, Theorem 6.5) and is satisfied for all p larger than the covariate dimension — a mild regularity condition that is implicit in the classical Sobolev embedding theory.

Proof sketch. The chain rule applied through the network layers gives $\hat{f}_n \in W^{2,p}(\Omega)$ with:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} \lesssim \prod_{\ell} \left\|W_\ell^{(n)}\right\|^2 \cdot \|\sigma''\|_\infty \tag{S4.2}$$

which is $O_p(1)$ by the weight norm assumption, the fitted weights being random. The L^p consistency condition and Lemma S2 then give the result. Full proof in Section S24 of the supplement. □

Remark S5. Theorem S1 is stated for smooth activations. ReLU networks require a modified argument because ReLU is not C^2 and the chain rule calculation does not apply directly. Extension to ReLU networks is possible through a distributional argument but complicates the presentation and is omitted here.

Theorem S2 (Gradient Consistency for Gradient Boosted Models)**.** *Let* $\hat{f}_n = \hat{F}_m = (I - (I - S_n)^{m+1})Y$ *be an* L_2*Boost estimator with a* fixed *symmetric linear learner* S_n *whose eigenvalues satisfy* $0 < \lambda_k^{(n)} \leq 1$ *for all* k*, and with base learners* $h_m = S_n u_m$ *lying in* $W^{2,p}$*. Suppose in addition that the spectral gap and the smoothing bound are* uniform in n*:*

$$\bar{\rho} := \sup_n \max_k \left(1 - \lambda_k^{(n)}\right) < 1, \qquad \sup_n \|S_n\|_{L^2 \to W^{2,p}} < \infty. \tag{S4.3}$$

Suppose $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$*. Then* $\hat{f}_n$ *is well-behaved in the Sobolev sense and* $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

The consistency hypothesis is in L^p, the same exponent as the conclusion and as the second condition of Definition S1; an L^1 hypothesis would not suffice, since on a bounded domain L^p consistency implies L^1 consistency and not the reverse. For $p \leq 2$ it is implied by L^2 consistency, again by Hölder on the bounded domain Ω, and L^2 is the exponent in which the risk results of Bühlmann and Yu (2003) for this estimator are stated.

Theorem S2 is deliberately narrow, and it is worth being explicit about what it does not reach. It applies to L_2Boost driven by a single fixed symmetric linear smoother — a smoothing spline at a fixed smoothing parameter is the canonical example — for which Bühlmann and Yu (2003) supply the representation and the spectral decay. It does *not* apply to XGBoost or LightGBM, the implementations used in Section 4. Those grow each tree greedily from the current residual, so there is no fixed operator S, the representation $\hat{F}_m = (I-(I-S)^{m+1})Y$ fails, and the geometric decay that the proof turns on is unavailable. Remark S13 sets out why no choice of tree base learner can be brought inside the hypotheses, and what is and is not known about those implementations in consequence.

The eigenvalue condition $0 < \lambda_k \leq 1$ is the key regularity condition on the base learner. Bühlmann and Yu (2003) impose this condition explicitly in their Theorem 1 and establish that under it the bias decays exponentially fast while the variance increases only by exponentially diminishing amounts — the exponential bias–variance trade-off that makes L_2Boost analytically tractable. The condition holds for smoothing spline learners and, generically, for tree stump learners whose splits have positive but incomplete explanatory power, which is the typical case in practice. Learners with any eigenvalue equal to zero are degenerate in the sense of Corollary 1 of Bühlmann and Yu (2003): $\mathcal{B}_m \equiv S$ for all m and boosting does not improve on the base learner. The functional gradient descent perspective of Mason et al. (2000) provides the conceptual framework for understanding the boosting path as gradient descent in function space.

Proof sketch. By Proposition 1 of Bühlmann and Yu (2003) the residual at step m is $u_m = (I - S_n)^m Y$, and by Proposition 2 and Theorem 1, under the eigenvalue condition, it decays geometrically:

$$\|u_m\|_{L^2} \leq C\rho^m, \qquad \rho = \max_k \left(1 - \lambda_k^{(n)}\right) \leq \bar{\rho} < 1, \qquad C = \|Y\|_{L^2}, \tag{S4.4}$$

in the normalized empirical L^2 norm, for which $C = O_p(1)$ under Assumption 1. Each base learner fit satisfies $\|h_m\|_{W^{2,p}} \leq \|S_n\|_{L^2 \to W^{2,p}} \|u_m\|_{L^2}$, so by linearity and the triangle

inequality

$$\left\|\hat{f}_n\right\|_{W^{2,p}} \leq \sum_{m\geq 1} \|h_m\|_{W^{2,p}} \leq \frac{C\,\bar{\rho}}{1-\bar{\rho}} \sup_n \|S_n\|_{L^2\to W^{2,p}} = O_p(1), \tag{S4.5}$$

the two suprema in (S4.3) making the bound free of n apart from the factor C. This is the first condition of Definition S1; the second is assumed, and Lemma S2 gives the conclusion. Full proof in Section S25 of the supplement. □

Theorem S3 (Gradient Consistency for Mondrian Forests)**.** *Let $\hat{f}_n$ be a Mondrian random forest estimator with budget $\lambda_n \asymp n^{1/(d+2)}$ and any fixed number $B \geq 1$ of trees, with f Lipschitz on $[0,1]^d$. Then $\hat{f}_n$ is well-behaved in the BV sense and:*

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \tag{S4.6}$$

Theorem S3 is the most technically demanding result in the paper. Random forests are genuinely different from neural networks and boosted models in a way that makes gradient consistency harder to establish — the partition structure is random and the response averages within leaves depend on the data in a complex way. The restriction to Mondrian forests, where splits are made independently of the data, is the key simplification that makes the proof tractable. For Mondrian forests the partition geometry and the response variation can be controlled separately, whereas for Breiman forests with data-dependent splits the two are coupled in a way that resists clean analysis. The L^2 consistency result under the budget $\lambda_n \asymp n^{1/(d+2)}$ follows from Mourtada et al. (2017), who establish the minimax rate $O(n^{-2/(d+2)})$ for Lipschitz regression functions, and L^1 consistency follows by Cauchy–Schwarz on the bounded domain Ω.

The budget condition $\lambda_n \asymp n^{1/(d+2)}$ is the only condition the theorem imposes, and it is the natural one: it balances the bias from coarse partitions against the variance from fine ones at the minimax optimal rate. What is perhaps surprising is that the same rate balances the two contributions to the *total variation* of the fitted tree. The jumps across leaf boundaries carry a signal part, which reflects the variation of f itself, and a noise part, which reflects the sampling error in the leaf means; at this budget both are

$O(1)$, and the partition may refine without limit without the total variation growing. No condition on the number of trees is needed.

Proof sketch. Each tree is piecewise constant and therefore in BV by Lemma S3, and the forest average is a convex combination, so $|D\hat{f}_n|(\Omega) \le B^{-1}\sum_b |DT_b|(\Omega)$. It therefore suffices to bound a single tree, and no cancellation between trees is required.

For one tree, the distributional gradient is carried by the split faces (Lemma S6), and the jump across a face splits into a signal part and a noise part. Reducing both to one-dimensional sections turns the signal part into the expected aspect ratio $\mathbb{E}[\mathrm{diam}(l(X))/s_1(l(X))]$ of the cell containing a uniform point, which Lemma S9 bounds by $2d-1+O(1/\lambda)$ using the Mondrian cell law of Mourtada et al. (2017) — a bound with no rate in it, because the variation of a Lipschitz function over a refining partition depends on the partition's shape and not on how fine it is. The noise part is bounded by Lemma S11 at $O(\lambda^{1+d/2}n^{-1/2})$, which is $O(1)$ precisely when $\lambda \asymp n^{1/(d+2)}$, since $(1+d/2)/(d+2) = 1/2$. Lemma S4 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. Full proof in Section S26. □

Remark S6. Extension of Theorem S3 to Breiman forests with data-dependent splits requires additional control over the interaction between partition geometry and response variation. The data-dependent splitting rule couples these two quantities in a way that makes the cross-term cancellation argument unavailable — splits are chosen precisely to maximize response differences across boundaries, which is the worst case for BV control. This extension is an important direction for future work and is discussed further in Section 6.

S5 Nuisance Estimation and Rates

Assumption 3 asks for two regression-type L^2 rates whose product is $o_p(n^{-1/2})$; the benchmark sufficient condition is that each nuisance converge faster than $n^{-1/4}$. This subsection records when the learners of Section 2 meet the benchmark for $\hat{f}$, and how the representer α_h is estimated without ever estimating a density.

For the regression function, the required rates are exactly the L^2 risk bounds of the modern nonparametric literature, and a single smoothness threshold organizes all three learner classes. For deep neural networks with ReLU activations, Schmidt-Hieber (2020) establishes the rate $n^{-s/(2s+d)}$ up to logarithmic factors for s-Hölder regression functions (with the effective d reduced under compositional structure), so the $n^{-1/4}$ benchmark is met precisely when:

$$\frac{s}{2s+d} > \frac{1}{4} \quad \Longleftrightarrow \quad s > \frac{d}{2} \tag{S5.1}$$

For Mondrian forests, the minimax-optimal rates of Mourtada et al. (2017) take the same form $n^{-s/(2s+d)}$ for Hölder smoothness $s \leq 2$, so the benchmark is again (S5.1): at Lipschitz smoothness $s = 1$ this requires $d < 2$, and exploiting forest smoothing up to $s = 2$ accommodates $d \leq 3$. For L_2Boost, Bühlmann and Yu (2003) establish minimax rates for smoothing-spline base learners with the boosting iteration acting as the regularization parameter, and the benchmark corresponds to smoothness exceeding one half in their univariate and additive settings — with additive structure, as usual, breaking the dependence on the ambient dimension. The reading we intend is not that these conditions are innocuous — (S5.1) is a real restriction, and in high dimensions it binds — but that the currency in which the conditions are denominated has changed: what inference requires of the learner is an L^2 error rate of the kind the learning-theory literature actually proves, rather than root-n convergence in a Sobolev norm, which no growing-complexity learner satisfies. For Breiman forests with data-dependent splits, L^2 consistency is available under structural assumptions (Scornet et al., 2015), and polynomial L^2 rates for the original algorithm — sample CART splits and column subsampling included — have been established under sufficient-impurity-decrease conditions by Chi et al. (2022); the resulting exponents can fall short of the $n^{-1/4}$ benchmark, but the product form of Assumption 3(b) permits pairing a slow forest rate with a fast representer estimate, and Proposition 2 covers the known-representer case with no forest rate at all. The open question relevant to this paper is therefore an L^2 rate question, a substantially more tractable target than the gradient-consistency question that Section S4 leaves open for that class.

The representer is estimated by Riesz regression (Chernozhukov et al., 2022), which exploits the same linearity that produced Lemma 2. By Lemma 1, for any candidate $\alpha \in L^2(P)$:

$$\mathbb{E}\left[\alpha(X)^2\right] - 2\,\mathbb{E}\left[w(X)\,\Delta_{j,h}\alpha(X)\right] = \|\alpha - \alpha_h\|_{P,2}^2 - \|\alpha_h\|_{P,2}^2 \tag{S5.2}$$

so the population loss on the left is minimized uniquely at α_h, and its sample analog:

$$\hat{L}(\alpha) = \frac{1}{n}\sum_{i=1}^{n}\left[\alpha(\mathbf{x}_i)^2 - 2\,w(\mathbf{x}_i)\,\frac{\alpha(\mathbf{x}_i + he_j) - \alpha(\mathbf{x}_i - he_j)}{2h}\right] \tag{S5.3}$$

is computable directly: minimizing $\hat{L}$ over a class of functions — penalized linear sieves, neural networks, or forests — requires only the ability to evaluate candidates at shifted points, and at no stage is the density p, let alone a derivative of it, estimated. Chernozhukov et al. (2022) provide L^2 convergence guarantees for such estimators over exactly these classes, with rates governed by the complexity of the class and the approximability of α_h. Since α_h obeys the explicit bound of Lemma 1, the minimizer is truncated at that bound, which enforces Assumption 3(c) by construction without affecting the L^2 error. Two further remarks. First, the product form of Assumption 3(b) permits asymmetry: a learner with a fast rate for f tolerates a slow Riesz estimate, and conversely. Second, in designed or simulated data where the covariate density is known — as in the Monte Carlo study of Section 5 — α_h is available in closed form and the representer error is identically zero; Proposition 2 then applies, and valid inference requires no rate — indeed no consistency — from the learner at all.

S6 Verifying a New Learner

The architecture above is deliberately modular, and it is worth stating plainly what a reader must check to bring a learner not treated here under the results. There are two obligations, and they are separate. Valid inference requires only an $L^2(P)$ convergence rate for $\hat{f}$ fast enough that its product with the representer rate is $o_p(n^{-1/2})$, the bench-

mark sufficient condition being $n^{-1/4}$ for each. Nothing else about the learner enters Theorem 1: not differentiability, not the existence of a gradient, not membership in any particular function space. Consistency of the learner's own gradient information — the object of Section S4 — is a separate question, required only if one wishes to interpret the uncorrected plug-in, and it is answered by identifying the natural function space of the estimator class and applying the appropriate compactness theorem.

The practical consequence is that adding a learner is usually a matter of locating an existing rate rather than proving a new theorem. Table S1 records the position for the classes treated here and for several that are not. Penalized linear estimators are the easiest case: the lasso attains L^2 rates of order $\sqrt{s \log d / n}$ under standard sparsity and restricted-eigenvalue conditions, comfortably inside the benchmark whenever $s \log d = o(\sqrt{n})$, and it is a linear estimator in a Sobolev space, so both obligations are discharged by citation. Smoothing splines and series estimators are similar on both counts, attaining the classical $n^{-s/(2s+1)}$ rate in the univariate case and living in $W^{1,p}$ by construction. Kernel ridge regression over a reproducing kernel Hilbert space — the function space generated by the chosen kernel, in which evaluation at a point is a continuous linear functional — has minimax L^2 rates of the same form. For a sufficiently smooth kernel that space embeds continuously in $W^{1,p}$, so the interpretation argument of Section S4 applies unchanged; for a kernel that is not smooth it need not, and the embedding is the thing to check. Generalized additive models inherit the univariate rate componentwise, which is what makes additive structure so useful here: it breaks the dependence on ambient dimension in (S5.1). For support vector regression with a smooth kernel and for Bayesian additive regression trees, L^2 rates are available under conditions in the respective literatures, though in the latter case posterior contraction rather than a frequentist risk bound is the natural object and the translation requires care. Any learner of bounded variation — which includes every tree-based method, however its splits are chosen — falls under the pairing identity of Theorem 3 without further argument, so what its window effect measures is characterized even when its rate is unknown.

Two cases in the table deserve separate mention because they are where the framework

is currently incomplete. Breiman forests have the inference obligation open and the interpretation obligation conjectured, for the reasons given above. Learners whose output is not a fixed function of the data — online or streaming estimators, for instance — fall outside the sampling framework of Assumption 1 altogether and would require a different treatment.

Table S1: What must be verified to apply the framework to a learner. The inference column is what Theorem 1 consumes; the interpretation column is what Section S4 requires in order to read the uncorrected plug-in as a gradient.

Learner	Inference: L^2 rate	Interpretation: function space
Linear and GLM	parametric	$W^{1,p}$, exact
Lasso, ridge	established	$W^{1,p}$
Splines, series	established	$W^{1,p}$
Kernel ridge	established	$W^{1,p}$ via smooth kernel
GAM	componentwise	$W^{1,p}$
Neural networks	established	$W^{1,p}$, proved here
L_2Boost, fixed smoother	established	$W^{1,p}$/BV, proved here
Tree boosting (XGBoost)	established	BV; gradient *open*
Mondrian forests	minimax optimal	BV, proved here
Breiman forests	*open*	BV; gradient *conjectured*
SVM (smooth kernel)	available	$W^{1,p}$ via kernel; not treated
BART	contraction rate	BV; not treated

S7 Implementation Details for the Empirical Applications

Across the three empirical applications we fit four model classes: logistic or linear regression as the parametric baseline, a random forest, XGBoost, and a feedforward neural network. The random forest uses 200 trees with a maximum depth of 8. XGBoost uses 200 boosting rounds with a maximum depth of 4 and a learning rate of 0.05. The neural network architecture and training schedule depend on the outcome type. For binary classification, the network is a feedforward architecture with two hidden layers of 64 and 32 units trained with the Adam optimizer at a learning rate of 10^{-3}, with early stopping on a 10% validation split with patience of 20 epochs. For continuous regression, the network is deeper — four hidden layers of 256, 128, 64, and 32 units — trained

at a learning rate of 3×10^{-4} with patience of 50 epochs, reflecting the larger capacity required to fit the continuous outcome surface and the longer training horizon needed for the optimization to converge. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates and predictions on the natural outcome scale. Estimation follows the debiased cross-fitted procedure of Section 2.4 with $K = 5$ folds: within each fold the learner and the Riesz representer are fit on the other four folds and evaluated on the held-out fold, and standard errors come from the sample variance of the orthogonal score. No learner is ever differenced at an observation it was trained on. Every model class is refit from scratch on each fold, including the neural networks, whose early stopping schedule is applied within the fold. The fit statistics reported at the foot of each table are computed from a single full-sample fit and are descriptive of the learner rather than of the estimator; they are not used in the estimation. Where applicable, predicted probabilities for classification models are extracted from `predict_proba(X)[:,1]` as discussed in Section 2.

Binary and categorical variables, and classifier outputs

For binary or categorical variables the partial derivative does not exist and the finite difference is replaced by a contrast. The marginal effect for observation i is:

$$m_{ij} = \hat{f}\left(\mathbf{x}_i^{(j,1)}\right) - \hat{f}\left(\mathbf{x}_i^{(j,0)}\right) \tag{S7.1}$$

which measures the change in predicted outcome when x_j is switched from 0 to 1 holding all other variables at their observed values. The sample average marginal effect is then $\widehat{\mathrm{AME}}_j = n^{-1} \sum_{i=1}^{n} m_{ij}$, with interpretation identical to the continuous case: a change in x_j from 0 to 1 is associated with a $\widehat{\mathrm{AME}}_j$ change in the predicted outcome, on average across the sample.

For binary classification models, $\hat{f}$ must return the estimated conditional probability $\hat{P}(Y = 1 \mid \mathbf{x})$ rather than the predicted class label. Class labels are discrete and their

finite differences are identically zero; the method requires a continuous output. This is the predicted probability of the positive class, available as `predict_proba(X)[:,1]` in scikit-learn and equivalent interfaces in other frameworks. The AME is then interpreted as the change in predicted probability associated with a one-unit change in x_j, on average across the sample.

Window settings and the reading of trimmed effects

The window estimand requires two settings to be fixed before estimation, and because $\theta_{j,h}$ is defined by them they are reported rather than treated as tuning. The step size is the default $h_j = \max(10^{-4}, 0.05\hat{\sigma}_j)$, floored at 0.5 for integer-valued features. The floor matters for count variables such as bedrooms, bathrooms, years of education and months delayed: a count has no mass strictly between its levels, so a window narrower than one unit has an empty interior, and for a piecewise constant learner the difference across it vanishes except where the window happens to straddle a split. At $h_j = 0.5$ the window is exactly the one-unit contrast, which is also the quantity of substantive interest for a count. The second setting is the trimming weight, which gives weight zero to observations lying within h_j of the boundary of the observed support of x_j, where $x \pm h_j e_j$ would require the learner to extrapolate. For most features this removes a negligible share of the sample, but for variables with mass sitting exactly on a boundary it does not: capital gains and capital losses are zero for over ninety percent of the Adult sample, and zero is the minimum of their observed support, so those features are retained for 7.7% and 4.7% of observations respectively. By (3.3) the reported effects are the effects among that small minority scaled by its population share, not the effects on the full sample and not the conditional effects among the retained; dividing by the retained share recovers the latter. They should not be read against the untrimmed features in the same column without that adjustment. Table S2 of the supplement reports h_j and the trimmed fraction for every feature in all three applications, and the discussion below flags the cases where the trimming is large enough to change the interpretation.

The model-refitting bootstrap as a diagnostic

The package additionally retains a nonparametric model-refitting bootstrap (Efron, 1979; Efron and Tibshirani, 1993) as a *diagnostic* for how far the fitted function moves when the sample is perturbed — variability that the asymptotic theory conditions away. The learner's settings are held fixed across replicates; if the learner is tuned, the tuning is part of what each replicate refits. For $b = 1, \ldots, B$: draw $\mathcal{D}^{(b)}$ by resampling $(\mathbf{x}_i, y_i)$ pairs jointly with replacement; refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ from a fresh initialization at the same settings; and recompute the plug-in average marginal effect. The dispersion of the replicates measures how sensitive the reported effects are to the joint variability of resampling and refitting, and large dispersion is a useful warning that the fitted model is unstable. It is not, however, a substitute for the influence function standard errors above, for reasons Section 3 makes precise: the refitting bootstrap recenters at the fitted model and is therefore blind to the regularization bias that the correction term removes — a blindness whose consequences for coverage are documented for random forests in Section 5 — and, because each replicate is evaluated at the original sample, it omits the component of sampling variance that comes from averaging a heterogeneous effect over a finite sample of covariate values.

S8 Specification Search for the Credit Default Application

Table S3 presents the specification search. The dominant finding is the importance of payment history and of the payment ratio. Months delayed enters at 0.035 for logistic regression and 0.030 for the neural network in specification A+B and remains at 0.030 and 0.025 in the full specification, while the payment ratio, once it is available in specifications A+C and A+B+C, is much the largest effect in the table: -0.265 for logistic regression and -0.194 for the neural network in the full specification. The demographic panel tells a different story from the Adult Income dataset of Section S10: age is statistically indistinguishable from zero in nearly every specification for both models, and the female

Table S2: Window settings for the empirical applications. Step size h_j and the fraction of observations receiving trimming weight zero.

Dataset	Feature	h_j	Trimmed
UCI Adult Income	age	0.6855	1.3%
	female	—	0.0%
	education_num	0.5	1.4%
	hours_per_week	0.6196	0.3%
	government	—	0.0%
	self_employed	—	0.0%
	white_collar	—	0.0%
	blue_collar	—	0.0%
	service	—	0.0%
	capital_gain	373	92.3%
	capital_loss	20.15	95.3%
	married	—	0.0%
UCI Credit Card Default	age	0.5	0.2%
	female	—	0.0%
	education	0.5	36.8%
	married	—	0.0%
	avg_pay_status	0.04911	7.0%
	months_delayed	0.5	70.9%
	avg_bill_amt	3,163	0.0%
	avg_pay_amt	507	11.7%
	pay_ratio	0.01839	18.4%
Ames Housing	GrLivArea	26.27	0.1%
	BedroomAbvGr	0.5	0.5%
	FullBath	0.5	2.9%
	HalfBath	0.5	63.4%
	OverallQual	0.5	1.4%
	OverallCond	0.5	1.6%
	YearBuilt	1.51	1.4%
	YearRemodAdd	1.032	14.5%
	LotArea	499	1.2%
	GarageArea	10.69	5.6%
	nbhd_high	—	0.0%
	nbhd_mid	—	0.0%

Notes: h_j is the default adaptive step size, $\max(10^{-4}, 0.05\hat{\sigma}_j)$, floored at 0.5 for integer-valued features so that a count is contrasted over a whole unit. Features declared binary use the level contrast $f(x \mid x_j = 1) - f(x \mid x_j = 0)$, for which h_j is undefined and no trimming applies. The trimmed fraction is the share of observations lying within h_j of the boundary of the observed support of x_j, which receive weight zero. The estimand is the *weighted* effect $\mathbb{E}[w(X)\Delta_{j,h}f(X)]$ of Definition 1, not the effect among retained observations: by (3.3) the two differ by the factor $\mathbb{E}[w(X)]$, which is one minus the trimmed fraction. For a feature trimmed at 92.3% the reported number is about one thirteenth of its conditional counterpart, so entries in this table are not comparable across features without that rescaling.

Table S3: UCI Credit Card Default: Specification Search. Average marginal effects on P(Default).

	Logistic				Neural Net			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Demographics								
age	-0.0015***	-0.0002	-0.0005	0.0001	-0.0026***	-0.0006*	-0.0010***	-0.0000
	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)	(0.0004)	(0.0003)
female	-0.0358***	-0.0206***	-0.0291***	-0.0198***	-0.0364***	-0.0205***	-0.0265***	-0.0176***
	(0.0050)	(0.0046)	(0.0049)	(0.0046)	(0.0050)	(0.0046)	(0.0049)	(0.0045)
education	-0.0001	-0.0027	-0.0054**	-0.0040*	-0.0008	-0.0035	-0.0048**	-0.0040*
	(0.0022)	(0.0022)	(0.0023)	(0.0022)	(0.0021)	(0.0021)	(0.0022)	(0.0022)
married	0.0253***	0.0189***	0.0303***	0.0206***	0.0260***	0.0152**	0.0269***	0.0186***
	(0.0068)	(0.0064)	(0.0066)	(0.0063)	(0.0068)	(0.0064)	(0.0067)	(0.0063)
Panel B: Payment History								
avg_pay_status	–	-0.0140***	–	-0.0218***	–	0.0073*	–	0.0040
		(0.0041)		(0.0075)		(0.0041)		(0.0072)
months_delayed	–	0.0347***	–	0.0301***	–	0.0298***	–	0.0254***
		(0.0011)		(0.0012)		(0.0011)		(0.0012)
Panel C: Bill and Payment Amounts								
avg_bill_amt	–	–	-0.0000***	-0.0000**	–	–	-0.0000	-0.0000*
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
avg_pay_amt	–	–	-0.0000***	0.0000	–	–	-0.0000***	-0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
pay_ratio	–	–	-0.3521***	-0.2651***	–	–	-0.2572***	-0.1943**
			(0.0731)	(0.0730)			(0.0792)	(0.0773)
McFadden R^2	0.003	0.130	0.031	0.137	0.010	0.155	0.055	0.166
Accuracy	0.779	0.803	0.779	0.803	0.779	0.808	0.779	0.808

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

indicator, while consistently negative and significant, shrinks substantially once payment history is controlled for, falling from -0.036 to -0.020 for logistic regression and from -0.036 to -0.018 for the neural network. This pattern of attenuation is cleanly legible in the AME framework because the estimates are on a common probability scale across specifications.

What is most striking in this table is how little separates the two columns. In specification A the female AME is -0.0358 for logistic regression and -0.0364 for the neural network; married is 0.0253 and 0.0260; in the full specification the pair is -0.0198 and -0.0176, and 0.0206 and 0.0186. A parametric model and a four-layer network, differing in McFadden R^2 by three percentage points, produce estimates that agree to the third decimal place across sixteen model-specification cells. This is what the orthogonal score is supposed to deliver and what the plug-in did not: since each column estimates the same $\theta_{j,h}$ and the correction absorbs the first-order effect of the fitted regression, the remaining gap between the two columns reflects only the second-order term and sampling noise. The standard errors are also now nearly identical between the two models

— 0.0046 and 0.0045 on the female indicator, 0.0012 and 0.0012 on months delayed — rather than differing by the order of magnitude the refitting bootstrap reported, where the network's standard error for months delayed was 0.032 against the logistic model's 0.002. Both figures were describing the stability of a particular fitting procedure under resampling rather than the sampling variability of an estimate of a fixed target.

S9 Comparison with SHAP and Partial Dependence for the Credit Default Application

Table S4 again shows the two quantities parting company. For the payment history variables the separation is large: the AME for months delayed is 0.030 for logistic regression while the mean signed SHAP value is −0.015, opposite in sign. The payment ratio illustrates the most consequential divergence: the random forest and XGBoost AMEs are −0.489 and −0.584, indicating that a higher payment ratio is associated with a large reduction in default probability, while the corresponding SHAP values are 0.0012 and 0.0005, near zero and of the opposite sign. This is again a difference of estimand rather than a defect: a feature can carry a large average derivative while contributing little on average to the displacement of predictions from their baseline. The practical concern is that a researcher reading the SHAP column as a marginal effect would conclude that the payment ratio is essentially irrelevant to default risk, and, with no standard errors attached to either the SHAP or the PDP column, would have no statistical basis on which to check that conclusion.

The partial dependence column separates from the AME column here in the same pattern as in the Ames application of Section S11, and for the same reason: a partial dependence slope is a plug-in average over a single full-sample fit, so it carries the regularization bias that the correction removes. The separation tracks the quality of the underlying fit. On the female indicator, where the logistic model is close to correctly specified, the AME of −0.0198 and the PDP slope of −0.0198 agree exactly; for the random forest on the same variable they are −0.0161 and −0.0062. On the payment

Table S4: UCI Credit Card Default: Method Comparison. AME, SHAP, and PDP estimates for P(Default), full specification (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0001	-0.0003	0.0002	-0.0002	0.0011	-0.0000	-0.0005	0.0004	-0.0002	-0.0000	-0.0010	0.0012
female	-0.0198	-0.0005	-0.0198	-0.0161	0.0001	-0.0062	-0.0175	0.0002	-0.0146	-0.0176	0.0009	-0.0200
education	-0.0040	0.0002	0.0002	-0.0018	0.0006	-0.0057	0.0034	-0.0005	-0.0356	-0.0040	-0.0016	0.0082
married	0.0206	-0.0038	0.0220	0.0222	0.0002	0.0054	0.0205	0.0002	0.0142	0.0186	0.0022	0.0217
avg_pay_status	-0.0218	-0.0144	-0.0246	0.0061	0.0029	0.0610	0.0907	0.0027	0.0404	0.0040	-0.0116	0.0097
months_delayed	0.0301	-0.0148	0.1107	0.0134	-0.0008	0.0456	0.0191	0.0001	0.0404	0.0254	-0.0154	0.0653
avg_bill_amt	-0.0000	-0.0001	0.0000	-0.0000	-0.0018	-0.0000	-0.0000	-0.0046	-0.0000	-0.0000	0.0036	0.0000
avg_pay_amt	0.0000	0.0026	-0.0000	-0.0000	-0.0006	-0.0000	-0.0000	-0.0005	-0.0000	-0.0000	-0.0025	-0.0000
pay_ratio	-0.2651	0.0100	-0.0444	-0.4890	0.0012	-0.0998	-0.5836	0.0005	-0.0953	-0.1943	0.0015	-0.0170

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

ratio, the hardest nuisance problem in the design, the gap is largest and always towards zero: -0.265 against -0.044 for logistic regression, -0.489 against -0.100 for the random forest, -0.584 against -0.095 for XGBoost, -0.194 against -0.017 for the neural network. Months delayed runs the other way, with PDP slopes of 0.111, 0.046, 0.040 and 0.065 against debiased AMEs of 0.030, 0.013, 0.019 and 0.025, which is what one expects when an uncorrected estimator distributes a shared effect between two correlated features differently than the corrected one does. Taken across all three applications, the attenuation is the dominant pattern rather than a universal one: restricting attention to the effects the debiased estimator resolves away from zero, the partial dependence slope is smaller in absolute value than the AME in 62 of 84 cases, with a median ratio of 0.47 on Ames Housing, 0.60 here and 0.95 on Adult Income. It is most pronounced for the random forest, the learner whose regularization bias the simulations isolate.

S10 UCI Adult Income

The first dataset is the UCI Adult Income dataset (Dua and Graff, 2019; Kohavi, 1996), a widely used benchmark for binary classification in the social sciences. We combine the training and test sets for a total of 48,842 observations. The outcome is whether an individual earns more than \$50,000 per year. Features are organized into three panels: demographics (age, sex, years of education), work characteristics (hours worked per week, occupation group, work class), and financial variables (capital gains, capital losses, marital status). The specification search considers logistic regression and XGBoost across four specifications, allowing the method to be evaluated both where the parametric model is approximately correctly specified and where the black-box learner is substantially better fitted. Years of education takes the floored step size $h_j = 0.5$, so its effect is the one-year contrast; age and hours per week have enough spread that the adaptive rule already exceeds the floor, at $h_j = 0.686$ and 0.620. Trimming is negligible for every feature except capital gains and capital losses, which lose 92.3% and 95.3% of the sample for the reason discussed below.

Table S5: UCI Adult Income: Specification Search. Average marginal effects on P(Income > $50k).

	Logistic				XGBoost			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Demographics								
age	0.0084***	0.0081***	0.0041***	0.0043***	0.0059***	0.0059***	0.0031***	0.0033***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
female	-0.1749***	-0.1670***	-0.0122***	-0.0153**	-0.1604***	-0.1507***	-0.0089**	-0.0128**
	(0.0034)	(0.0045)	(0.0046)	(0.0061)	(0.0033)	(0.0041)	(0.0042)	(0.0056)
education_num	0.0464***	0.0351***	0.0371***	0.0284***	0.0606***	0.0458***	0.0417***	0.0325***
	(0.0007)	(0.0008)	(0.0006)	(0.0007)	(0.0007)	(0.0008)	(0.0006)	(0.0007)
Panel B: Work Characteristics								
hours_per_week	–	0.0035***	–	0.0027***	–	0.0084***	–	0.0038***
		(0.0002)		(0.0001)		(0.0002)		(0.0001)
government	–	0.0059	–	-0.0004	–	-0.0072	–	-0.0060
		(0.0067)		(0.0061)		(0.0066)		(0.0055)
self_employed	–	0.0218***	–	-0.0085**	–	0.0192***	–	-0.0080**
		(0.0058)		(0.0043)		(0.0056)		(0.0040)
white_collar	–	0.0649***	–	0.0561***	–	0.0480***	–	0.0478***
		(0.0053)		(0.0049)		(0.0051)		(0.0045)
blue_collar	–	-0.0246***	–	-0.0248***	–	-0.0482***	–	-0.0300***
		(0.0036)		(0.0031)		(0.0034)		(0.0028)
service	–	-0.0744***	–	-0.0456***	–	-0.0795***	–	-0.0342***
		(0.0024)		(0.0023)		(0.0024)		(0.0020)
Panel C: Financial								
capital_gain	–	–	0.0000***	0.0000***	–	–	0.0000***	0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
capital_loss	–	–	0.0000***	0.0000***	–	–	-0.0001***	-0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
married	–	–	0.2842***	0.2785***	–	–	0.2442***	0.2423***
			(0.0048)	(0.0054)			(0.0045)	(0.0047)
McFadden R^2	0.199	0.233	0.376	0.400	0.257	0.296	0.467	0.492
Accuracy	0.797	0.806	0.841	0.847	0.805	0.818	0.864	0.872

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

Table S5 presents the specification search results for the UCI Adult Income dataset. This is the largest of the three applications at 48,842 observations, and the estimates are correspondingly the most precisely determined in the paper. The demographic panel tells a consistent story across all specifications and both model classes. Age and education have positive effects on the probability of earning above $50,000, and the female indicator is large and negative in the specifications that omit marital status, at -0.175 for logistic regression and -0.160 for XGBoost, collapsing to -0.012 and -0.009 once marital status enters in specification A+C. That collapse is the single most dramatic omitted variable pattern in the paper, and it is legible only because the AME puts every specification on the same probability scale. Education runs from 0.046 to 0.028 for logistic regression and 0.061 to 0.033 for XGBoost as controls accumulate. Marital status itself carries much the largest effect in the table, 0.278 and 0.242 in the full specification.

The two model classes track each other closely throughout, and their standard errors are now of the same order everywhere — 0.0061 and 0.0056 on the female indicator in the full specification, 0.0007 and 0.0007 on education. Under the refitting bootstrap the black-box standard errors were sometimes an order of magnitude smaller than the parametric ones, which the earlier draft read as an efficiency gain from the better-fitting model. That reading was mistaken. The two models estimate the same $\theta_{j,h}$ at the same sample size, so their standard errors ought to be similar, and the earlier gap measured how stable each fitting procedure was under resampling rather than how precisely either pinned down the target. The influence-function standard errors do behave as sampling variability should: they shrink with n, they are largest for the features carrying the least independent variation, and they do not reward a model for being reproducibly wrong.

The financial panel illustrates the limits of what this design can say about capital gains and losses. Both variables are measured in raw dollars, so the reported effect is the response to a one-dollar change, and it displays as zero to four decimals in every cell but one — as it did under the plug-in. The estimates are nonetheless significant at the one percent level, because the standard errors are smaller still; with nearly fifty thousand observations a substantively negligible effect can be sharply distinguished from

zero, and this panel is a reminder that the stars answer a different question from the one the reader usually cares about. They also carry by far the heaviest trimming in the paper. Both are zero for the overwhelming majority of the sample and zero is the minimum of their observed support, so 92.3% and 95.3% of observations receive weight zero and the reported effect is identified only among filers who have nonzero gains or losses. The sign reversal between logistic regression and XGBoost that the earlier draft flagged in specification A+C does not survive the correction; both are zero to the displayed precision. Nothing here should be read as evidence about capital income, and a study interested in that question would want the variable re-expressed on a scale where a unit change is meaningful.

Table S6 compares all four model classes on the full specification, and with nearly fifty thousand observations it is the cleanest test in the paper of whether the correction delivers a common estimand. It does. The education AME is 0.028, 0.034, 0.033 and 0.028 for logistic regression, random forest, XGBoost and the neural network, and marital status is 0.279, 0.250, 0.242 and 0.244 — a spread of 0.005 and 0.036 respectively across four architectures with McFadden R^2 ranging from 0.400 to 0.492. The female indicator, which the plug-in spread from -0.027 down to -0.009 and left insignificant for the neural network, is -0.015, -0.021, -0.013 and -0.017 after correction, all four significant and all four within a standard error or two of one another. The variable the earlier draft singled out as sensitive to model specification turns out not to have been.

The standard errors tell the same story from the other side. For a common target at a common sample size, four estimators using different nuisances should report standard errors of similar magnitude, and here they do: 0.0056 to 0.0061 on the female indicator, 0.0047 to 0.0054 on marital status, 0.0020 to 0.0023 on the service indicator. Under the plug-in they did not. The random forest reported 0.0001 on the female indicator and 0.0006 on marital status, one to two orders of magnitude below every other model, which the earlier draft attributed to ensemble averaging producing a smoother prediction surface and therefore more stable finite differences across bootstrap replicates. That description was accurate as a description of the bootstrap and wrong as a statement about precision:

Table S6: UCI Adult Income: Model Comparison. Average marginal effects on P(Income > \$50k), full specification (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0043***	0.0056***	0.0033***	0.0041***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
female	-0.0153**	-0.0210***	-0.0128**	-0.0166***
	(0.0061)	(0.0058)	(0.0056)	(0.0059)
education_num	0.0284***	0.0336***	0.0325***	0.0284***
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
hours_per_week	0.0027***	0.0077***	0.0038***	0.0049***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
government	-0.0004	-0.0025	-0.0060	-0.0060
	(0.0061)	(0.0056)	(0.0055)	(0.0060)
self_employed	-0.0085**	-0.0103**	-0.0080**	-0.0071*
	(0.0043)	(0.0041)	(0.0040)	(0.0042)
white_collar	0.0561***	0.0606***	0.0478***	0.0383***
	(0.0049)	(0.0047)	(0.0045)	(0.0047)
blue_collar	-0.0248***	-0.0316***	-0.0300***	-0.0533***
	(0.0031)	(0.0030)	(0.0028)	(0.0031)
service	-0.0456***	-0.0262***	-0.0342***	-0.0758***
	(0.0023)	(0.0021)	(0.0020)	(0.0023)
capital_gain	0.0000***	0.0000***	0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
capital_loss	0.0000***	-0.0000***	-0.0000***	0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
married	0.2785***	0.2504***	0.2423***	0.2438***
	(0.0054)	(0.0048)	(0.0047)	(0.0048)
McFadden R^2	0.400	0.450	0.492	0.449
Accuracy	0.847	0.863	0.872	0.858

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

the replicates agreed with each other because every one of them was regularized the same way, not because the quantity was well determined. The convergence of the four standard-error columns to a common magnitude is what the orthogonal score is supposed to produce, and it is the more convincing for happening at the sample size where the asymptotics have the best claim to apply.

Table S7 compares AMEs against SHAP values and partial dependence plot slopes across all four models. The SHAP column measures something else and is reported here to make the difference concrete rather than to adjudicate between the two. Mean signed SHAP values are on a different scale entirely and in several cases carry the opposite sign: for logistic regression the SHAP value for education is -0.119 against an AME of 0.028, and for marital status it is -0.077 against an AME of 0.279 — opposite in sign on the two largest effects in the application. The SHAP value for age is negative for three of the four models while every AME is positive. The most striking entry is capital gains, whose SHAP value of -0.263 for logistic regression is the largest SHAP magnitude anywhere in the table, attached to a variable whose AME is zero to four decimals and 92% of whose observations are trimmed. None of this indicates an error in SHAP or in its implementation. A Shapley value is an additive attribution of a prediction relative to a reference distribution, and its average over the sample is not an estimate of the average rate of change of the prediction in a feature. The two quantities answer different questions and there is no reason to expect them to agree in magnitude or in sign. The point of the comparison is interpretive rather than evaluative: SHAP summaries are frequently read as though larger values meant larger effects on the outcome, and that reading is not supported by what the quantity measures.

The partial dependence column behaves as it did in the other two applications: it agrees with the debiased AME where the underlying plug-in is nearly unbiased and falls short of it elsewhere. For logistic regression, the best-specified model here, agreement is close on the large effects — marital status 0.279 against 0.275, the white collar indicator 0.056 against 0.054, service -0.046 against -0.043, education 0.028 against 0.027. For the two tree ensembles the same features separate substantially: the random forest education

Table S7: UCI Adult Income: Method Comparison. AME, SHAP, and PDP estimates for P(Income > $50k), full specification (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0043	-0.0270	0.0028	0.0056	-0.0068	0.0024	0.0033	-0.0097	0.0036	0.0041	0.0044	0.0033
female	-0.0153	0.0022	-0.0270	-0.0210	0.0011	-0.0218	-0.0128	0.0015	-0.0122	-0.0166	-0.0001	-0.0222
education_num	0.0284	-0.1186	0.0269	0.0336	-0.0077	0.0137	0.0325	-0.0093	0.0170	0.0284	-0.0051	0.0207
hours_per_week	0.0027	-0.0691	0.0033	0.0077	-0.0054	0.0020	0.0038	-0.0080	0.0028	0.0049	-0.0032	0.0028
government	-0.0004	-0.0009	0.0026	-0.0025	-0.0003	0.0051	-0.0060	-0.0008	-0.0003	-0.0060	0.0002	0.0082
self_employed	-0.0085	0.0016	-0.0265	-0.0103	-0.0003	0.0032	-0.0080	-0.0002	-0.0055	-0.0071	-0.0008	-0.0045
white_collar	0.0561	0.0127	0.0538	0.0606	0.0031	0.0600	0.0478	0.0025	0.0474	0.0383	0.0007	0.0312
blue_collar	-0.0248	0.0136	-0.0273	-0.0316	0.0010	-0.0240	-0.0300	0.0004	-0.0251	-0.0533	-0.0008	-0.0531
service	-0.0456	-0.0027	-0.0427	-0.0262	0.0003	-0.0176	-0.0342	0.0005	-0.0343	-0.0758	0.0011	-0.0558
capital_gain	0.0000	-0.2627	0.0000	0.0000	-0.0224	0.0000	0.0000	-0.0264	-0.0001	0.0000	0.0042	0.0000
capital_loss	0.0000	0.0027	–	-0.0000	-0.0034	–	-0.0000	-0.0062	–	0.0000	-0.0115	–
married	0.2785	-0.0768	0.2749	0.2504	-0.0064	0.2446	0.2423	-0.0049	0.2357	0.2438	0.0149	0.1958

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

AME of 0.034 sits against a PDP slope of 0.014, and its hours-per-week AME of 0.008 against a slope of 0.002. A piecewise constant fit has little local slope to report between splits, and the partial dependence average inherits that flatness while the orthogonal correction does not. Partial dependence slopes for capital losses are unavailable for every model, a limitation of the grid construction when almost all mass sits at a single value. As with SHAP, no standard errors accompany either column, so a researcher working from them has point estimates of unknown precision and no way to test whether any variable has a statistically distinguishable effect.

S11 Ames Housing

The third dataset is the Ames Housing dataset (De Cock, 2011), comprising approximately 1,460 observations of residential home sales in Ames, Iowa between 2006 and 2010. The outcome is sale price in dollars, so average marginal effects are interpreted in dollar units — a one-unit change in a continuous feature is associated with a $\widehat{\theta}_{j,h}$ change in predicted sale price, on average across the untrimmed sample. Features are organized into physical characteristics (above-ground living area, number of bedrooms, number of bathrooms), quality and condition (overall quality rating, condition rating, year built, year remodeled), and lot and location (lot area, garage area, neighborhood tier indicators). The specification search considers linear regression and a random forest, covering the regression case and completing the sweep across outcome types. Six of the ten continuous features here are integer counts or ratings and so take the floored step size $h_j = 0.5$, which makes their reported effects one-unit contrasts: one more bedroom, one more point of overall quality. One feature requires a narrower reading than the rest. Half bathrooms take only the values zero, one and two, and both outer levels lie on the boundary of the observed support, so 63.4% of the sample is trimmed and the half-bath effect is identified only among homes that have exactly one. The remaining features in this application are trimmed at 15% or below.

Table S8 presents the specification search. The regression setting allows AMEs to

Table S8: Ames Housing: Specification Search. Average marginal effects on Sale Price (USD).

	Linear				Random Forest			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Physical Characteristics								
GrLivArea	138.96***	106.16***	84.82***	68.05***	128.77***	103.42***	90.19***	81.50***
	(20.51)	(15.58)	(14.25)	(12.14)	(13.06)	(8.98)	(8.97)	(8.77)
BedroomAbvGr	-33209.66***	-14046.40*	-24282.55***	-26454.60*	-28632.01***	-11415.18***	-11292.79**	-9212.02
	(5594.51)	(7435.20)	(9027.62)	(14881.54)	(3382.73)	(3920.43)	(5576.82)	(8925.07)
FullBath	18589.00***	-10533.55**	7261.14	-10578.42	16236.01***	-6616.40*	5652.46	7858.42
	(5990.54)	(4912.27)	(4987.91)	(9969.12)	(3779.18)	(3670.56)	(3754.82)	(8988.79)
HalfBath	-11228.94**	-8180.81**	-9092.88**	1028.31	-7189.23*	-6638.05***	-11578.88**	-10317.96**
	(4624.11)	(3242.49)	(4451.09)	(5697.41)	(4264.04)	(2525.05)	(4601.22)	(4713.92)
Panel B: Quality and Condition								
OverallQual	–	15819.18***	–	5436.28	–	11911.79***	–	16255.82***
		(1589.02)		(4964.23)		(1342.87)		(4829.24)
OverallCond	–	6477.48*	–	2890.60	–	8281.97***	–	9190.11***
		(3687.03)		(5496.46)		(2434.71)		(3492.01)
YearBuilt	–	443.39	–	-141.08	–	1033.46***	–	1323.39***
		(277.71)		(404.21)		(190.55)		(419.89)
YearRemodAdd	–	192.69	–	1743.45**	–	556.20***	–	-311.44
		(161.82)		(740.09)		(173.98)		(753.53)
Panel C: Lot and Location								
LotArea	–	–	5.51	23.02**	–	–	-7.53	-12.51
			(3.93)	(11.37)			(8.06)	(11.14)
GarageArea	–	–	40.08*	-102.78	–	–	103.31***	90.33
			(20.94)	(71.57)			(34.64)	(59.74)
nbhd_high	–	–	128801.76***	89812.31**	–	–	66978.55***	43512.42*
			(34103.92)	(38393.43)			(16110.93)	(24635.87)
nbhd_mid	–	–	77339.38**	74336.31*	–	–	42305.81***	33767.43
			(33663.73)	(39182.81)			(15387.40)	(26676.67)
R^2	0.584	0.758	0.739	0.802	0.829	0.937	0.926	0.953
RMSE	51,243.9	39,080.1	40,596.4	35,355.5	32,830.9	19,881.6	21,534.2	17,244.4

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

be interpreted directly in dollar units, which makes the substantive findings immediately legible. Above-ground living area carries an AME of \$139 per square foot in the physical-characteristics specification for linear regression, falling to \$106, \$85 and finally \$68 as quality, condition, and location controls are added — a pattern of positive omitted variable bias that makes intuitive sense, since larger homes tend to be in better neighborhoods and of higher quality. The random forest traces the same path from \$129 to \$81. That the two model classes track each other this closely across the sequence is the first thing to note about the debiased estimates: because the orthogonal correction removes the first-order dependence on the fitted regression, the two columns are estimating the same window effect rather than reporting the slope each model happens to believe, and the linear column is no longer simply the ordinary least squares coefficient. The standard errors are now of comparable size across the two model classes — \$12 for linear regression and \$9 for the random forest on GrLivArea in the full specification — rather than differing by the order of magnitude that the refitting bootstrap produced, where the ensemble's smooth prediction surface made every replicate agree with every other and drove the reported standard error to \$1.93. Those intervals were narrow around a biased centre; the influence-function intervals are wider and honest.

The cost of that honesty is visible in the quality panel. The overall quality rating is estimated at \$15,819 per point for linear regression and \$11,912 for the random forest in specification A+B, both precisely determined, but in the full specification the linear estimate falls to \$5,436 with a standard error of \$4,964 and is no longer distinguishable from zero, while the random forest gives \$16,256 with a standard error of \$4,829. With $n = 1{,}460$ observations, twelve features and a second-order sieve for the representer, the full specification is close to the limit of what this sample can support: the estimator reports that limit rather than concealing it. The neighbourhood tier indicators show the same attenuation as before — the high neighbourhood effect falls from \$128,802 to \$89,812 for linear regression and from \$66,979 to \$43,512 for the random forest as quality controls are added — but with standard errors near \$25,000 to \$38,000, the attenuation is a pattern in the point estimates rather than a difference the data can resolve.

Table S9: Ames Housing: Model Comparison. Average marginal effects on Sale Price (USD), full specification (A+B+C).

Feature	Linear	Random Forest	XGBoost	Neural Net
GrLivArea	68.05***	81.50***	70.27***	66.20***
	(12.14)	(8.77)	(6.60)	(6.24)
BedroomAbvGr	-26454.60*	-9212.02	-3464.31	-12638.60***
	(14881.54)	(8925.07)	(4508.19)	(3830.14)
FullBath	-10578.42	7858.42	4761.27	-5055.91
	(9969.12)	(8988.79)	(7661.72)	(5080.88)
HalfBath	1028.31	-10317.96**	-10906.00***	-2175.44
	(5697.41)	(4713.92)	(3940.81)	(3286.03)
OverallQual	5436.28	16255.82***	14913.45***	8524.39***
	(4964.23)	(4829.24)	(4271.98)	(2394.41)
OverallCond	2890.60	9190.11***	12102.59***	7234.42**
	(5496.46)	(3492.01)	(2519.57)	(2920.49)
YearBuilt	-141.08	1323.39***	1352.73***	230.04
	(404.21)	(419.89)	(366.04)	(267.61)
YearRemodAdd	1743.45**	-311.44	-289.89	942.55***
	(740.09)	(753.53)	(673.39)	(351.83)
LotArea	23.02**	-12.51	-10.83	10.04**
	(11.37)	(11.14)	(9.91)	(4.81)
GarageArea	-102.78	90.33	92.29*	-6.11
	(71.57)	(59.74)	(49.36)	(27.32)
nbhd_high	89812.31**	43512.42*	33476.93**	35458.39**
	(38393.43)	(24635.87)	(16103.49)	(17705.63)
nbhd_mid	74336.31*	33767.43	17468.12	36951.86**
	(39182.81)	(26676.67)	(18286.73)	(17998.56)
R^2	0.802	0.953	0.954	0.932
RMSE	35,355.5	17,244.4	17,036.6	20,699.9

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are the debiased cross-fitted window AME with $K = 5$ folds; standard errors from the orthogonal score in parentheses. Step sizes h_j and trimmed fractions are reported in Table S2.

Table S9 compares all four models on the full specification. All four are well fitted, with R^2 ranging from 0.802 for linear regression to 0.954 for XGBoost, the random forest at 0.953 and the neural network at 0.932 with normalization. The GrLivArea AME is \$68, \$81, \$70, and \$66 for linear regression, random forest, XGBoost, and neural network respectively — a range of \$15 across four models that differ fundamentally in their functional form, and all four intervals overlap comfortably. Under the debiased estimator this agreement means something different from what it meant under the plug-in. Every column is now an estimate of the same $\theta_{j,h}$, so agreement is evidence that the four nuisance estimates are all good enough for the correction to do its work, not evidence that four different models happen to have similar slopes. Disagreement, correspondingly, is now diagnostic of nuisance quality rather than of functional form.

The most consequential change from the plug-in results is in the standard errors, and it runs in the opposite direction to the one the refitting bootstrap suggested. Under that bootstrap the neural network's standard errors were implausibly tight — on the order of single dollars for several variables, \$7 for bedrooms and \$8 for overall quality — because resampling propagated uncertainty through the model fitting but the smooth differentiable surface produced nearly identical finite differences across replicates. The influence-function standard errors for the same quantities are \$3,830 and \$2,394. The random forest shows the same correction: the high-neighborhood standard error rises from \$256 to \$24,636. Those earlier figures were not precision but the bootstrap's inability to see a bias that every replicate shared, which is the failure mode of Remark S4, and their disappearance is the clearest single indication in the empirical results that the orthogonal score is doing what it was constructed to do.

Read with those intervals attached, the bedroom variable is more equivocal than the plug-in made it look. The AME is −\$26,455 for linear regression, −\$9,212 for the random forest, −\$3,464 for XGBoost, and −\$12,639 for the neural network. All four are negative, consistent with the well-known hedonic finding that holding square footage fixed, more bedrooms means smaller rooms, but only the neural network estimate is significant at conventional levels and the linear estimate only marginally so. The plug-in reported all

four as significant with standard errors between \$7 and \$3,042. The honest reading is that the sign is stable across architectures while this sample cannot pin down the magnitude, which is a weaker conclusion than the paper could previously state and a better supported one.

Table S10 makes the difference between the two quantities easiest to see, since the units are dollars and the magnitudes are large enough to be concrete. SHAP values are unavailable for the neural network in this table, reflecting a limitation of the GradientExplainer implementation for this architecture. For the other three models the two columns are far apart. The SHAP value for GrLivArea is \$3,423 for the linear model against an AME of \$68, roughly fifty times larger — which is what one should expect, since the SHAP figure is the average absolute displacement a typical house's living area contributes to its predicted price, not the price change associated with one additional square foot. The SHAP values for garage area (−\$884), lot area (−\$183), and the high neighborhood indicator (−\$506) are negative for the random forest while the AME estimates are positive for all three. And as before, no standard errors attach to the SHAP column, so a reader who did want to treat those figures as effects would have no way to gauge their precision.

The partial dependence column requires a different reading than it did under the plug-in, and the change is itself informative. A partial dependence slope is computed by perturbing a single full-sample fit and averaging, which makes it a plug-in average of exactly the kind the orthogonal correction is designed to repair; it carries no correction term and no standard error. Where the plug-in is nearly unbiased the two therefore continue to agree closely: on above-ground living area, the best supported and smoothest feature in the data, the AME is \$68 and the PDP slope \$66 for linear regression, \$66 and \$59 for the neural network. Where the regularization bias bites, they separate, and they separate in the direction the theory predicts — towards zero for the uncorrected quantity. The bedroom AME of −\$26,455 for linear regression sits against a PDP slope of −\$6,189; the high-neighbourhood AME of \$43,512 for the random forest sits against a PDP slope of \$5,059, and the mid-neighbourhood AME of \$33,767 against \$5,950.

Table S10: Ames Housing: Method Comparison. AME, SHAP, and PDP estimates for Sale Price (USD), full specification (A+B+C).

	Linear			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
GrLivArea	68.05	3423.49	65.57	81.50	61.69	38.64	70.27	-971.88	39.06	66.20	–	58.62
BedroomAbvGr	-26454.60	145.82	-6189.02	-9212.02	90.87	-1272.27	-3464.31	158.44	-4456.61	-12638.60	–	-9294.43
FullBath	-10578.42	31.01	-6287.42	7858.42	34.06	543.63	4761.27	54.86	193.04	-5055.91	–	2427.62
HalfBath	1028.31	-298.76	-6967.90	-10317.96	50.46	-125.75	-10906.00	30.53	-1495.20	-2175.44	–	-2723.69
OverallQual	5436.28	1635.24	16465.22	16255.82	596.70	20776.14	14913.45	713.13	20130.92	8524.39	–	10802.73
OverallCond	2890.60	980.44	6311.48	9190.11	161.26	2030.44	12102.59	-481.60	5071.47	7234.42	–	6365.50
YearBuilt	-141.08	-415.01	440.47	1323.39	288.29	209.05	1352.73	569.00	397.33	230.04	–	421.27
YearRemodAdd	1743.45	86.86	101.51	-311.44	373.04	118.06	-289.89	499.84	152.00	942.55	–	561.16
LotArea	23.02	267.44	0.61	-12.51	-183.50	1.33	-10.83	216.28	2.38	10.04	–	2.35
GarageArea	-102.78	93.22	38.68	90.33	-884.31	23.77	92.29	-704.59	28.98	-6.11	–	41.48
nbhd_high	89812.31	77.50	51433.02	43512.42	-506.36	5058.64	33476.93	50.70	20307.34	35458.39	–	12399.98
nbhd_mid	74336.31	266.38	7508.07	33767.43	40.68	5949.97	17468.12	-134.71	9744.97	36951.86	–	11392.44

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

These are the count and binary contrasts, where the fitted surface is flattest between levels and where the plug-in has the most to lose. The pattern that once read as mutual corroboration between two estimators now reads as a measurement of the bias separating them, which is the same quantity Simulation 2 isolates in the coverage of the refitting bootstrap. The Ames Housing results complete the sweep across outcome types: the framework produces dollar-denominated marginal effects with valid standard errors for a continuous regression outcome, using exactly the same estimation procedure as the binary classification examples, with no modification required.

S12 Detailed simulation results

This section reports the three simulation studies in full: every sample size, every data generating process, every feature, and the sieve-representer and bounded-support arms of Simulation 3. The designs and outcome types are described in Section 5 of the main paper, the learners and their hyperparameters are set out immediately below, and Table 2 there summarizes the results that follow.

Learners and hyperparameters

The simulation study fits four model classes with hyperparameters chosen to balance computational tractability against representativeness of typical applied use. The random forest uses 100 trees with a maximum depth of 6, and XGBoost uses 100 boosting rounds with a maximum depth of 4 and a learning rate of 0.05 — conservative settings that prevent overfitting on the small sample sizes considered while remaining flexible enough to capture the nonlinear and interaction structures in the data generating processes. The neural network is a feedforward architecture with two hidden layers of 32 and 16 units, trained for 50 epochs with batch size 64 and learning rate 10^{-3} using the Adam optimizer. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2 to ensure scale-invariant gradient updates. Logistic and linear regression are included as baselines and use no hyperparameter

tuning. All model fits are performed independently on each Monte Carlo iteration with no warm starting between iterations, so the variance estimates reflect the full sampling distribution of the AME estimator including model fitting variability. The Monte Carlo ground truth AMEs are computed once at the start of the study using n = 1,000,000 observations drawn from the true prediction function directly, with the resulting AMEs treated as known quantities for the bias and coverage calculations. Simulation 2 uses 1,000 bootstrap resamples per Monte Carlo iteration and 1,000 Monte Carlo iterations per cell for the linear, logistic, random forest and XGBoost learners. The neural network uses 200 resamples and 500 iterations: every resample is a fresh training run, which makes the network roughly nine tenths of the total cost of the study, and the calibration result the argument rests on is the forest's. Percentile intervals are used throughout, which is why the resample count is set well above the couple of hundred that would suffice for a standard error alone. Simulation 3 uses 1,000 Monte Carlo iterations per cell for every learner; it needs no resampling, since its standard errors come from the influence function.

S12.1 Simulation 1: Recovering the Window Effect

Table S11 reports bias and RMSE for the linear classification DGP, where the true AMEs of x_1 and x_2 are 0.199 and 0.298 — these are not 2.0 and 3.0 because the sigmoid compresses the scale. Logistic regression recovers the true AMEs with negligible bias at all sample sizes, and RMSE declines from around 0.020 at $n = 250$ to 0.005 at $n = 5{,}000$ at the expected $O(n^{-1/2})$ rate. The neural network behaves similarly, with small bias at $n = 250$ that disappears by $n = 500$. XGBoost shows larger bias at small samples — around 0.06 for both x_1 and x_2 at $n = 250$ — reflecting the well-known tendency of gradient boosted models to underfit at small sample sizes, but bias falls sharply and RMSE converges by $n = 2{,}500$. Random forests show a different pattern: bias is small at $n = 250$ and $n = 500$ but turns negative and persistent at $n = 2{,}500$ and $n = 5{,}000$, reaching around -0.016 for both signal features at the largest sample size. This reflects the regularization bias inherent in random forests — the tree depth and subsampling constraints impose shrinkage that does not vanish with sample size at the default tuning

Table S11: Simulation 1: AME Recovery — Linear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.199	-0.0058	-0.0034	-0.0014	-0.0008	-0.0006
		(0.0201)	(0.0146)	(0.0106)	(0.0064)	(0.0048)
x2	0.298	-0.0109	-0.0057	-0.0026	-0.0010	-0.0005
		(0.0229)	(0.0153)	(0.0105)	(0.0066)	(0.0046)
x3	0.000	0.0012	0.0006	0.0006	0.0003	0.0002
		(0.0198)	(0.0136)	(0.0099)	(0.0063)	(0.0045)
x4	0.000	0.0008	-0.0001	-0.0008	-0.0001	0.0003
		(0.0197)	(0.0136)	(0.0100)	(0.0062)	(0.0045)
Panel: Random Forest						
x1	0.199	0.0126	0.0044	-0.0018	-0.0111	-0.0157
		(0.0412)	(0.0239)	(0.0160)	(0.0142)	(0.0168)
x2	0.298	0.0199	0.0113	-0.0008	-0.0102	-0.0158
		(0.0500)	(0.0309)	(0.0174)	(0.0143)	(0.0172)
x3	0.000	0.0007	0.0010	0.0005	0.0002	0.0002
		(0.0284)	(0.0171)	(0.0104)	(0.0056)	(0.0034)
x4	0.000	-0.0008	-0.0013	-0.0012	-0.0001	0.0002
		(0.0267)	(0.0177)	(0.0106)	(0.0055)	(0.0034)
Panel: XGBoost						
x1	0.199	0.0590	0.0265	0.0107	-0.0016	-0.0049
		(0.0782)	(0.0401)	(0.0223)	(0.0108)	(0.0083)
x2	0.298	0.0643	0.0315	0.0085	-0.0015	-0.0046
		(0.0893)	(0.0490)	(0.0218)	(0.0116)	(0.0083)
x3	0.000	0.0008	-0.0002	0.0006	0.0002	0.0002
		(0.0332)	(0.0193)	(0.0110)	(0.0056)	(0.0033)
x4	0.000	0.0008	-0.0009	-0.0006	0.0000	0.0002
		(0.0344)	(0.0198)	(0.0116)	(0.0054)	(0.0032)
Panel: Neural Net						
x1	0.199	-0.0078	-0.0007	0.0005	0.0003	-0.0002
		(0.0222)	(0.0156)	(0.0115)	(0.0074)	(0.0058)
x2	0.298	-0.0162	-0.0017	0.0004	0.0004	0.0002
		(0.0276)	(0.0174)	(0.0116)	(0.0073)	(0.0052)
x3	0.000	0.0014	0.0010	0.0007	0.0003	0.0004
		(0.0204)	(0.0144)	(0.0103)	(0.0069)	(0.0057)
x4	0.000	0.0009	-0.0003	-0.0007	-0.0001	0.0004
		(0.0203)	(0.0143)	(0.0105)	(0.0070)	(0.0055)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

settings. The noise features x_3 and x_4 have bias and RMSE near zero throughout for all model classes, confirming that the estimator does not spuriously attribute effects to irrelevant features.

Table S12 reports results for the nonlinear classification DGP, where $P(y = 1) = \sigma(2x_1^2 + 3x_2)$. The true AME of x_1 is zero despite the quadratic term, because the sigmoid and quadratic effects cancel in expectation at these parameter values as confirmed by the Monte Carlo ground truth. All four model classes recover this near-zero AME accurately, with bias well below 0.005 across all sample sizes. The more important result is x_2, which has true AME of 0.298. Logistic regression misspecifies the DGP — the true relationship is nonlinear in x_1 — but still recovers the x_2 AME accurately because x_2 enters linearly and the misspecification in x_1 does not propagate to the x_2 marginal effect. This is an important practical result: AMEs from a misspecified parametric model can still be reliable for variables that enter the true DGP in a manner consistent with the model's functional form. The black-box models show the same pattern as the linear DGP: XGBoost has elevated RMSE at small samples, random forests develop negative bias at large samples, and neural networks perform comparably to logistic regression throughout.

Table S13 reports results for the interaction DGP, where $P(y = 1) = \sigma(2x_1 + 3x_2 + 2x_1x_2)$. The true AMEs of x_1 and x_2 remain 0.199 and 0.298 — the same as the linear DGP — because the interaction term $2x_1x_2$ has zero expectation under $N(0, 1)$ and the Monte Carlo ground truth confirms near-identical values. The key finding is that all four model classes recover these AMEs with bias and RMSE patterns essentially identical to the linear DGP, despite the true DGP including a multiplicative interaction. This is by design of the AME estimand: the average marginal effect averages over the joint distribution of the features, so the interaction term is absorbed into the average derivative rather than requiring explicit modelling. A logistic regression that omits the interaction term recovers the correct AME because the interaction's contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2. This is an important robustness result — AMEs are not sensitive to interaction misspecification when the interactions involve zero-mean features, which is the common case after centring.

Table S12: Simulation 1: AME Recovery — Nonlinear DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.000	0.0002	0.0015	0.0009	0.0002	-0.0001
		(0.0230)	(0.0166)	(0.0123)	(0.0078)	(0.0055)
x2	0.298	-0.0077	-0.0042	-0.0017	-0.0007	-0.0005
		(0.0215)	(0.0152)	(0.0109)	(0.0065)	(0.0048)
x3	0.000	0.0003	0.0001	-0.0002	-0.0001	0.0003
		(0.0229)	(0.0158)	(0.0112)	(0.0074)	(0.0052)
x4	0.000	-0.0012	0.0005	0.0006	-0.0001	-0.0000
		(0.0252)	(0.0159)	(0.0108)	(0.0075)	(0.0051)
Panel: Random Forest						
x1	0.000	0.0014	0.0001	0.0000	-0.0003	-0.0000
		(0.0407)	(0.0256)	(0.0173)	(0.0114)	(0.0082)
x2	0.298	0.0177	0.0060	-0.0040	-0.0120	-0.0155
		(0.0521)	(0.0278)	(0.0182)	(0.0157)	(0.0171)
x3	0.000	0.0019	-0.0000	0.0003	0.0001	0.0003
		(0.0285)	(0.0166)	(0.0100)	(0.0052)	(0.0031)
x4	0.000	-0.0026	-0.0007	0.0005	0.0000	-0.0001
		(0.0295)	(0.0165)	(0.0099)	(0.0054)	(0.0031)
Panel: XGBoost						
x1	0.000	0.0039	0.0003	0.0004	-0.0000	-0.0003
		(0.0549)	(0.0324)	(0.0209)	(0.0123)	(0.0079)
x2	0.298	0.0471	0.0221	0.0066	-0.0028	-0.0054
		(0.0787)	(0.0396)	(0.0208)	(0.0114)	(0.0087)
x3	0.000	0.0038	0.0007	0.0007	0.0002	0.0003
		(0.0307)	(0.0172)	(0.0108)	(0.0051)	(0.0028)
x4	0.000	-0.0016	0.0010	0.0006	0.0001	0.0001
		(0.0329)	(0.0184)	(0.0105)	(0.0052)	(0.0027)
Panel: Neural Net						
x1	0.000	0.0006	0.0015	0.0002	0.0001	0.0000
		(0.0234)	(0.0189)	(0.0140)	(0.0094)	(0.0069)
x2	0.298	-0.0132	-0.0026	0.0002	0.0005	0.0001
		(0.0249)	(0.0155)	(0.0112)	(0.0077)	(0.0064)
x3	0.000	0.0001	0.0002	0.0007	0.0003	0.0003
		(0.0215)	(0.0138)	(0.0101)	(0.0068)	(0.0053)
x4	0.000	-0.0007	0.0004	0.0005	-0.0001	0.0002
		(0.0230)	(0.0144)	(0.0100)	(0.0070)	(0.0056)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

Table S13: Simulation 1: AME Recovery — Interaction DGP (Classification). Mean bias and RMSE (in parentheses) across 500 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.199	-0.0065	-0.0042	-0.0021	-0.0008	-0.0007
		(0.0219)	(0.0153)	(0.0112)	(0.0071)	(0.0051)
x2	0.298	-0.0083	-0.0035	-0.0018	-0.0008	-0.0007
		(0.0222)	(0.0154)	(0.0102)	(0.0071)	(0.0050)
x3	0.000	0.0009	-0.0002	0.0005	0.0003	-0.0001
		(0.0206)	(0.0155)	(0.0107)	(0.0070)	(0.0046)
x4	0.000	0.0007	-0.0000	-0.0001	0.0001	0.0002
		(0.0212)	(0.0159)	(0.0106)	(0.0066)	(0.0047)
Panel: Random Forest						
x1	0.199	0.0114	0.0051	-0.0031	-0.0110	-0.0154
		(0.0423)	(0.0264)	(0.0169)	(0.0144)	(0.0167)
x2	0.298	0.0178	0.0104	-0.0023	-0.0101	-0.0140
		(0.0498)	(0.0320)	(0.0200)	(0.0148)	(0.0158)
x3	0.000	0.0019	0.0004	0.0002	0.0002	-0.0000
		(0.0276)	(0.0170)	(0.0104)	(0.0059)	(0.0034)
x4	0.000	-0.0005	-0.0011	-0.0001	0.0001	-0.0000
		(0.0267)	(0.0177)	(0.0111)	(0.0057)	(0.0033)
Panel: XGBoost						
x1	0.199	0.0511	0.0225	0.0079	-0.0014	-0.0046
		(0.0737)	(0.0392)	(0.0219)	(0.0105)	(0.0082)
x2	0.298	0.0496	0.0228	0.0042	-0.0021	-0.0047
		(0.0796)	(0.0439)	(0.0227)	(0.0125)	(0.0091)
x3	0.000	0.0004	0.0005	0.0004	0.0000	0.0000
		(0.0310)	(0.0205)	(0.0111)	(0.0064)	(0.0035)
x4	0.000	-0.0012	-0.0015	-0.0004	0.0001	0.0001
		(0.0338)	(0.0205)	(0.0118)	(0.0060)	(0.0037)
Panel: Neural Net						
x1	0.199	-0.0089	-0.0013	0.0001	0.0000	-0.0004
		(0.0233)	(0.0158)	(0.0117)	(0.0079)	(0.0061)
x2	0.298	-0.0114	-0.0002	0.0007	0.0005	-0.0002
		(0.0257)	(0.0170)	(0.0131)	(0.0085)	(0.0067)
x3	0.000	0.0008	0.0004	0.0008	0.0001	0.0000
		(0.0193)	(0.0148)	(0.0099)	(0.0070)	(0.0053)
x4	0.000	0.0005	-0.0002	0.0000	-0.0001	0.0006
		(0.0206)	(0.0151)	(0.0103)	(0.0069)	(0.0057)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

Table S14: Simulation 1: AME Recovery — Linear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.0017	-0.0001	-0.0000	-0.0006	-0.0003
		(0.0659)	(0.0445)	(0.0322)	(0.0207)	(0.0137)
x2	3.000	-0.0043	-0.0017	0.0007	-0.0004	-0.0003
		(0.0636)	(0.0453)	(0.0316)	(0.0197)	(0.0140)
x3	0.000	-0.0001	-0.0006	-0.0005	0.0012	-0.0003
		(0.0650)	(0.0457)	(0.0312)	(0.0199)	(0.0140)
x4	0.000	-0.0034	0.0006	-0.0002	-0.0003	-0.0001
		(0.0642)	(0.0452)	(0.0315)	(0.0196)	(0.0139)
Panel: Random Forest						
x1	2.000	-0.0973	-0.0332	-0.0387	-0.0703	-0.0911
		(0.2064)	(0.1189)	(0.0850)	(0.0824)	(0.0956)
x2	3.000	-0.0215	-0.0105	-0.0433	-0.0797	-0.0953
		(0.2338)	(0.1456)	(0.0997)	(0.0958)	(0.1025)
x3	0.000	-0.0015	-0.0005	-0.0005	0.0001	0.0000
		(0.0562)	(0.0290)	(0.0137)	(0.0036)	(0.0011)
x4	0.000	-0.0021	-0.0010	-0.0005	-0.0002	0.0000
		(0.0574)	(0.0297)	(0.0138)	(0.0037)	(0.0011)
Panel: XGBoost						
x1	2.000	0.4114	0.1780	0.0366	-0.0337	-0.0467
		(0.4644)	(0.2227)	(0.0917)	(0.0554)	(0.0547)
x2	3.000	0.3122	0.0948	-0.0172	-0.0558	-0.0601
		(0.4103)	(0.1884)	(0.0947)	(0.0754)	(0.0677)
x3	0.000	0.0007	0.0016	-0.0003	0.0011	-0.0001
		(0.1090)	(0.0550)	(0.0313)	(0.0153)	(0.0083)
x4	0.000	-0.0080	0.0002	0.0011	-0.0006	0.0001
		(0.1077)	(0.0597)	(0.0314)	(0.0149)	(0.0081)
Panel: Neural Net						
x1	2.000	-0.0659	-0.0208	-0.0063	-0.0021	-0.0008
		(0.1161)	(0.0544)	(0.0344)	(0.0293)	(0.0329)
x2	3.000	-0.1231	-0.0349	-0.0102	-0.0031	-0.0006
		(0.1765)	(0.0638)	(0.0361)	(0.0324)	(0.0372)
x3	0.000	0.0014	-0.0006	-0.0003	0.0009	-0.0006
		(0.0788)	(0.0484)	(0.0330)	(0.0250)	(0.0280)
x4	0.000	-0.0002	0.0008	-0.0003	-0.0004	-0.0017
		(0.0752)	(0.0496)	(0.0336)	(0.0247)	(0.0287)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

Table S14 turns to the regression setting with the linear DGP, $y = 2x_1 + 3x_2 + \varepsilon$. Linear regression recovers the true AMEs of 2.0 and 3.0 with negligible bias across all sample sizes, and RMSE declines from around 0.066 at $n = 250$ to 0.014 at $n = 5{,}000$, consistent with $O(n^{-1/2})$ convergence. The regression setting reveals the regularization bias of random forests more starkly than the classification setting: bias reaches -0.091 for x_1 and -0.095 for x_2 at $n = 5{,}000$, and RMSE is not declining at the largest sample sizes, indicating that the bias is not vanishing with n at default tuning. This is a known property of random forests in the regression case — the ensemble averaging and subsampling impose shrinkage towards zero that does not vanish at these sample sizes. XGBoost shows large positive bias at small samples — 0.41 for x_1 at $n = 250$ — but converges quickly, with bias falling below 0.05 by $n = 2{,}500$. The neural network shows moderate negative bias at small samples that falls towards zero by $n = 2{,}500$, though RMSE does not decline monotonically at the largest sample sizes, suggesting residual variance from the optimization landscape. The noise features have RMSE near zero for linear regression and declining rapidly for the black-box models, confirming that the estimator correctly attributes near-zero effects to irrelevant features in the regression setting.

Table S15 reports results for the nonlinear regression DGP, $y = 2x_1^2 + 3x_2 + \varepsilon$, where the true AME of x_1 is zero by symmetry of $N(0, 1)$ and the true AME of x_2 is 3.0. Linear regression recovers the x_2 AME accurately with RMSE declining at the standard rate, and correctly attributes a near-zero AME to x_1 despite misspecifying its functional form — the same robustness result observed in the classification case. However, the RMSE for x_1 under linear regression is substantially larger than in the linear DGP — 0.387 at $n = 250$ versus 0.066 — reflecting increased estimation variance from the misspecified model. The most striking result is random forests: the bias for x_2 reaches -0.185 at $n = 5{,}000$ and is not declining, indicating that the regularization bias is amplified in the nonlinear setting where the forest must capture the quadratic relationship in x_1 while simultaneously estimating the linear effect of x_2. XGBoost and the neural network both show elevated RMSE relative to the linear DGP but declining bias, suggesting that these models are better able to adapt to the nonlinear DGP with sufficient data. The x_1 AME

Table S15: Simulation 1: AME Recovery — Nonlinear DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	0.000	0.0241	0.0261	0.0148	0.0048	0.0015
		(0.3868)	(0.2820)	(0.1966)	(0.1272)	(0.0904)
x2	3.000	-0.0145	-0.0041	-0.0012	-0.0002	0.0001
		(0.1921)	(0.1276)	(0.0930)	(0.0572)	(0.0417)
x3	0.000	-0.0025	0.0001	-0.0018	-0.0020	-0.0009
		(0.1889)	(0.1343)	(0.0941)	(0.0576)	(0.0414)
x4	0.000	-0.0114	-0.0003	0.0023	0.0006	-0.0018
		(0.1857)	(0.1328)	(0.0927)	(0.0594)	(0.0430)
Panel: Random Forest						
x1	0.000	0.0071	0.0097	0.0053	0.0006	0.0007
		(0.3433)	(0.2547)	(0.1836)	(0.1185)	(0.0903)
x2	3.000	0.0142	-0.1152	-0.1701	-0.1867	-0.1849
		(0.2704)	(0.2008)	(0.2004)	(0.1974)	(0.1916)
x3	0.000	0.0001	0.0008	0.0003	-0.0002	0.0000
		(0.0549)	(0.0274)	(0.0138)	(0.0055)	(0.0027)
x4	0.000	-0.0009	-0.0007	-0.0001	-0.0000	-0.0000
		(0.0573)	(0.0278)	(0.0138)	(0.0054)	(0.0025)
Panel: XGBoost						
x1	0.000	0.0552	0.0038	-0.0117	-0.0137	-0.0057
		(0.4662)	(0.2744)	(0.1829)	(0.1040)	(0.0735)
x2	3.000	0.2725	0.0816	-0.0183	-0.0702	-0.0794
		(0.3937)	(0.1981)	(0.1058)	(0.0885)	(0.0862)
x3	0.000	-0.0040	-0.0022	-0.0020	-0.0011	-0.0006
		(0.1124)	(0.0704)	(0.0446)	(0.0191)	(0.0097)
x4	0.000	-0.0052	-0.0028	-0.0021	0.0003	0.0006
		(0.1121)	(0.0636)	(0.0381)	(0.0189)	(0.0096)
Panel: Neural Net						
x1	0.000	0.0201	0.0178	0.0065	-0.0003	-0.0013
		(0.3086)	(0.1925)	(0.1390)	(0.0911)	(0.0760)
x2	3.000	-0.1545	-0.0202	-0.0150	-0.0098	-0.0022
		(0.2601)	(0.0710)	(0.0461)	(0.0343)	(0.0395)
x3	0.000	0.0017	0.0004	0.0007	0.0009	-0.0004
		(0.1419)	(0.0634)	(0.0400)	(0.0293)	(0.0321)
x4	0.000	-0.0028	-0.0001	-0.0013	-0.0004	-0.0016
		(0.1410)	(0.0626)	(0.0406)	(0.0281)	(0.0323)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

of zero is recovered accurately by all model classes at moderate sample sizes, which is a reassuring result for practitioners who want to test whether nonlinear features have nonzero average effects.

Table S16 reports results for the interaction regression DGP, $y = 2x_1 + 3x_2 + 2x_1x_2 + \varepsilon$, with true AMEs of 2.0 and 3.0 for x_1 and x_2. Linear regression recovers these AMEs accurately despite omitting the interaction term, confirming the robustness result from the classification case: the interaction contribution to the marginal effect of x_1 is $\mathbb{E}[2x_2] = 0$ and similarly for x_2, so the misspecified linear model nevertheless recovers the correct AME. The RMSE for the noise features under linear regression is elevated relative to the linear DGP — around 0.14 at $n = 250$ — because the interaction term inflates the residual variance and increases estimation uncertainty for all coefficients. Random forests again show persistent negative bias that grows rather than shrinks with sample size, reaching -0.080 and -0.114 for x_1 and x_2 respectively at $n = 5{,}000$. XGBoost shows the same large positive bias at small samples followed by convergence, but the bias does not fully vanish at $n = 5{,}000$ — around -0.043 for x_1 and -0.059 for x_2 — suggesting that the interaction structure requires larger samples for the gradient boosting to fully adapt. The neural network performs well, with bias below 0.003 for both signal features at $n = 2{,}500$ and $n = 5{,}000$, and RMSE declining steadily. Taken together the six bias and RMSE tables establish that the AME estimator is consistent for logistic regression, neural networks, and XGBoost across all three DGPs and both outcome types, with random forests showing persistent regularization bias in the regression setting that practitioners should be aware of when reporting AMEs from default-tuned forest models.

S12.2 Simulation 2: The Refitting Bootstrap

Table S17 reports coverage of nominal 95% intervals for the linear classification design. Two distinct failures are visible, and Section 3.4 predicts both.

The first is the one the paper is built around. Under the random forest, coverage at $n = 5{,}000$ falls to 0.186 for x_1 and 0.268 for x_2 while remaining near nominal for the noise features, whose true effect is zero and whose regularization bias is therefore also zero. This

Table S16: Simulation 1: AME Recovery — Interaction DGP (Regression). Mean bias and RMSE (in parentheses) across 1,000 Monte Carlo iterations.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.0041	0.0005	0.0000	0.0006	0.0011
		(0.2284)	(0.1531)	(0.1129)	(0.0715)	(0.0495)
x2	3.000	0.0040	0.0087	0.0075	0.0012	0.0009
		(0.2194)	(0.1627)	(0.1135)	(0.0727)	(0.0514)
x3	0.000	-0.0114	-0.0088	-0.0054	0.0010	-0.0009
		(0.1388)	(0.1002)	(0.0725)	(0.0450)	(0.0329)
x4	0.000	-0.0073	0.0004	-0.0006	0.0013	0.0005
		(0.1443)	(0.1018)	(0.0690)	(0.0453)	(0.0321)
Panel: Random Forest						
x1	2.000	-0.0755	-0.0310	-0.0433	-0.0724	-0.0804
		(0.2566)	(0.1663)	(0.1169)	(0.0979)	(0.0925)
x2	3.000	-0.0634	-0.0690	-0.0922	-0.1112	-0.1143
		(0.2806)	(0.1913)	(0.1497)	(0.1306)	(0.1244)
x3	0.000	-0.0031	-0.0004	-0.0009	0.0003	-0.0001
		(0.0612)	(0.0352)	(0.0182)	(0.0067)	(0.0029)
x4	0.000	-0.0023	0.0004	-0.0001	-0.0002	0.0000
		(0.0665)	(0.0349)	(0.0184)	(0.0069)	(0.0030)
Panel: XGBoost						
x1	2.000	0.3184	0.1016	-0.0009	-0.0406	-0.0429
		(0.4539)	(0.2184)	(0.1163)	(0.0758)	(0.0602)
x2	3.000	0.3672	0.1135	-0.0083	-0.0572	-0.0589
		(0.4920)	(0.2373)	(0.1227)	(0.0898)	(0.0746)
x3	0.000	-0.0031	0.0011	-0.0017	0.0005	-0.0001
		(0.1040)	(0.0551)	(0.0319)	(0.0151)	(0.0078)
x4	0.000	-0.0082	0.0003	-0.0003	-0.0004	-0.0001
		(0.1028)	(0.0564)	(0.0314)	(0.0149)	(0.0080)
Panel: Neural Net						
x1	2.000	-0.0842	-0.0083	-0.0056	0.0003	0.0022
		(0.2380)	(0.1114)	(0.0779)	(0.0573)	(0.0536)
x2	3.000	-0.1231	-0.0068	-0.0052	0.0004	0.0026
		(0.2467)	(0.1120)	(0.0783)	(0.0579)	(0.0533)
x3	0.000	-0.0026	-0.0016	-0.0024	0.0004	-0.0004
		(0.0986)	(0.0551)	(0.0383)	(0.0293)	(0.0316)
x4	0.000	-0.0079	0.0015	0.0002	-0.0005	-0.0019
		(0.0981)	(0.0565)	(0.0387)	(0.0284)	(0.0331)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with n = 1,000,000 observations. Noise features x3 and x4 have true AME of zero.

Table S17: Simulation 2: Bootstrap SE Calibration — Linear DGP (Classification). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

		Coverage (CI Width)		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Logistic				
x1	0.199	0.881	0.878	0.886
		(0.0652)	(0.0334)	(0.0150)
x2	0.298	0.726	0.766	0.781
		(0.0505)	(0.0257)	(0.0115)
x3	0.000	0.944	0.939	0.946
		(0.0752)	(0.0386)	(0.0174)
x4	0.000	0.933	0.954	0.944
		(0.0754)	(0.0386)	(0.0174)
Panel: Random Forest				
x1	0.199	0.968	0.933	0.186
		(0.1343)	(0.0553)	(0.0205)
x2	0.298	0.961	0.958	0.268
		(0.1500)	(0.0614)	(0.0219)
x3	0.000	0.962	0.942	0.949
		(0.0976)	(0.0386)	(0.0132)
x4	0.000	0.969	0.964	0.953
		(0.0978)	(0.0386)	(0.0132)
Panel: XGBoost				
x1	0.199	0.926	0.947	0.877
		(0.1864)	(0.0702)	(0.0230)
x2	0.298	0.908	0.954	0.910
		(0.2136)	(0.0794)	(0.0239)
x3	0.000	0.976	0.961	0.960
		(0.1139)	(0.0436)	(0.0141)
x4	0.000	0.973	0.969	0.959
		(0.1149)	(0.0436)	(0.0142)
Panel: Neural Net				
x1	0.199	0.870	0.870	0.958
		(0.0679)	(0.0351)	(0.0203)
x2	0.298	0.658	0.774	0.894
		(0.0586)	(0.0287)	(0.0154)
x3	0.000	0.936	0.936	0.986
		(0.0764)	(0.0391)	(0.0225)
x4	0.000	0.938	0.936	0.982
		(0.0772)	(0.0392)	(0.0225)

Notes: Coverage rate and mean 95% CI width (in parentheses). Intervals are bootstrap percentile intervals. Logistic, random forest and XGBoost use 1,000 Monte Carlo iterations and 1,000 resamples; the neural network uses 500 and 200, since every one of its resamples is a fresh training run. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

is the mechanism of Remark S4: the refitting bootstrap is consistent for the dispersion of the plug-in average around its own expectation, but that expectation is displaced from $\theta_{j,h}$ by $\mathbb{E}[\alpha_h(\hat{f} - f)]$, and resampling cannot see the displacement because every replicate recentres at a similarly regularized fit. The interval shrinks at the usual rate; the bias does not shrink at all; coverage follows the ratio down. XGBoost, whose bias does vanish with n, keeps coverage between 0.877 and 0.976 throughout, so the failure is specific to learners whose regularization bias persists rather than general to flexible learners.

The second failure is milder, and it affects even a correctly specified model. Logistic regression covers at 0.881, 0.878 and 0.886 for x_1 and at 0.726, 0.766 and 0.781 for x_2, with no tendency to improve as the sample grows, while covering at roughly the nominal rate for the two noise features. The pattern orders the features by coefficient size — $\beta_2 = 3$, $\beta_1 = 2$, and zero for x_3 and x_4 — and that ordering identifies the cause. As Section 3.4 notes, each replicate is evaluated at the observed sample rather than at the resample, so the bootstrap distribution omits the sampling variability of the average of the derivative over the covariates. That omitted component is identically zero when the fitted derivative does not vary with x and grows with how much it does, and under a logistic link the derivative varies most for the feature with the largest coefficient. The estimated standard errors are correspondingly too small: averaged over cells the ratio of the bootstrap standard error to the Monte Carlo standard deviation is 0.86 for logistic regression, 0.89 for the network and 0.90 for the forest. Neither failure is detectable from the bootstrap output alone, which is the practical point: both produce intervals that look unremarkable.

Table S18 reports the corresponding regression results, where the two mechanisms separate cleanly. Linear regression covers between 0.928 and 0.957 at every sample size and every feature, and its bootstrap standard error matches the Monte Carlo standard deviation to within one percent. That is the second mechanism switching off exactly where the theory says it must: the fitted derivative of a linear model is constant in x, so the variance component that the evaluation point omits is identically zero, and the understatement that afflicted every classification panel disappears. Comparing the two

Table S18: Simulation 2: Bootstrap SE Calibration — Linear DGP (Regression). Nominal coverage target: 95%. Coverage rate and mean 95% CI width (in parentheses) across 500 Monte Carlo iterations.

		Coverage (CI Width)		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Linear				
x1	2.000	0.928	0.945	0.954
		(0.2478)	(0.1235)	(0.0553)
x2	3.000	0.955	0.948	0.956
		(0.2489)	(0.1239)	(0.0552)
x3	0.000	0.951	0.947	0.948
		(0.2495)	(0.1234)	(0.0552)
x4	0.000	0.949	0.948	0.957
		(0.2489)	(0.1234)	(0.0552)
Panel: Random Forest				
x1	2.000	0.870	0.886	0.092
		(0.7447)	(0.2990)	(0.1170)
x2	3.000	0.967	0.962	0.323
		(0.9327)	(0.3632)	(0.1441)
x3	0.000	0.996	0.996	0.999
		(0.2555)	(0.0707)	(0.0082)
x4	0.000	0.992	0.995	1.000
		(0.2566)	(0.0708)	(0.0083)
Panel: XGBoost				
x1	2.000	0.849	0.982	0.521
		(0.8174)	(0.3009)	(0.1041)
x2	3.000	0.917	0.990	0.503
		(0.9709)	(0.3552)	(0.1184)
x3	0.000	0.988	0.969	0.971
		(0.3863)	(0.1217)	(0.0382)
x4	0.000	0.978	0.975	0.969
		(0.3778)	(0.1210)	(0.0382)
Panel: Neural Net				
x1	2.000	0.950	0.978	1.000
		(0.3442)	(0.1497)	(0.1938)
x2	3.000	0.936	0.974	1.000
		(0.4000)	(0.1537)	(0.2004)
x3	0.000	0.972	0.970	1.000
		(0.3075)	(0.1471)	(0.1887)
x4	0.000	0.968	0.972	1.000
		(0.3116)	(0.1474)	(0.1869)

Notes: Coverage rate and mean 95% CI width (in parentheses). Intervals are bootstrap percentile intervals. Logistic, random forest and XGBoost use 1,000 Monte Carlo iterations and 1,000 resamples; the neural network uses 500 and 200, since every one of its resamples is a fresh training run. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

tables therefore isolates the cause more sharply than either does alone.

The first mechanism is correspondingly starker. The forest covers at 0.092 for x_1 and 0.323 for x_2 at $n = 5{,}000$, and XGBoost — which was well behaved in classification — falls to 0.521 and 0.503. Here the forest's bootstrap standard error is if anything conservative, averaging 1.24 times the Monte Carlo standard deviation, so the coverage failure cannot be attributed to an understated standard error. It is bias alone. The neural network sits at the opposite extreme, covering at 1.000 for every feature at $n = 5{,}000$: its estimate is dominated by optimization noise rather than sampling noise at that sample size, and because each replicate re-draws the initialization independently of the point fit, the replicate distribution centres near the truth rather than near the estimate. Its intervals are therefore wide enough to be useless rather than wrong, which the bootstrap output again does not reveal.

S12.3 Simulation 3: The Debiased Estimator

Simulation 3 repeats the design of Simulation 2 for the debiased cross-fitted estimator of Section 3.4. The covariates are independent standard normals, so the Riesz representer of Lemma 1 is available in closed form, $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, and the estimator requires no representer estimation at all; a second arm reported at the end of this subsection replaces it with a sieve estimate, to show what is lost when the covariate density is unknown. Cross-fitting uses $K = 5$ folds. The data generating processes, learners, hyperparameters, sample sizes, window h and Monte Carlo ground truth are all held at the values used in the two preceding studies, so any difference between Simulation 3 and its predecessors is attributable to the orthogonal correction and to nothing else. Trimming is inactive throughout because the covariate support is unbounded and the trimming weight of Section 2.1 is therefore identically one. For the same reason the design satisfies Assumption 2 rather than the density bounds of Assumption 1: a Gaussian density is not bounded below on $\mathbb{R}^d$, and the representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$ is unbounded.

It is square integrable, with

$$\mathbb{E}\left[\alpha_h(X)^2\right] = \frac{e^{-h^2}}{h^2}\,\mathbb{E}\left[\sinh^2(hX_j)\right] = \frac{e^{-h^2}\left(e^{2h^2}-1\right)}{2h^2} < \infty, \tag{S12.1}$$

and all higher moments finite as well; the shift ratio $\exp(hu_j - h^2/2)$ has moments of every order, so both conditions of Assumption 2 hold for every r and q. The theorems demonstrated here are therefore the ones proved under that assumption set, and the closed-form representer used below is the exact one for this design rather than a bounded approximation to it.

Table S19 reports bias and RMSE for the linear regression DGP. The result to compare against is the random forest panel of Table S14, where the plug-in's bias was -0.097 at $n = 250$ and, rather than shrinking, grew to -0.091 for x_1 and -0.095 for x_2 at $n = 5{,}000$. Under the correction that bias is -0.002 at $n = 250$ and 0.000 at $n = 5{,}000$, and the RMSE now falls monotonically from 0.171 to 0.031, at the $O(n^{-1/2})$ rate the plug-in never achieved. The correction does not merely shrink the bias at a given sample size; it restores the rate, which is what inference requires. The other learners were already close to unbiased on this design and are essentially unchanged, as they should be: the orthogonal score corrects a first-order term that is near zero for them.

Table S20 places the two inference procedures side by side on the same designs and the same ground truth. The bootstrap columns range from 0.092 to 1.000; the influence-function columns range from 0.930 to 0.964. Across all 300 cells of the closed-form arm — three data generating processes, two outcome types, five learners, five sample sizes and four features — empirical coverage has mean 0.948 and lies between 0.920 and 0.970, with 98.7% of cells inside $[0.93, 0.97]$. No learner is an exception: mean coverage by model runs from 0.947 to 0.949. The forest cell that motivates the paper moves from 0.092 to 0.946.

The forest panel carries a second reading. These forests are capped at depth six, and a depth-capped forest does not converge to the regression function; it converges to the best depth-limited approximation of it. The estimator is therefore consistent

Table S19: Simulation 3: debiased estimator, linear regression DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.002	-0.000	-0.000	-0.001	-0.000
		(0.067)	(0.044)	(0.032)	(0.021)	(0.014)
x2	3.000	-0.004	-0.002	0.001	-0.000	-0.000
		(0.064)	(0.045)	(0.032)	(0.020)	(0.014)
x3	0.000	-0.000	-0.001	-0.000	0.001	-0.000
		(0.065)	(0.046)	(0.031)	(0.020)	(0.014)
x4	0.000	-0.003	0.001	-0.000	-0.000	-0.000
		(0.065)	(0.045)	(0.031)	(0.020)	(0.014)
Panel: Random Forest						
x1	2.000	-0.002	0.002	-0.002	0.000	0.000
		(0.171)	(0.106)	(0.074)	(0.044)	(0.031)
x2	3.000	-0.001	0.010	0.003	0.001	-0.000
		(0.228)	(0.147)	(0.094)	(0.056)	(0.038)
x3	0.000	0.002	0.001	-0.001	0.001	-0.000
		(0.087)	(0.054)	(0.035)	(0.022)	(0.015)
x4	0.000	-0.002	0.000	-0.001	-0.001	-0.000
		(0.089)	(0.056)	(0.036)	(0.021)	(0.015)
Panel: XGBoost						
x1	2.000	-0.002	0.001	-0.001	-0.002	0.000
		(0.230)	(0.142)	(0.088)	(0.048)	(0.030)
x2	3.000	-0.002	0.007	0.001	0.001	0.001
		(0.280)	(0.173)	(0.101)	(0.054)	(0.034)
x3	0.000	-0.001	0.000	-0.000	0.001	-0.000
		(0.095)	(0.058)	(0.036)	(0.022)	(0.015)
x4	0.000	-0.004	0.001	0.000	-0.000	-0.000
		(0.099)	(0.061)	(0.036)	(0.021)	(0.014)
Panel: Neural Net						
x1	2.000	-0.002	-0.001	0.000	-0.001	-0.000
		(0.084)	(0.051)	(0.034)	(0.022)	(0.014)
x2	3.000	-0.004	-0.002	0.001	-0.001	-0.000
		(0.090)	(0.055)	(0.036)	(0.021)	(0.015)
x3	0.000	0.000	-0.001	-0.000	0.001	-0.000
		(0.079)	(0.049)	(0.033)	(0.021)	(0.014)
x4	0.000	-0.003	0.001	-0.001	0.000	-0.000
		(0.080)	(0.051)	(0.033)	(0.020)	(0.014)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

Table S20: Coverage of nominal 95% confidence intervals, linear regression DGP: refitting bootstrap against the influence function.

		Refitting bootstrap			Influence function		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Linear							
x1	2.000	0.928	0.945	0.954	0.930	0.951	0.957
x2	3.000	0.955	0.948	0.956	0.955	0.946	0.960
x3	0.000	0.951	0.947	0.948	0.944	0.949	0.950
x4	0.000	0.949	0.948	0.957	0.941	0.949	0.956
Panel: Random Forest							
x1	2.000	0.870	0.886	0.092	0.936	0.939	0.946
x2	3.000	0.967	0.962	0.323	0.936	0.952	0.952
x3	0.000	0.996	0.996	0.999	0.953	0.963	0.950
x4	0.000	0.992	0.995	1.000	0.940	0.950	0.964
Panel: XGBoost							
x1	2.000	0.849	0.982	0.521	0.932	0.942	0.938
x2	3.000	0.917	0.990	0.503	0.948	0.946	0.941
x3	0.000	0.988	0.969	0.971	0.957	0.952	0.952
x4	0.000	0.978	0.975	0.969	0.940	0.957	0.962
Panel: Neural Net							
x1	2.000	0.950	0.978	1.000	0.940	0.948	0.956
x2	3.000	0.936	0.974	1.000	0.950	0.939	0.959
x3	0.000	0.972	0.970	1.000	0.947	0.951	0.948
x4	0.000	0.968	0.972	1.000	0.949	0.952	0.946

Notes: Empirical coverage of nominal 95% intervals. The bootstrap columns resample and refit the learner, centring on the plug-in average; the influence-function columns use the debiased cross-fitted estimator with $K = 5$ folds and the closed-form representer. Both are evaluated against the same window estimand $\theta_{j,h}$ and the same Monte Carlo ground truth. Noise features x_3 and x_4 have a true effect of zero.

for neither nuisance in the usual sense, and yet its intervals cover at the nominal rate. This is Proposition 2 operating exactly as stated: with the representer known, $\sqrt{n}$-normality survives an outcome model that converges to the wrong function, and the learner's approximation error is paid in the width of the interval rather than in its location. The forest's RMSE at $n = 5{,}000$ is 0.031 against the linear model's 0.014 — wider, and correctly centred. That is the practical content of double robustness here, and it is why the choice of learner becomes a question of efficiency rather than of validity.

Tables S21 and S22 report the corresponding classification results, where the same pattern holds on a compressed scale: the forest's coverage at $n = 5{,}000$ moves from 0.186 to 0.944 for x_1 and from 0.268 to 0.946 for x_2, and the debiased intervals are centred for every learner at every sample size. The nonlinear and interaction designs behave identically and are reported in the supplement.

The closed-form representer used above is a luxury of the simulation design, and the natural question is what the estimator costs when α_h must itself be estimated. The second arm answers it by replacing the closed form with a sieve estimate, holding everything else fixed. Coverage survives, with a mild conservatism: over the same cells the sieve arm has mean coverage 0.958 against the closed form's 0.948. The cost appears where theory says it should, in the product bias of Assumption 3(b) and in efficiency, and it vanishes with sample size. Mean absolute bias is 3.3 times the closed form's at $n = 250$, then 1.6, 1.2, 1.1, and 0.9 at $n = 5{,}000$ — indistinguishable by the largest sample size. Standard errors are 15% wider at $n = 250$ and 1% wider at $n = 5{,}000$. Estimating the representer costs something real at small samples and asymptotically nothing, which is the behaviour the product structure of the bias predicts: neither nuisance need converge quickly on its own, so long as the product of their errors is $o(n^{-1/2})$.

One design remains to be accounted for. The arms reported so far draw independent standard normal covariates, and a Gaussian density is not bounded below on $\mathbb{R}^d$, so those designs satisfy Assumption 2 rather than the density bounds of Assumption 1. That is a real gap between what the theorems assume and what the simulation exhibits, and the natural way to close it is a design that meets the density-bounded hypotheses literally.

Table S21: Simulation 3: debiased estimator, linear classification DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.1986	0.0019	0.0005	0.0006	0.0000	-0.0002
		(0.0199)	(0.0144)	(0.0105)	(0.0064)	(0.0048)
x2	0.2978	0.0004	-0.0001	0.0003	0.0002	0.0002
		(0.0206)	(0.0142)	(0.0102)	(0.0065)	(0.0046)
x3	0.0000	0.0013	0.0007	0.0007	0.0003	0.0001
		(0.0208)	(0.0139)	(0.0100)	(0.0063)	(0.0045)
x4	0.0000	0.0008	-0.0002	-0.0008	-0.0001	0.0003
		(0.0206)	(0.0139)	(0.0101)	(0.0062)	(0.0045)
Panel: Random Forest						
x1	0.1986	0.0034	-0.0009	0.0008	0.0003	-0.0003
		(0.0339)	(0.0223)	(0.0152)	(0.0091)	(0.0060)
x2	0.2978	-0.0003	0.0024	0.0004	0.0006	0.0004
		(0.0437)	(0.0269)	(0.0176)	(0.0101)	(0.0063)
x3	0.0000	0.0003	0.0009	0.0003	0.0003	0.0003
		(0.0246)	(0.0162)	(0.0111)	(0.0068)	(0.0047)
x4	0.0000	-0.0001	-0.0010	-0.0010	-0.0002	0.0002
		(0.0249)	(0.0168)	(0.0113)	(0.0065)	(0.0046)
Panel: XGBoost						
x1	0.1986	0.0050	-0.0025	0.0013	0.0001	-0.0003
		(0.0495)	(0.0304)	(0.0193)	(0.0107)	(0.0069)
x2	0.2978	-0.0003	0.0035	0.0005	0.0010	0.0003
		(0.0630)	(0.0395)	(0.0220)	(0.0119)	(0.0069)
x3	0.0000	0.0011	0.0007	0.0004	0.0002	0.0002
		(0.0260)	(0.0171)	(0.0112)	(0.0068)	(0.0047)
x4	0.0000	0.0004	-0.0003	-0.0004	-0.0001	0.0003
		(0.0268)	(0.0174)	(0.0117)	(0.0065)	(0.0046)
Panel: Neural Net						
x1	0.1986	0.0019	0.0005	0.0004	0.0001	-0.0001
		(0.0217)	(0.0158)	(0.0115)	(0.0068)	(0.0049)
x2	0.2978	0.0004	-0.0003	0.0000	0.0004	0.0002
		(0.0229)	(0.0171)	(0.0117)	(0.0072)	(0.0048)
x3	0.0000	0.0012	0.0010	0.0005	0.0004	0.0002
		(0.0216)	(0.0144)	(0.0101)	(0.0063)	(0.0045)
x4	0.0000	0.0008	-0.0005	-0.0008	-0.0001	0.0003
		(0.0216)	(0.0141)	(0.0105)	(0.0063)	(0.0045)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2} \sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

Table S22: Coverage of nominal 95% confidence intervals, linear classification DGP: refitting bootstrap against the influence function.

		Refitting bootstrap			Influence function		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Logistic							
x1	0.199	0.881	0.878	0.886	0.944	0.934	0.940
x2	0.298	0.726	0.766	0.781	0.924	0.956	0.954
x3	0.000	0.944	0.939	0.946	0.934	0.950	0.952
x4	0.000	0.933	0.954	0.944	0.948	0.950	0.944
Panel: Random Forest							
x1	0.199	0.968	0.933	0.186	0.942	0.956	0.944
x2	0.298	0.961	0.958	0.268	0.932	0.952	0.946
x3	0.000	0.962	0.942	0.949	0.944	0.960	0.948
x4	0.000	0.969	0.964	0.953	0.944	0.944	0.944
Panel: XGBoost							
x1	0.199	0.926	0.947	0.877	0.932	0.952	0.950
x2	0.298	0.908	0.954	0.910	0.934	0.968	0.972
x3	0.000	0.976	0.961	0.960	0.964	0.948	0.946
x4	0.000	0.973	0.969	0.959	0.954	0.938	0.946
Panel: Neural Net							
x1	0.199	0.870	0.870	0.958	0.940	0.930	0.942
x2	0.298	0.658	0.774	0.894	0.934	0.950	0.958
x3	0.000	0.936	0.936	0.986	0.932	0.946	0.954
x4	0.000	0.938	0.936	0.982	0.946	0.932	0.944

Notes: Empirical coverage of nominal 95% intervals. The bootstrap columns resample and refit the learner, centring on the plug-in average; the influence-function columns use the debiased cross-fitted estimator with $K = 5$ folds and the closed-form representer. Both are evaluated against the same window estimand $\theta_{j,h}$ and the same Monte Carlo ground truth. Noise features x_3 and x_4 have a true effect of zero.

The final arm does so: the same linear predictor, with the covariates drawn independently and uniformly on $[-\sqrt{3}, \sqrt{3}]$ so that each coordinate again has unit variance and the two designs are directly comparable. There the density is constant, and the representer follows from Lemma 1 with the density cancelling entirely,

$$\alpha_h(u) = \frac{w(u - he_j) - w(u + he_j)}{2h} = \frac{\mathbb{1}\{u_j \geq a + 2h\} - \mathbb{1}\{u_j \leq b - 2h\}}{2h} \tag{S12.2}$$

on $[a, b] = [-\sqrt{3}, \sqrt{3}]$: the step function taking $-1/(2h)$ on the lower trimming shell, $+1/(2h)$ on the upper one, and zero between them. It is bounded by $1/(2h)$ and integrates to zero, so Assumption 1 and Assumption 3(c) hold as originally stated. The trimming weight is also active here rather than vacuous, which the Gaussian arms never exercise: the true effect of x_1 is 1.943 rather than 2, being exactly $2\,(1-2h/(b-a))$, so the boundary shell that the weight excises is visible in the target itself.

Table S23 reports bias and RMSE. The estimator behaves as it does on the unbounded designs: bias is at most 0.002 on the signal features at any sample size and for any learner, and RMSE declines at the $O(n^{-1/2})$ rate. Coverage is the quantity of interest, and across the 160 combinations of learner, sample size and feature in this arm it has mean 0.944 and lies between 0.890 and 0.970, against a mean of 0.948 over the matched Gaussian cells. The two assumption sets therefore deliver the same behaviour to within four thousandths. Since the two designs share a predictor and a coordinate variance and differ only in the covariate law, that agreement isolates boundedness as making no material difference: what the moment conditions buy is coverage of the Gaussian case, not a weaker guarantee within it. The influence-function standard errors are honest on both, with a mean ratio to the Monte Carlo standard deviation of 0.99 here.

The bounded design is nonetheless the slightly harder of the two at small samples, and the pattern is worth recording rather than averaging away. Fourteen of the 160 cells fall below 0.93; thirteen are classification and eleven have $n \leq 500$. The weakest is the logistic coefficient on x_2, whose coverage runs 0.912, 0.890, 0.908, 0.948 and 0.958 across the five sample sizes while the ratio of the estimated standard error to the Monte Carlo standard

Table S23: Simulation 3: debiased estimator, bounded-support linear regression DGP. Bias and RMSE (in parentheses) by sample size.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	1.943	0.005	0.002	-0.004	-0.001	-0.001
		(0.151)	(0.111)	(0.075)	(0.049)	(0.034)
x2	2.913	-0.001	-0.002	-0.002	-0.001	-0.001
		(0.161)	(0.113)	(0.077)	(0.049)	(0.034)
x3	0.000	0.004	0.001	0.003	0.002	0.001
		(0.159)	(0.107)	(0.076)	(0.048)	(0.033)
x4	0.000	-0.002	0.001	-0.002	0.000	0.001
		(0.156)	(0.105)	(0.079)	(0.049)	(0.034)
Panel: Random Forest						
x1	1.943	0.003	0.002	-0.005	0.001	-0.000
		(0.252)	(0.162)	(0.107)	(0.064)	(0.044)
x2	2.913	-0.008	-0.002	0.001	-0.001	-0.000
		(0.288)	(0.188)	(0.124)	(0.070)	(0.048)
x3	0.000	0.004	-0.001	0.002	0.002	0.001
		(0.184)	(0.123)	(0.082)	(0.051)	(0.034)
x4	0.000	-0.006	0.002	-0.003	-0.000	0.001
		(0.180)	(0.120)	(0.086)	(0.051)	(0.036)
Panel: XGBoost						
x1	1.943	-0.002	-0.002	-0.003	-0.001	-0.000
		(0.302)	(0.175)	(0.111)	(0.065)	(0.042)
x2	2.913	-0.004	0.002	0.001	-0.002	0.000
		(0.334)	(0.200)	(0.127)	(0.068)	(0.043)
x3	0.000	0.006	-0.002	0.000	0.001	0.001
		(0.196)	(0.127)	(0.085)	(0.051)	(0.035)
x4	0.000	-0.004	0.002	-0.003	0.000	0.001
		(0.193)	(0.125)	(0.085)	(0.051)	(0.036)
Panel: Neural Net						
x1	1.943	0.008	0.003	-0.004	-0.001	-0.002
		(0.192)	(0.120)	(0.079)	(0.050)	(0.035)
x2	2.913	-0.002	-0.003	-0.001	-0.000	-0.001
		(0.202)	(0.119)	(0.080)	(0.050)	(0.034)
x3	0.000	0.003	0.000	0.004	0.002	0.000
		(0.184)	(0.112)	(0.078)	(0.050)	(0.034)
x4	0.000	-0.002	0.003	-0.003	0.000	0.001
		(0.182)	(0.114)	(0.081)	(0.050)	(0.035)

Notes: Debiased cross-fitted estimator with $K = 5$ folds and the closed-form Riesz representer $\alpha_h(u) = e^{-h^2/2}\sinh(hu_j)/h$, exact for independent standard normal covariates. Trimming is inactive because the support is unbounded. Bias is the mean of $\hat{\theta}_{j,h} - \theta_{j,h}$ and RMSE its root mean square, over the Monte Carlo iterations. True values computed by Monte Carlo integration at n = 1,000,000. Noise features x_3 and x_4 have a true effect of zero.

deviation runs 0.87, 0.91, 0.96, 1.00 and 1.07. This is finite-sample understatement of the standard error converging to correctness, not a failure of the assumption set: the mean ratio over all cells at $n = 250$ is 0.958 and at $n = 5{,}000$ is 1.015. The trimming weight is the plausible source, since it discards a shell of probability $O(h)$ and so costs effective sample size exactly where there is least of it to spare.

S13 Proof of Lemma 1 (Riesz Representation)

Proof. Recall $q = wp$ extended by zero outside Ω, so that q is defined on all of $\mathbb{R}^d$, $\|q\|_\infty \le p_{\max}$, and $q(x) \neq 0$ implies $x \in \operatorname{supp}(w) \subseteq \Omega_{j,h}$. Fix $g \in L^1(P)$ and consider the positive-shift half of the finite difference:

$$\mathbb{E}\left[w(X)\, g(X + he_j)\right] = \int_{\mathbb{R}^d} q(x)\, g(x + he_j)\, dx = \int_{\mathbb{R}^d} q(u - he_j)\, g(u)\, du \tag{S13.1}$$

by the substitution $u = x + he_j$. The manipulation is legitimate because on the set where the integrand is nonzero we have $x \in \Omega_{j,h}$, hence $u = x + he_j \in \Omega$, so g is evaluated only on Ω; and the integral is absolutely convergent since:

$$\int q(u - he_j)\ |g(u)|\ du \le p_{\max} \int_\Omega |g(u)|\ du \le \frac{p_{\max}}{p_{\min}}\, \mathbb{E}\,|g(X)| < \infty \tag{S13.2}$$

The negative-shift half is identical with $u = x - he_j$. Subtracting and dividing by $2h$:

$$\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] = \int_\Omega \frac{q(u - he_j) - q(u + he_j)}{2h}\, g(u)\, du = \mathbb{E}\left[\alpha_h(X)\, g(X)\right] \tag{S13.3}$$

where the restriction of the domain of integration to Ω in the second step uses that g is supported there, and the final step multiplies and divides by $p(u) \ge p_{\min} > 0$. The uniform bound follows from $|q(u - he_j) - q(u + he_j)| \le 2p_{\max}$ and $p(u) \ge p_{\min}$. Applying the representation to $g \equiv 1$ gives $\mathbb{E}[\alpha_h(X)] = \mathbb{E}[w(X)\Delta_{j,h}1] = 0$ since the finite difference of a constant vanishes. □

S14 Proof of Proposition 1 (Recovery of the Average Derivative)

Proof. Let $f \in W^{1,1}(\Omega)$ and fix $h \le h_0$, so that $\mathrm{supp}(w) \subseteq \Omega_{j,h_0} \subseteq \Omega_{j,h}$ and every segment $\{x + te_j : |t| \le h\}$ with $x \in \mathrm{supp}(w)$ lies in Ω. Sobolev functions are absolutely continuous on almost every line parallel to a coordinate axis, with the classical derivative along the line agreeing almost everywhere with the weak derivative (Evans and Gariepy, 2015, Section 4.9). Hence for almost every $x \in \mathrm{supp}(w)$:

$$f(x + he_j) - f(x - he_j) = \int_{-h}^{h} \partial_j f(x + te_j)\, dt \tag{S14.1}$$

and therefore:

$$\theta_h(f) - \mathbb{E}\left[w(X)\, \partial_j f(X)\right] = \int_{\mathbb{R}^d} q(x)\, \frac{1}{2h} \int_{-h}^{h} \left[\partial_j f(x + te_j) - \partial_j f(x)\right] dt\, dx \tag{S14.2}$$

Bounding q by $p_{\max}$, applying Tonelli, and extending $\partial_j f$ by zero outside Ω:

$$|\theta_h(f) - \mathbb{E}\left[w\, \partial_j f\right]| \le p_{\max} \sup_{|t| \le h} \int_{\mathbb{R}^d} |\partial_j f(x + te_j) - \partial_j f(x)|\, \mathbf{1}\{x \in \mathrm{supp}(w)\}\, dx \tag{S14.3}$$

where the extension introduces no error because $x + te_j \in \Omega$ for $x \in \mathrm{supp}(w)$ and $|t| \le h \le h_0$. Continuity of translation in $L^1(\mathbb{R}^d)$ applied to the extension of $\partial_j f$ sends the right side to zero as $h \downarrow 0$. For the second claim, $|w_m \partial_j f| \le |\partial_j f|$, which is P-integrable because $\partial_j f \in L^1(\Omega)$ and $p \le p_{\max}$; dominated convergence gives $\mathbb{E}[w_m \partial_j f] \to \mathbb{E}[\partial_j f] = \mathrm{AME}_j$. □

S15 Proof of Lemma 2 (Exact Product Bias)

Proof. Conditioning on X and using $\mathbb{E}[Y \mid X] = f(X)$:

$$\mathbb{E}\left[\tilde{\alpha}(X)\,(Y - \tilde{f}(X))\right] = \mathbb{E}\left[\tilde{\alpha}(X)\,(f(X) - \tilde{f}(X))\right] \tag{S15.1}$$

By linearity of $\Delta_{j,h}$ and Lemma 1 applied to $g = \tilde{f} - f \in L^1(P)$:

$$\mathbb{E}\left[w(X)\,\Delta_{j,h}\tilde{f}(X)\right] - \theta_h(f) = \mathbb{E}\left[w(X)\,\Delta_{j,h}(\tilde{f} - f)(X)\right] = \mathbb{E}\left[\alpha_h(X)\,(\tilde{f}(X) - f(X))\right] \tag{S15.2}$$

Summing the two displays gives:

$$\mathbb{E}\left[\psi(X, Y; \tilde{f}, \tilde{\alpha}, \theta_h(f))\right] = \mathbb{E}\left[(\alpha_h(X) - \tilde{\alpha}(X))(\tilde{f}(X) - f(X))\right] \tag{S15.3}$$

and the bound is Cauchy–Schwarz. □

S16 Proof of Theorem 1 (Asymptotic Linearity and Normality)

S16.1 Step 0: The Trimmed Difference Operator is Bounded on $L^2(P)$

For any $g \in L^2(\Omega)$, using $(a-b)^2 \le 2a^2 + 2b^2$ and the change of variables from the proof of Lemma 1:

$$\begin{aligned}\mathbb{E}\left[\left(w(X)\,\Delta_{j,h}g(X)\right)^2\right] &\le \frac{1}{2h^2}\left(\mathbb{E}\left[w(X)^2 g(X + he_j)^2\right] + \mathbb{E}\left[w(X)^2 g(X - he_j)^2\right]\right) \\ &\le \frac{1}{h^2}\cdot\frac{p_{\max}}{p_{\min}}\,\mathbb{E}\left[g(X)^2\right]\end{aligned} \tag{S16.1}$$

which establishes (3.12) with constant $\kappa = h^{-1}(p_{\max}/p_{\min})^{1/2}$. This is the only property of the finite difference used below; no derivative of g appears anywhere.

S16.2 Step 1: Decomposition by Folds

Write $\theta = \theta_h(f)$, $\psi_i = \psi(\mathbf{x}_i, y_i; f, \alpha_h, \theta)$, and for $i \in I_k$ let $\hat{\psi}_i = \psi(\mathbf{x}_i, y_i; \hat{f}^{(-k)}, \hat{\alpha}^{(-k)}, \theta)$. Then:

$$\sqrt{n}\,(\hat{\theta}_h - \theta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + \sum_{k=1}^{K}\sqrt{\frac{n_k}{n}}\left(\sqrt{n_k}\,A_k + \sqrt{n_k}\,B_k\right) \tag{S16.2}$$

where, with $\mathbb{E}_k$ denoting expectation over a fresh draw (X, Y) holding the fold-k nuisances fixed:

$$A_k = \mathbb{E}_k\left[\hat{\psi} - \psi\right], \qquad B_k = \frac{1}{n_k}\sum_{i \in I_k}\left(\hat{\psi}_i - \psi_i - A_k\right) \tag{S16.3}$$

Since K is fixed and $n_k/n \to 1/K$, it suffices that $\sqrt{n_k}A_k = o_p(1)$ and $\sqrt{n_k}B_k = o_p(1)$ for each k.

S16.3 Step 2: The Bias Term

By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (\hat{f}^{(-k)}, \hat{\alpha}^{(-k)})$:

$$|\sqrt{n_k}\, A_k| \le \sqrt{n_k}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} \cdot \left\|\hat{f}^{(-k)} - f\right\|_{P,2} = o_p(1) \tag{S16.4}$$

by Assumption 3(b). This is the only place the product rate is used, and it is used without any expansion: the bound is exact.

S16.4 Step 3: The Empirical Process Term

Conditional on the out-of-fold data, the summands of B_k are i.i.d. and mean zero, so:

$$\mathbb{E}\left[(\sqrt{n_k}\, B_k)^2 \mid \text{out-of-fold}\right] \le \mathbb{E}_k\left[(\hat{\psi} - \psi)^2\right] \tag{S16.5}$$

Decompose, adding and subtracting $\hat{\alpha}^{(-k)}(x)(y - f(x))$:

$$\hat{\psi} - \psi = w\, \Delta_{j,h}(\hat{f}^{(-k)} - f) + (\hat{\alpha}^{(-k)} - \alpha_h)(y - f) - \hat{\alpha}^{(-k)}\,(\hat{f}^{(-k)} - f) \tag{S16.6}$$

and bound each piece in $L^2(P)$. The middle piece is bounded by $\bar{\sigma}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}$ after conditioning on X under either assumption set, since it uses only the conditional variance bound of Assumption 1. The first and third pieces are where the two assumption sets diverge.

Under the density bounds of Assumption 1, the first is bounded by $\kappa \left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ via (S16.1) and the third by $\bar{A} \left\|\hat{f}^{(-k)} - f\right\|_{P,2}$ using Assumption 3(c). All three are then

$o_p(1)$ by Assumption 3(a).

Under Assumption 2 neither bound is available: the representer is unbounded, and the finite-difference operator is genuinely unbounded on $L^2(P)$ rather than merely unbounded by this argument. For independent standard normal covariates, taking $g_a(x) = e^{ax_j}$ gives $\|g_a\|_{P,2}^2 = e^{2a^2}$ and $\|g_a(\cdot + he_j)\|_{P,2}^2 = e^{2ah}e^{2a^2}$, so the operator norm is $\sup_a e^{ah} = \infty$. Both pieces are instead bounded by Hölder. Writing $\Delta = \hat{f}^{(-k)} - f$ and letting r be as in (3.1) with conjugate r', a change of variables gives

$$\mathbb{E}\left[\Delta(X \pm he_j)^2\right] = \mathbb{E}\left[\Delta(X)^2 \frac{p(X \mp he_j)}{p(X)}\right] \leq \|\Delta\|_{P,2r'}^2 \left(\mathbb{E}\left[\left(\frac{p(X \mp he_j)}{p(X)}\right)^r\right]\right)^{1/r} \tag{S16.7}$$

which is $o_p(1)$ by Assumption 3(a) in its $L^{2+\delta}$ form, with $2r' = 2+\delta$; the first piece follows from (S16.7) and $|w| \leq 1$. For the third, $\mathbb{E}[(\hat{\alpha}^{(-k)})^2\Delta^2] \leq \left\|\hat{\alpha}^{(-k)}\right\|_{P,q}^2 \|\Delta\|_{P,2s'}^2$ by Hölder with $s = q/2$, which is $o_p(1)$ by Assumption 3(c) in its moment form together with (a).

Either way $\mathbb{E}_k[(\hat{\psi} - \psi)^2] = o_p(1)$, and conditional Chebyshev gives $\sqrt{n_k}\, B_k = o_p(1)$. Note that only consistency of the nuisances is used here, never a rate; the rate enters at Step 2 alone.

S16.5 Step 4: Central Limit Theorem and Variance Estimation

The ψ_i are i.i.d., mean zero by Lemma 2 at $(\tilde{f}, \tilde{\alpha}) = (f, \alpha_h)$, with variance:

$$V_h = \mathbb{E}[\psi_i^2] \leq 2\,\mathbb{E}\left[(w\Delta_{j,h}f)^2\right] + 2\,\|\alpha_h\|_\infty^2\,\mathbb{E}[\sigma^2(X)] < \infty \tag{S16.8}$$

by (S16.1) and Lemma 1 under the density bounds of Assumption 1. Under Assumption 2 the same finiteness follows from (S16.7) applied to f, which is bounded, together with $\alpha_h \in L^2(P)$ from Assumption 2(c) and $\mathbb{E}[Y^2] < \infty$; for the Gaussian design the representer's second moment is (S12.1). Either way $\mathrm{Var}(\psi) < \infty$ and the classical CLT applies. For the variance estimator, write $\hat{\psi}_i(\hat{\theta}_h) = \hat{\psi}_i - (\hat{\theta}_h - \theta)$; then:

$$\hat{V}_h = \frac{1}{n}\sum_i \hat{\psi}_i^2 - 2(\hat{\theta}_h - \theta)\,\frac{1}{n}\sum_i \hat{\psi}_i + (\hat{\theta}_h - \theta)^2 \tag{S16.9}$$

The last two terms vanish in probability since $\hat{\theta}_h - \theta = o_p(1)$ and $n^{-1}\sum_i \hat{\psi}_i = O_p(n^{-1/2})$ by Steps 1–3. For the first term:

$$\left|\frac{1}{n}\sum_i \hat{\psi}_i^2 - \frac{1}{n}\sum_i \psi_i^2\right| \le \frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2 + 2\left(\frac{1}{n}\sum_i (\hat{\psi}_i - \psi_i)^2\right)^{1/2}\left(\frac{1}{n}\sum_i \psi_i^2\right)^{1/2} \tag{S16.10}$$

by Cauchy–Schwarz; $n^{-1}\sum_i(\hat{\psi}_i - \psi_i)^2 = o_p(1)$ by the fold-wise conditional Markov inequality and the Step 3 bound, and $n^{-1}\sum_i \psi_i^2 \xrightarrow{p} V_h$ by the law of large numbers. Hence $\hat{V}_h \xrightarrow{p} V_h$. □

S17 Proof of Proposition 2 (Double Robustness)

The argument retraces the proof of Theorem 1 with the pseudo-true influence function in place of the efficient one; we record the steps that change.

S17.1 Step 1: The Pseudo-True Influence Function

Define $\psi_{g,i} = w(\mathbf{x}_i)\,\Delta_{j,h}g(\mathbf{x}_i) + \alpha_h(\mathbf{x}_i)\,(y_i - g(\mathbf{x}_i)) - \theta_h(f)$. By Lemma 2 with $(\tilde{f}, \tilde{\alpha}) = (g, \alpha_h)$:

$$\mathbb{E}[\psi_{g,i}] = \mathbb{E}\big[(\alpha_h - \alpha_h)(g - f)\big] = 0 \tag{S17.1}$$

for every $g \in L^2(\Omega)$, which also establishes $\mathbb{E}[m_g(X)] = \theta_h(f)$ after conditioning on X. Conditioning likewise gives $\mathbb{E}[\psi_{g,i} \mid X] = m_g(X) - \theta_h(f)$ and $\mathrm{Var}(\psi_{g,i} \mid X) = \alpha_h(X)^2\sigma^2(X)$, so $\mathrm{Var}(\psi_{g,i}) = V_h(g)$ by the law of total variance; finiteness follows from (S16.1), $g \in L^2(\Omega)$, the bound on α_h from Lemma 1, and $\mathbb{E}[Y^2] < \infty$ under the density bounds; under Assumption 2 it follows from (S16.7), $\alpha_h \in L^2(P)$, and $\mathbb{E}[Y^2] < \infty$.

S17.2 Step 2: The Bias Term

With $A_k = \mathbb{E}_k[\hat{\psi} - \psi_g] = \mathbb{E}_k[\hat{\psi}]$, Lemma 2 gives $A_k = \mathbb{E}[(\alpha_h - \hat{\alpha}^{(-k)})(\hat{f}^{(-k)} - f)]$, so:

$$\begin{aligned} |\sqrt{n_k}\, A_k| &\le \sqrt{n_k}\, \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2} \left(\left\|\hat{f}^{(-k)} - g\right\|_{P,2} + \|g - f\|_{P,2} \right) \\ &= \sqrt{n_k}\, o_p(n^{-1/2}) \cdot O_p(1) = o_p(1) \end{aligned} \tag{S17.2}$$

by conditions (a) and (b); the second factor is $O_p(1)$ rather than $o_p(1)$ because g need not equal f, and it is exactly here that condition (b) must be strengthened to $o_p(n^{-1/2})$ relative to Assumption 3(b). When $\hat{\alpha}^{(-k)} = \alpha_h$, $A_k = 0$ identically at every n; averaging the fold-wise identity $\mathbb{E}[\hat{\psi}_i + \theta_h(f) \mid \text{out-of-fold data}] = \theta_h(f)$, which holds because the observations in fold k are independent of $\hat{f}^{(-k)}$, gives the exact unbiasedness claim.

S17.3 Step 3: The Empirical Process Term

Decompose, for $i \in I_k$:

$$\hat{\psi}_i - \psi_{g,i} = w\, \Delta_{j,h}(\hat{f}^{(-k)} - g) + (\hat{\alpha}^{(-k)} - \alpha_h)(y_i - \hat{f}^{(-k)}) + \alpha_h\, (g - \hat{f}^{(-k)}) \tag{S17.3}$$

The first and third terms have conditional $L^2(P)$ norms bounded by $\kappa \left\|\hat{f}^{(-k)} - g\right\|_{P,2}$ and $\|\alpha_h\|_\infty \left\|\hat{f}^{(-k)} - g\right\|_{P,2}$ respectively, both $o_p(1)$ by condition (a). For the middle term, write $y - \hat{f}^{(-k)} = (y - f) + (f - g) + (g - \hat{f}^{(-k)})$ and bound the three resulting pieces: the first by $\bar{\sigma}^2 \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,2}^2 = o_p(1)$ after conditioning on X; the third by $(\bar{A} + \|\alpha_h\|_\infty)^2 \left\|\hat{f}^{(-k)} - g\right\|_{P,2}^2 = o_p(1)$; and the second piece, $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f - g)^2]$, tends to zero. Under the density bounds this is dominated convergence along subsequences: the factor $(\hat{\alpha}^{(-k)} - \alpha_h)^2$ is bounded by $(\bar{A} + \|\alpha_h\|_\infty)^2$ and converges to zero in P-measure by condition (b), while $(f - g)^2 \in L^1(P)$ dominates. Under Assumption 2 the domination is unavailable and Hölder is used instead: $\mathbb{E}[(\hat{\alpha}^{(-k)} - \alpha_h)^2 (f - g)^2] \le \left\|\hat{\alpha}^{(-k)} - \alpha_h\right\|_{P,q}^2 \|f - g\|_{P,2t'}^2$ with $t = q/2$, which requires the fixed approximation error $f - g$ to lie in $L^{2t'}(P)$ — a mild condition, and one that holds whenever g and f are bounded. Conditional Chebyshev then gives $\sqrt{n_k}\, B_k = o_p(1)$ exactly as in Step 3 of the

proof of Theorem 1.

S17.4 Step 4: Conclusion

The $\psi_{g,i}$ are i.i.d., mean zero, with finite positive variance $V_h(g)$, so the classical CLT applies to their normalized sum, and Steps 2–3 identify $\sqrt{n}(\hat{\theta}_h - \theta_h(f))$ with that sum up to $o_p(1)$. The consistency of $\hat{V}_h$ for $V_h(g)$ repeats Step 4 of the proof of Theorem 1 verbatim with ψ_g in place of ψ, using the Step 3 bound just established. Finally, when $g \neq f$ the influence function ψ_g differs from the efficient influence function ψ of Theorem 2; since in the nonparametric model every regular asymptotically linear estimator has influence function equal to the efficient one, the estimator is not regular in this case, and no conflict with the convolution theorem arises. □

S18 Proof of Theorem 2 (Semiparametric Efficiency)

Proof. Consider one-dimensional submodels $dP_t = (1 + ts)\, dP$ indexed by bounded functions $s \in L_0^2(P) = \{s : \mathbb{E}[s(X, Y)] = 0\}$, which are well defined for $|t| < 1/\|s\|_\infty$ and differentiable in quadratic mean at $t = 0$ with score s; bounded mean-zero functions are dense in $L_0^2(P)$, so the tangent space of the nonparametric model is all of $L_0^2(P)$ (van der Vaart, 1998, Chapter 25). Write $\bar{s}_X(x) = \mathbb{E}[s(X, Y) \mid X = x]$ for the induced score of the covariate marginal and $f_t(x) = \mathbb{E}_t[Y \mid X = x]$ for the regression function along the path. Differentiating the Bayes formula for f_t at $t = 0$ gives, for P-almost every x:

$$\dot{f}(x) := \frac{\partial}{\partial t} f_t(x)\Big|_{t=0} = \mathbb{E}\left[(Y - f(X))\, s(X, Y) \mid X = x\right] \tag{S18.1}$$

where differentiation under the integral is justified by $\mathbb{E}[Y^2] < \infty$ and the boundedness of s. The functional along the path is $\theta(t) = \mathbb{E}_t[w(X)\, \Delta_{j,h} f_t(X)]$, and the product rule gives:

$$\frac{d\theta}{dt}\Big|_{t=0} = \mathbb{E}\left[w\, \Delta_{j,h} f(X) \cdot \bar{s}_X(X)\right] + \mathbb{E}\left[w\, \Delta_{j,h} \dot{f}(X)\right] \tag{S18.2}$$

For the first term, note $\mathbb{E}[\bar{s}_X] = 0$, so the term equals $\mathbb{E}[(w\Delta_{j,h}f(X) - \theta_h(f))\,\bar{s}_X(X)] = \mathbb{E}[(w\Delta_{j,h}f(X) - \theta_h(f))\,s(X,Y)]$, the last equality because $w\Delta_{j,h}f(X) - \theta_h(f)$ is X-measurable. For the second term, $\dot{f} \in L^1(P)$ (indeed L^2) since s is bounded, so Lemma 1 applies:

$$\mathbb{E}\left[w\,\Delta_{j,h}\dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\,\dot{f}(X)\right] = \mathbb{E}\left[\alpha_h(X)\,(Y - f(X))\,s(X,Y)\right] \tag{S18.3}$$

by the tower property. Summing:

$$\left.\frac{d\theta}{dt}\right|_{t=0} = \mathbb{E}\left[\psi(X,Y;f,\alpha_h,\theta_h(f))\;s(X,Y)\right] \tag{S18.4}$$

for every score s, so θ_h is pathwise differentiable with influence function ψ; since $\psi \in L_0^2(P)$ lies in the tangent space, it is the efficient influence function and the bound is $\mathbb{E}[\psi^2]$ (van der Vaart, 1998, Theorem 25.20). The cross term in $\mathbb{E}[\psi^2]$ vanishes by conditioning on X, yielding (3.15). Attainment follows from Theorem 1, whose expansion exhibits $\hat{\theta}_h$ as asymptotically linear with influence function ψ, hence regular and efficient (van der Vaart, 1998, Lemma 25.23). □

S19 Proof of Lemma S1 (Derivative Existence, Sobolev)

S19.1 What Needs to Be Shown

That $\hat{f}_n \in W^{2,p}(\Omega)$ for $p > d$ implies $\nabla\hat{f}_n \in W^{1,p}(\Omega)^d$ with the norm bound:

$$\left\|\nabla\hat{f}_n\right\|_{L^p} \le \left\|\hat{f}_n\right\|_{W^{1,p}} \le \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S19.1}$$

Remark S7. The condition $p > d$ is required by the proof. It is the condition under which Theorem 6.5 of Evans and Gariepy (2015) guarantees pointwise differentiability of $W^{1,p}_{loc}$ functions. This is slightly stronger than the condition $p > 1$ stated in the main text and is an explicit assumption of the lemma.

S19.2 Step 1: Existence of the Gradient as an L^p Function

Since $\hat{f}_n \in W^{2,p}(\Omega)$ and $p > d$, we have in particular that $\hat{f}_n \in W^{1,p}(\Omega)$. Extending $\hat{f}_n$ to a $W^{1,p}_{loc}(\mathbb{R}^d)$ function by a standard extension operator (Evans and Gariepy, 2015, Section 4.4), we may apply Theorem 6.5 of Evans and Gariepy (2015):

> *Theorem 6.5 (Evans and Gariepy, 2015).* Assume $f \in W^{1,p}_{loc}(\mathbb{R}^n)$ for some $n < p \leq \infty$. Then f is differentiable $\mathcal{L}^n$-a.e., and its classical derivative equals its weak derivative $\mathcal{L}^n$-a.e.

Applying this theorem to $\hat{f}_n$ with $n = d$ and $p > d$, we conclude that $\hat{f}_n$ is classically differentiable at $\mathcal{L}^d$-almost every $x \in \Omega$, and:

$$\nabla \hat{f}_n(x) = D\hat{f}_n(x) \quad \mathcal{L}^d\text{-a.e.} \tag{S19.2}$$

where $D\hat{f}_n$ denotes the weak gradient of $\hat{f}_n$. Since $\hat{f}_n \in W^{1,p}(\Omega)$, the weak gradient $D\hat{f}_n \in L^p(\Omega)^d$ by definition of the Sobolev space. Therefore $\nabla \hat{f}_n \in L^p(\Omega)^d$.

Applying the same argument to each first-order weak derivative $\partial_j \hat{f}_n \in W^{1,p}(\Omega)$ — which belongs to $W^{1,p}(\Omega)$ because $\hat{f}_n \in W^{2,p}(\Omega)$ — we conclude that each $\partial_j \hat{f}_n$ is classically differentiable $\mathcal{L}^d$-a.e. with classical second derivatives in $L^p(\Omega)$. Therefore $\nabla \hat{f}_n \in W^{1,p}(\Omega)^d$.

S19.3 Step 2: The Norm Bound

We establish the chain of inequalities:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S19.3}$$

First Inequality

By definition of the $W^{1,p}(\Omega)$ norm:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \sum_{j=1}^{d} \left\|\partial_j \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{S19.4}$$

Since $\left\|\hat{f}_n\right\|_{L^p} \geq 0$:

$$\left\|\hat{f}_n\right\|_{W^{1,p}} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} \geq \left\|\nabla \hat{f}_n\right\|_{L^p} \tag{S19.5}$$

Second Inequality

By definition of the $W^{2,p}(\Omega)$ norm using multi-index notation:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \tag{S19.6}$$

The multi-indices with $|\alpha| \leq 1$ form a strict subset of those with $|\alpha| \leq 2$, and all terms are nonnegative, so:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \sum_{|\alpha| \leq 2} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} \geq \sum_{|\alpha| \leq 1} \left\|\partial^\alpha \hat{f}_n\right\|_{L^p} = \left\|\hat{f}_n\right\|_{W^{1,p}} \tag{S19.7}$$

S19.4 Conclusion

For $p > d$, membership of $\hat{f}_n$ in $W^{2,p}(\Omega)$ implies via Theorem 6.5 of Evans and Gariepy (2015) that $\nabla \hat{f}_n$ exists $\mathcal{L}^d$-almost everywhere as a classical gradient, coincides with the weak gradient, and belongs to $W^{1,p}(\Omega)^d \subset L^p(\Omega)^d$. The gradient operator $\nabla : W^{2,p}(\Omega) \to W^{1,p}(\Omega)^d$ is bounded with:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n\right\|_{W^{1,p}} \leq \left\|\hat{f}_n\right\|_{W^{2,p}} \tag{S19.8}$$

□

S20 Proof of Lemma S2 (Derivative Consistency, Sobolev)

S20.1 What Needs to Be Shown

That well-behavedness in the Sobolev sense implies $\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0$.

S20.2 Proof Strategy

Compactness is a statement about deterministic sequences and $\hat{f}_n$ is random, so the two are kept apart. We first prove a deterministic lemma: on a ball of $W^{2,p}(\Omega)$, closeness to f in L^p forces closeness of the gradients in L^p, with a modulus depending only on the radius of the ball. The probabilistic conclusion then follows in a few lines, the tightness condition (S2.4) supplying the radius.

The order matters. Applying Rellich–Kondrachov directly to $\{\hat{f}_n\}$ extracts a subsequence that depends on the realization, and an almost sure statement along a realization-dependent subsequence does not combine with a probability bound along a deterministic one. Separating the compactness from the randomness avoids the difficulty rather than managing it, and has the further advantage of needing only tightness of the $W^{2,p}(\Omega)$ norms, which is what Definition S1 provides and all that the estimators of Section S4 deliver.

S20.3 Step 1: A Deterministic Gradient Modulus

Lemma S5 (Gradient modulus on a Sobolev ball)**.** *Let Ω be bounded and Lipschitz, let $p > 1$, and let $f \in W^{1,p}(\Omega)$. For every $M < \infty$ and $\delta > 0$ there is an $\epsilon = \epsilon(M, \delta) > 0$ such that every $g \in W^{2,p}(\Omega)$ with*

$$\|g\|_{W^{2,p}} \leq M \qquad \text{and} \qquad \|g - f\|_{L^p} \leq \epsilon \tag{S20.1}$$

satisfies $\|\nabla g - \nabla f\|_{L^p} \leq \delta$.

Proof. Suppose not. Then for some M and some δ there is a sequence $\{g_k\} \subset W^{2,p}(\Omega)$ with $\|g_k\|_{W^{2,p}} \leq M$ and $\|g_k - f\|_{L^p} \to 0$, but $\|\nabla g_k - \nabla f\|_{L^p} > \delta$ for every k: negating the statement for a fixed M and δ produces, for each $\epsilon = 1/k$, a witness g_k.

This is a deterministic bounded sequence in $W^{2,p}(\Omega)$. By the Rellich–Kondrachov theorem the embedding $W^{2,p}(\Omega) \hookrightarrow\hookrightarrow W^{1,p}(\Omega)$ is compact (Evans, 2010, Theorem 5, Section 5.7), so some subsequence $\{g_{k_j}\}$ converges strongly in $W^{1,p}(\Omega)$ to a limit $h \in W^{1,p}(\Omega)$. Strong $W^{1,p}(\Omega)$ convergence implies strong L^p convergence, and $g_k \to f$ in L^p by con-

struction, so $h = f$ $\mathcal{L}^d$-almost everywhere by uniqueness of limits in $L^p(\Omega)$. Hence $\|g_{k_j} - f\|_{W^{1,p}} \to 0$, and since $g_{k_j} - f \in W^{1,p}(\Omega)$ with $\|\nabla g - \nabla f\|_{L^p} \le \|g - f\|_{W^{1,p}}$, this gives $\|\nabla g_{k_j} - \nabla f\|_{L^p} \to 0$, contradicting $\|\nabla g_{k_j} - \nabla f\|_{L^p} > \delta$. □

Being a compactness statement, the lemma carries no rate: $\epsilon(M, \delta)$ is not quantified and cannot be without further assumptions on f. That is sufficient here, because what is being established is consistency and not a rate of convergence. It is also where the extra degree of differentiability is spent — the hypothesis is a bound in $W^{2,p}(\Omega)$ and the conclusion is about gradients in L^p, a gap of one derivative that the compact embedding closes and the continuous embedding of Lemma S1 does not.

S20.4 Step 2: The Probabilistic Conclusion

Fix $\delta > 0$ and $\eta > 0$. By the tightness condition (S2.4) there is an $M < \infty$ with $\sup_n \mathbb{P}(\|\hat{f}_n\|_{W^{2,p}} > M) \le \eta/2$; let $\epsilon = \epsilon(M, \delta) > 0$ be the radius supplied by Lemma S5 for this M and δ. On the event

$$A_n = \left\{ \|\hat{f}_n\|_{W^{2,p}} \le M \right\} \cap \left\{ \|\hat{f}_n - f\|_{L^p} \le \epsilon \right\} \tag{S20.2}$$

the lemma applies pointwise in ω — for each such ω the realization $\hat{f}_n(\omega)$ is a fixed element of $W^{2,p}(\Omega)$ satisfying both hypotheses — and yields $\|\nabla \hat{f}_n - \nabla f\|_{L^p} \le \delta$ on A_n. Taking complements,

$$\mathbb{P}\left(\|\nabla \hat{f}_n - \nabla f\|_{L^p} > \delta \right) \le \mathbb{P}(A_n^c) \le \frac{\eta}{2} + \mathbb{P}\left(\|\hat{f}_n - f\|_{L^p} > \epsilon \right). \tag{S20.3}$$

The remaining term tends to zero by the L^p consistency condition of Definition S1, with ϵ held fixed, so the left-hand side is at most η for all n sufficiently large. As δ and η were arbitrary,

$$\|\nabla \hat{f}_n - \nabla f\|_{L^p} \xrightarrow{p} 0 \tag{S20.4}$$

which is the conclusion of Lemma S2. □

Remark S8. The key step is the application of Rellich–Kondrachov in Step 1. The

theorem requires Ω to be bounded and Lipschitz — both of which are assumed throughout — and $p > 1$, which is assumed in the lemma statement. The compact embedding $W^{2,p} \hookrightarrow\hookrightarrow W^{1,p}$ is strictly stronger than the continuous embedding $W^{2,p} \hookrightarrow W^{1,p}$: the latter follows from the norm bound in Lemma S1, while the former provides the compactness needed to extract convergent subsequences. It is this compactness, not merely the norm bound, that allows the interchange of limit and gradient in the proof.

S21 Proof of Lemma S3 (Derivative Existence, BV)

S21.1 What Needs to Be Shown

Two things: that BV membership implies $D\hat{f}_n$ exists as a Radon measure with finite total variation, and the explicit formula for piecewise constant functions.

S21.2 Step 1: Existence of $D\hat{f}_n$ as a Radon Measure

We establish existence in three steps, using the Riesz representation theorem and the Radon-Nikodym theorem.

Substep 1a: The Distributional Gradient as a Linear Functional

For any $\hat{f}_n \in BV(\Omega)$, the distributional gradient is defined as the linear functional $\Lambda : C_c^1(\Omega)^d \to \mathbb{R}$ given by:

$$\Lambda(\phi) = -\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx \quad \forall \phi \in C_c^1(\Omega)^d \tag{S21.1}$$

By the definition of $BV(\Omega)$, this functional has finite total variation:

$$|\Lambda|(\Omega) = \sup\left\{\Lambda(\phi) : \phi \in C_c^1(\Omega)^d,\ \|\phi\|_{L^\infty} \leq 1\right\} < \infty \tag{S21.2}$$

This is precisely the condition that $\hat{f}_n \in BV(\Omega)$ imposes — the distributional gradient is a bounded linear functional on $C_c^1(\Omega)^d$ with respect to the supremum norm.

Substep 1b: Riesz Representation

Since Λ is a bounded linear functional on $C_c^1(\Omega)^d$ with finite total variation, the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) guarantees the existence of a vector-valued Radon measure $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ such that:

$$\Lambda(\phi) = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{S21.3}$$

That is:

$$-\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = \int_\Omega \phi \cdot d(D\hat{f}_n) \quad \forall \phi \in C_c^1(\Omega)^d \tag{S21.4}$$

This is the distributional gradient equation, and $D\hat{f}_n$ is the distributional gradient of $\hat{f}_n$ represented as a Radon measure.

Substep 1c: Finiteness of Total Variation

By the Radon-Nikodym theorem for Radon measures (Evans and Gariepy, 2015, Theorem 1.29), the derivative $D_{\mathcal{L}^d}(D\hat{f}_n)$ exists and is finite $\mathcal{L}^d$-almost everywhere on Ω. Since Ω is bounded and the total variation of Λ is finite by the BV assumption, we conclude:

$$|D\hat{f}_n|(\Omega) = |\Lambda|(\Omega) < \infty \tag{S21.5}$$

Therefore $D\hat{f}_n \in \mathcal{M}(\Omega)^d$ with $|D\hat{f}_n|(\Omega) < \infty$, as claimed.

S21.3 Step 2: Explicit Formula for Piecewise Constant Functions

Let $\hat{f}_n = \sum_j \bar{y}_j \mathbf{1}_{R_j}$ where $\{R_j\}$ is a finite partition of Ω into sets with Lipschitz boundaries, and $\bar{y}_j$ denotes the response mean on leaf R_j. We derive the explicit formula for $D\hat{f}_n$ by applying the Gauss–Green theorem to each leaf.

Substep 2a: Applying the Definition

For any test function $\phi \in C_c^1(\Omega)^d$, the distributional gradient equation from Substep 1b gives:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\int_\Omega \hat{f}_n \operatorname{div} \phi \, dx = -\sum_j \bar{y}_j \int_{R_j} \operatorname{div} \phi \, dx \tag{S21.6}$$

where the second equality uses the piecewise constant structure of $\hat{f}_n$ and linearity of the integral.

Substep 2b: Gauss–Green on Each Leaf

Since each R_j has Lipschitz boundary, the Gauss–Green theorem for BV functions (Evans and Gariepy, 2015, Theorem 5.16) applies to each leaf:

$$\int_{R_j} \operatorname{div} \phi \, dx = \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{S21.7}$$

where ν_j denotes the outward unit normal to ∂R_j and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional Hausdorff measure on the boundary.

Substep 2c: Cancellation on Interior Boundaries

Substituting the Gauss–Green formula into the expression from Substep 2a:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} \tag{S21.8}$$

Every interior boundary $\partial R_j \cap \partial R_k$ between adjacent leaves j and k appears twice in this sum — once with outward normal ν_j for leaf R_j and once with outward normal $\nu_k = -\nu_j$ for leaf R_k. Collecting these paired contributions and writing $\nu_{jk} = \nu_j$ for the unit normal

pointing from R_j into R_k:

$$\begin{aligned} -\sum_j \bar{y}_j \int_{\partial R_j} \phi \cdot \nu_j \, d\mathcal{H}^{d-1} &= -\sum_{j\sim k} \bar{y}_j \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &\quad + \sum_{j\sim k} \bar{y}_k \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \\ &= \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \end{aligned} \tag{S21.9}$$

Boundary terms on $\partial\Omega$ vanish because $\phi \in C_c^1(\Omega)^d$ has compact support in Ω.

Substep 2d: Identifying the Measure

Since the equation:

$$\int_\Omega \phi \cdot d(D\hat{f}_n) = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \int_{\partial R_j \cap \partial R_k} \phi \cdot \nu_{jk} \, d\mathcal{H}^{d-1} \tag{S21.10}$$

holds for all $\phi \in C_c^1(\Omega)^d$, the uniqueness part of the Riesz representation theorem (Evans and Gariepy, 2015, Theorem 1.38) identifies $D\hat{f}_n$ uniquely as the measure:

$$D\hat{f}_n = \sum_{j\sim k} (\bar{y}_k - \bar{y}_j) \cdot \nu_{jk} \cdot \mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k} \tag{S21.11}$$

where $\mathcal{H}^{d-1} \lfloor_{\partial R_j \cap \partial R_k}$ denotes the $(d-1)$-dimensional Hausdorff measure restricted to the shared boundary between leaves j and k. This is the formula stated in the lemma. □

S22 Proof of Lemma S4 (Derivative Consistency, BV)

S22.1 What Needs to Be Shown

That well-behavedness in the BV sense implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$, and the strict convergence upgrade under the additional condition $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$.

S22.2 Proof Strategy

The proof mirrors the subsequence argument of Lemma S2, replacing the Rellich–Kondrachov compact embedding with the BV compactness theorem and replacing strong L^p convergence with weak-* convergence of measures. We show that every subsequence of $\{D\hat{f}_n\}$ has a further subsequence converging weak-* to Df. By the metric space characterization of convergence in probability, this implies $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ in probability.

Suppose for contradiction that $D\hat{f}_n$ does not converge weak-* to Df in probability. Then there exist $\epsilon > 0$, $\delta > 0$, a test function $\phi \in C_0(\Omega)^d$ with $\|\phi\|_\infty \leq 1$, and a subsequence $\{n_k\}$ such that:

$$\mathbb{P}\left(\left|\int_\Omega \phi \cdot d(D\hat{f}_{n_k}) - \int_\Omega \phi \cdot d(Df)\right| > \epsilon\right) > \delta \quad \forall k \tag{S22.1}$$

We derive a contradiction by extracting a further subsequence along which weak-* convergence holds almost surely.

S22.3 Part I: Weak-* Convergence

Step 1: Tightness of the BV Norms

The well-behaved condition in the BV sense controls $|D\hat{f}_n|(\Omega)$ in expectation, and an expectation bound does not make the sequence bounded almost surely: the total variation may be arbitrarily large on an event of vanishing but positive probability for every n. What the condition does deliver is uniform tightness, and tightness is what the subsequence architecture needs. Fix ϵ, δ, ϕ and $\{n_k\}$ as in the contradiction hypothesis and write

$$A_k = \left\{\left|\int_\Omega \phi \cdot d(D\hat{f}_{n_k}) - \int_\Omega \phi \cdot d(Df)\right| > \epsilon\right\}, \qquad \mathbb{P}(A_k) > \delta. \tag{S22.2}$$

Since $\left\|\hat{f}_n - f\right\|_{L^1} \xrightarrow{p} 0$, pass once to a subsequence of $\{n_k\}$ along which $\left\|\hat{f}_{n_k} - f\right\|_{L^1} \to 0$ almost surely. The contradiction hypothesis is inherited by every subsequence, so nothing is lost and we continue to write $\{n_k\}$. Let Ω_0 be the probability-one event on which that convergence holds. Securing it here, rather than by a second extraction later,

is what keeps the argument free of a subsequence that depends on the sample point.

For the total variation term, $C_1 = \sup_n \mathbb{E}[|D\hat{f}_n|(\Omega)] < \infty$, so Markov's inequality gives $\mathbb{P}(|D\hat{f}_n|(\Omega) > M) \le C_1/M$ uniformly in n; fix M with $C_1/M < \delta/8$. For the L^1 term, $\left\|\hat{f}_n\right\|_{L^1} \le \left\|\hat{f}_n - f\right\|_{L^1} + \|f\|_{L^1}$ with $\|f\|_{L^1} < \infty$ because $f \in W^{1,p}(\Omega) \subset L^1(\Omega)$, so there are $C_2 < \infty$ and K_0 with $\mathbb{P}(\left\|\hat{f}_{n_k}\right\|_{L^1} > C_2) < \delta/8$ for $k \ge K_0$. Setting

$$E_k = \left\{|D\hat{f}_{n_k}|(\Omega) \le M\right\} \cap \left\{\left\|\hat{f}_{n_k}\right\|_{L^1} \le C_2\right\}, \tag{S22.3}$$

we have $\mathbb{P}(E_k) \ge 1-\delta/4$ for $k \ge K_0$, and on E_k the realized BV norm satisfies $\left\|\hat{f}_{n_k}\right\|_{BV} \le R := C_2 + M$.

Step 2: A Positive-Probability Event of Bounded Realizations

For $k \ge K_0$ we have $\mathbb{P}(A_k \cap E_k) \ge \delta - \delta/4 = 3\delta/4$. Because $\mathbb{P}(\limsup_k B_k) \ge \limsup_k \mathbb{P}(B_k)$ for any sequence of events B_k in a probability space, the event

$$G = \left(\limsup_{k\to\infty} A_k \cap E_k\right) \cap \Omega_0 \tag{S22.4}$$

satisfies $\mathbb{P}(G) \ge 3\delta/4 > 0$. Fix $\omega \in G$. By construction ω lies in $A_k \cap E_k$ for infinitely many k; let $\{k_j\}$ enumerate those indices. Along them the realized functions $\hat{f}_{n_{k_j}}(\omega)$ are a deterministic sequence bounded in BV by R, so the BV compactness theorem (Evans and Gariepy, 2015, Theorem 5.5) applies to it as a statement about functions, with no measurable selection required: there are a further subsequence $\{k_{j_i}\}$ and $g \in BV(\Omega)$ with

1. $\hat{f}_{n_{k_{j_i}}}(\omega) \to g$ strongly in $L^1(\Omega)$,
2. $D\hat{f}_{n_{k_{j_i}}}(\omega) \xrightarrow{w*} Dg$ in $\mathcal{M}(\Omega)^d$.

This is the BV analogue of Rellich–Kondrachov: one extraction delivers both the L^1 convergence of the functions and the weak-* convergence of their gradient measures.

Step 3: Identify the Limit

Because $\omega \in \Omega_0$, $\left\| \hat{f}_{n_k}(\omega) - f \right\|_{L^1} \to 0$ along the whole index set and hence along $\{k_{j_i}\}$. Limits in $L^1(\Omega)$ are unique, so $g = f$ $\mathcal{L}^d$-almost everywhere on Ω, $Dg = Df$ as Radon measures, and item (2) of Step 2 gives

$$\int_\Omega \phi \cdot d\Big(D\hat{f}_{n_{k_{j_i}}}(\omega)\Big) \longrightarrow \int_\Omega \phi \cdot d(Df). \tag{S22.5}$$

Step 4: Derive the Contradiction

But $\omega \in A_{k_j}$ for every j, so

$$\left| \int_\Omega \phi \cdot d\Big(D\hat{f}_{n_{k_j}}(\omega)\Big) - \int_\Omega \phi \cdot d(Df) \right| > \epsilon \tag{S22.6}$$

for every j, in particular along $\{k_{j_i}\}$, contradicting the convergence just established. As $\omega \in G$ was arbitrary and $\mathbb{P}(G) > 0$, the contradiction hypothesis is untenable.

Conclusion of Part I

The test function $\phi \in C_0(\Omega)^d$ in the contradiction hypothesis was arbitrary, so for every such ϕ the integrals $\int_\Omega \phi \cdot d(D\hat{f}_n)$ converge in probability to $\int_\Omega \phi \cdot d(Df)$, which is to say

$$D\hat{f}_n \xrightarrow{w*} Df \quad \text{in } \mathcal{M}(\Omega)^d \quad \text{in probability.} \tag{S22.7}$$

S22.4 Part II: Strict Convergence Upgrade

We now prove the additional claim: if $\mathbb{E}[|D\hat{f}_n|(\Omega)] \to |Df|(\Omega)$ then $D\hat{f}_n \to Df$ strictly in $\mathcal{M}(\Omega)^d$, meaning weak-* convergence holds and $|D\hat{f}_n|(\Omega) \to |Df|(\Omega)$.

Step 1: Lower Semicontinuity Gives One Direction

The total variation functional $\mu \mapsto |\mu|(\Omega)$ is lower semicontinuous with respect to weak-* convergence of Radon measures (Evans and Gariepy, 2015, Theorem 5.2). Since $D\hat{f}_n \xrightarrow{w*}$

Df in $\mathcal{M}(\Omega)^d$ by Part I:

$$|Df|(\Omega) \leq \liminf_{n\to\infty} |D\hat{f}_n|(\Omega) \tag{S22.8}$$

This gives the lower bound on the total variation of the limit.

Step 2: The Additional Condition Gives the Other Direction

The additional assumption constrains a sequence of expectations and does not by itself bound $\limsup_n |D\hat{f}_n|(\Omega)$ pathwise; a sequence of random variables can have convergent means while fluctuating without limit. The two directions combine through Fatou's lemma instead. Write $Z_n = |D\hat{f}_n|(\Omega) - |Df|(\Omega)$ and take any subsequence. By Part I it has a further subsequence along which $D\hat{f}_n \xrightarrow{w*} Df$ almost surely, and along that further subsequence lower semicontinuity gives $\liminf_n Z_n \geq 0$ almost surely. Hence the negative part satisfies $Z_n^- \to 0$ almost surely, and since $0 \leq Z_n^- \leq |Df|(\Omega) < \infty$ the bounded convergence theorem gives $\mathbb{E}[Z_n^-] \to 0$. The assumption gives $\mathbb{E}[Z_n] \to 0$, so $\mathbb{E}[Z_n^+] = \mathbb{E}[Z_n] + \mathbb{E}[Z_n^-] \to 0$, whence $Z_n^+ \to 0$ in L^1 and therefore in probability. Both parts vanish, so $Z_n \to 0$ in probability along that further subsequence; as the initial subsequence was arbitrary,

$$|D\hat{f}_n|(\Omega) \longrightarrow |Df|(\Omega) \quad \text{in probability.} \tag{S22.9}$$

Step 3: Conclude Strict Convergence

Part I gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$ in probability and Step 2 gives convergence of the total masses in probability. Together these are precisely strict convergence in $\mathcal{M}(\Omega)^d$ (Ambrosio et al., 2000, Definition 3.13). □

Remark S9. The parallel between Lemma S2 and Lemma S4 is close but not exact. In both cases uniform boundedness in a strong space gives compactness via a compact embedding or compactness theorem, the consistency condition identifies the limit, and the subsequence argument promotes deterministic convergence to convergence in probability. The first difference is in the function spaces and the topology of convergence — strong L^p in the Sobolev case, weak-* of measures in the BV case — reflecting the fundamental

difference in the inductive biases of smooth and piecewise constant estimators. The second is in the mode of the boundedness hypothesis, and it is a difference of form rather than of substance. Definition S1 asks for tightness of $\left\|\hat{f}_n\right\|_{W^{2,p}}$, and Definition S2 asks for a bound on $\mathbb{E}[|D\hat{f}_n|(\Omega)]$, which gives tightness by Markov's inequality. Neither makes the sequence almost surely bounded by a constant, so in neither case may a compactness theorem be applied to $\{\hat{f}_n\}$ directly: a subsequence extracted from a random sequence depends on the realization, and an almost sure statement along such a subsequence will not combine with a probability bound along a deterministic one. Both proofs therefore work on an event where the realized norms are bounded and apply compactness there — the Sobolev argument through the deterministic modulus of Lemma S5, the BV argument through an event of positive probability.

Remark S10. The strict convergence upgrade requires the additional assumption that $\mathbb{E}[|D\hat{f}_n|(\Omega)]$ converge to $|Df|(\Omega)$. This is not implied by weak-* convergence alone. This is because weak-* convergence of measures does not in general imply convergence of total masses — mass can escape to the boundary or concentrate in ways invisible to continuous test functions. The lower semicontinuity of total variation (Evans and Gariepy, 2015, Theorem 5.2) gives the liminf inequality for free; pinning the limsup requires the additional assumption. In applications where only weak-* convergence is needed — as in the bootstrap corollaries — the additional assumption is not required.

S23 Proof of Theorem 3 (Pairing Identity)

Proof. Write points as $x = (x_{-j}, x_j)$ and for fixed x_{-j} let $\Omega^{x_{-j}} = \{\tau \in \mathbb{R} : (x_{-j}, \tau) \in \Omega\}$ denote the one-dimensional section of the domain and $g^{x_{-j}}(\tau) = g(x_{-j}, \tau)$ the section of g. By the one-dimensional section theory of BV functions (Ambrosio et al., 2000, Section 3.11), for $\mathcal{H}^{d-1}$-almost every x_{-j} the section $g^{x_{-j}}$ of the precise representative is a good representative of a function of bounded variation on $\Omega^{x_{-j}}$, its distributional

derivative $Dg^{x_{-j}}$ is the corresponding section of $D_j g$, and the total variations disintegrate:

$$\int_\Omega \phi \, dD_j g = \int \left(\int_{\Omega^{x_{-j}}} \phi(x_{-j}, \tau) \, dDg^{x_{-j}}(\tau) \right) d\mathcal{H}^{d-1}(x_{-j}) \tag{S23.1}$$

for every bounded Borel ϕ. For a good representative of a BV function of one variable, the fundamental theorem of calculus holds in the form $g^{x_{-j}}(b) - g^{x_{-j}}(a) = Dg^{x_{-j}}\big((a,b]\big)$ for all $a < b$ in $\Omega^{x_{-j}}$ that are not atoms of $Dg^{x_{-j}}$; since each section has at most countably many atoms, for Lebesgue-almost every x_j the pair $(x_j - h, x_j + h)$ avoids them. Hence for almost every $x \in \operatorname{supp}(w)$ (where the whole segment lies in Ω by the trimming condition):

$$g(x + he_j) - g(x - he_j) = Dg^{x_{-j}}\big((x_j - h,\ x_j + h]\big) \tag{S23.2}$$

Multiplying by $q(x)/2h$, integrating, and applying Tonelli on each section (justified since $|Dg^{x_{-j}}|$ is finite for a.e. section and q is bounded):

$$\begin{aligned}
&\mathbb{E}\left[w(X)\, \Delta_{j,h} g(X)\right] \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int dx_j \, \frac{q(x_{-j}, x_j)}{2h} \int \mathbf{1}\{\tau \in (x_j - h, x_j + h]\} \, dDg^{x_{-j}}(\tau) \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int dDg^{x_{-j}}(\tau) \, \frac{1}{2h} \int \mathbf{1}\{x_j \in [\tau - h, \tau + h)\} \, q(x_{-j}, x_j) \, dx_j \\
&= \int d\mathcal{H}^{d-1}(x_{-j}) \int (K_h \star q)(x_{-j}, \tau) \, dDg^{x_{-j}}(\tau) \\
&= \int_\Omega (K_h \star q) \, dD_j g
\end{aligned} \tag{S23.3}$$

using $\tau \in (x_j - h, x_j + h] \Leftrightarrow x_j \in [\tau - h, \tau + h)$ in the second line, the definition of $K_h \star q$ (half-open versus closed windows agree up to Lebesgue-null sets) in the third, and (S23.1) in the last step. The piecewise constant case follows by substituting the explicit jump formula of Lemma S3 for $D_j g$.

For the corollary: if q extends to a uniformly continuous function of compact support on $\mathbb{R}^d$, then $K_h \star q \to q$ uniformly as $h \downarrow 0$, so:

$$\left| \int (K_h \star q) \, dD_j g - \int q \, dD_j g \right| \le \|K_h \star q - q\|_\infty \cdot |D_j g|\,(\Omega) \to 0 \tag{S23.4}$$

since $|D_j g|(\Omega) < \infty$. When $g \in W^{1,1}(\Omega)$, the measure $D_j g$ is absolutely continuous with density $\partial_j g$, so $\int q \, dD_j g = \int w \, \partial_j g \, p \, dx = \mathbb{E}[w \, \partial_j g(X)]$. □

S24 Proof of Theorem S1 (Neural Networks)

S24.1 Setup and Notation

Let $\hat{f}_n : \Omega \to \mathbb{R}$ be a feedforward neural network with L layers, weight matrices $W_\ell^{(n)} \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, bias vectors $b_\ell^{(n)} \in \mathbb{R}^{d_\ell}$, and activation function $\sigma \in C^2(\mathbb{R})$. The network computes:

$$\hat{f}_n(x) = W_L^{(n)} z_{L-1} + b_L^{(n)} \tag{S24.1}$$

where the intermediate representations are defined recursively by:

$$z_\ell = \sigma\left(W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}\right), \quad z_0 = x \tag{S24.2}$$

with σ applied elementwise. Denote the pre-activation at layer ℓ by $a_\ell = W_\ell^{(n)} z_{\ell-1} + b_\ell^{(n)}$ and the diagonal matrix of activation derivatives by $\Sigma'_\ell = \mathrm{diag}(\sigma'(a_\ell^{(1)}), \ldots, \sigma'(a_\ell^{(d_\ell)}))$ and similarly $\Sigma''_\ell = \mathrm{diag}(\sigma''(a_\ell^{(1)}), \ldots, \sigma''(a_\ell^{(d_\ell)}))$.

Define the product of weight matrices from layer k to layer ℓ as:

$$M_{\ell:k} = W_\ell^{(n)} \Sigma'_{\ell-1} W_{\ell-1}^{(n)} \Sigma'_{\ell-2} \cdots W_{k+1}^{(n)} \Sigma'_k \quad \text{for } \ell > k \tag{S24.3}$$

with the convention $M_{\ell:\ell} = \Sigma'_\ell$. This notation captures the Jacobian of the composition of layers from k to ℓ.

The proof proceeds in two steps. Step 1 establishes uniform $W^{2,p}(\Omega)$ boundedness. Step 2 applies Lemma S2 to conclude.

S24.2 Step 1: $W^{2,p}$ Tightness

Claim: $\left\|\hat{f}_n\right\|_{W^{2,p}(\Omega)} = O_p(1)$.

Substep 1a: Bound on $\hat{f}_n$ in L^p

Since $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ and $f \in W^{1,p}(\Omega) \subset L^p(\Omega)$, the triangle inequality gives:

$$\left\|\hat{f}_n\right\|_{L^p} \leq \left\|\hat{f}_n - f\right\|_{L^p} + \|f\|_{L^p} \tag{S24.4}$$

The first term converges to zero in probability and the second is a finite constant since $f \in L^p(\Omega)$. Hence $\left\|\hat{f}_n\right\|_{L^p} = O_p(1)$, which is the form of boundedness the claim asserts; no appeal to a subsequence or to an auxiliary condition on the network output is needed, because tightness is preserved under addition and a term converging to zero in probability is tight.

Substep 1b: First Derivative Bound

Differentiating $\hat{f}_n$ with respect to the input x via the chain rule through all L layers, the gradient at $x \in \Omega$ is:

$$\nabla \hat{f}_n(x) = \left(W_L^{(n)} M_{L-1:1}\right)^\top \tag{S24.5}$$

where $M_{L-1:1} = W_{L-1}^{(n)} \Sigma'_{L-2} W_{L-2}^{(n)} \cdots \Sigma'_1 W_1^{(n)}$ is the product of weight-activation Jacobians from layer 1 to layer $L-1$. More explicitly, for the j-th input coordinate:

$$\frac{\partial \hat{f}_n}{\partial x_j}(x) = \left(W_L^{(n)}\right)_{:} \cdot \prod_{\ell=L-1}^{1} \left(W_\ell^{(n)} \Sigma'_{\ell-1}\right) \cdot e_j \tag{S24.6}$$

where e_j is the j-th standard basis vector.

Taking the operator norm of this product and applying submultiplicativity:

$$\left|\frac{\partial \hat{f}_n}{\partial x_j}(x)\right| \leq \left\|W_L^{(n)}\right\| \cdot \prod_{\ell=1}^{L-1} \left\|W_\ell^{(n)}\right\| \cdot \|\Sigma'_\ell\| \leq \prod_{\ell=1}^{L} \left\|W_\ell^{(n)}\right\| \cdot \|\sigma'\|_\infty^{L-1} \tag{S24.7}$$

using that $\|\Sigma'_\ell\| = \max_i |\sigma'(a_\ell^{(i)})| \leq \|\sigma'\|_\infty < \infty$ since $\sigma \in C^2$. This bound holds pointwise for every $x \in \Omega$, so integrating over Ω:

$$\left\|\nabla \hat{f}_n\right\|_{L^p} \leq \left\|\nabla \hat{f}_n\right\|_{L^\infty} \cdot |\Omega|^{1/p} \leq \prod_{\ell} \left\|W_\ell^{(n)}\right\| \cdot \|\sigma'\|_\infty^{L-1} \cdot |\Omega|^{1/p} \tag{S24.8}$$

Substep 1c: Second Derivative Bound

Differentiating the gradient formula once more with respect to x_i using the product rule and chain rule, the (i,j) entry of the Hessian $\nabla^2 \hat{f}_n(x)$ is:

$$\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x) = \sum_{\ell=1}^{L-1} \left(W_L^{(n)} M_{L-1:\ell+1}\right)_{:} \cdot \Sigma_\ell'' \cdot (M_{\ell:1} e_j) \cdot (M_{\ell:1} e_i)^\top \cdot \left(W_L^{(n)} M_{L-1:\ell+1}\right)_{:}^\top \quad \text{(S24.9)}$$

where the sum runs over all layers ℓ at which the second derivative of σ is applied. The structure is: the forward pass from the input to layer ℓ contributes $M_{\ell:1} e_j$ for the j-th coordinate direction, the activation second derivative at layer ℓ contributes Σ_ℓ'', and the backward pass from layer ℓ to the output contributes $W_L^{(n)} M_{L-1:\ell+1}$.

Taking absolute values and applying submultiplicativity of operator norms:

$$\begin{aligned}
\left|\frac{\partial^2 \hat{f}_n}{\partial x_i \partial x_j}(x)\right| &\le \sum_{\ell=1}^{L-1} \left\|W_L^{(n)} M_{L-1:\ell+1}\right\|^2 \cdot \|\Sigma_\ell''\| \cdot \|M_{\ell:1}\|^2 \\
&\le \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L} \left\|W_k^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \left(\prod_{k=1}^{\ell} \left\|W_k^{(n)}\right\|\right)^2 \cdot \|\sigma'\|_\infty^{2(L-2)} \\
&\le (L-1) \cdot \left(\prod_{\ell=1}^{L} \left\|W_\ell^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)} \qquad \text{(S24.10)}
\end{aligned}$$

using $\|\Sigma_\ell''\| \le \|\sigma''\|_\infty < \infty$ since $\sigma \in C^2$, and collecting all weight norm factors into a single product. This bound holds pointwise for every $x \in \Omega$, so:

$$\left\|\nabla^2 \hat{f}_n\right\|_{L^p} \le \left\|\nabla^2 \hat{f}_n\right\|_{L^\infty} \cdot |\Omega|^{1/p} \le (L-1) \cdot \left(\prod_{\ell} \left\|W_\ell^{(n)}\right\|\right)^2 \cdot \|\sigma''\|_\infty \cdot \|\sigma'\|_\infty^{2(L-2)} \cdot |\Omega|^{1/p} \quad \text{(S24.11)}$$

Combining Substeps 1a–1c

Summing the three bounds:

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = \left\|\hat{f}_n\right\|_{L^p} + \left\|\nabla \hat{f}_n\right\|_{L^p} + \left\|\nabla^2 \hat{f}_n\right\|_{L^p} \le C\left(\prod_{\ell} \left\|W_\ell^{(n)}\right\|, \|\sigma'\|_\infty, \|\sigma''\|_\infty, L, |\Omega|\right) \quad \text{(S24.12)}$$

where C is continuous and increasing in its first argument and depends on the remaining quantities but not on n or x. The weight product $\prod_\ell \left\|W_\ell^{(n)}\right\|$ is a function of the fitted weights and so is random; by assumption it is $O_p(1)$ — which formalizes the effect of L^2 regularization during training (Bartlett et al., 2017) — and $\|\sigma'\|_\infty$, $\|\sigma''\|_\infty$ are finite constants since $\sigma \in C^2$. A continuous increasing function of a tight sequence is tight, so

$$\left\|\hat{f}_n\right\|_{W^{2,p}} = O_p(1), \tag{S24.13}$$

which is the first condition of Definition S1. This completes Step 1.

S24.3 Step 2: L^p Consistency

The L^p consistency condition $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$ is the standard universal approximation result for smooth neural networks. Hornik et al. (1990) establish that multilayer feedforward networks with smooth activation functions are dense in $W^{1,p}(\Omega)$ with respect to the $W^{1,p}$ norm, implying in particular density in $L^p(\Omega)$. Combined with standard empirical risk minimization results establishing that the training procedure converges to the best approximation in the function class, this gives $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$.

S24.4 Step 3: Applying Lemma S2

By Step 1, $\left\|\hat{f}_n\right\|_{W^{2,p}} = O_p(1)$, the bound being a fixed multiple of the random quantity $\prod_\ell \left\|W_\ell^{(n)}\right\|$, which is $O_p(1)$ by hypothesis. By Step 2, $\left\|\hat{f}_n - f\right\|_{L^p} \xrightarrow{p} 0$. These are precisely the two conditions of the well-behaved definition in the Sobolev sense (Definition S1). Lemma S2 then gives:

$$\left\|\nabla \hat{f}_n - \nabla f\right\|_{L^p} \xrightarrow{p} 0 \tag{S24.14}$$

which is the conclusion of Theorem S1. □

Remark S11. The weight norm assumption $\sup_n \prod_\ell \left\|W_\ell^{(n)}\right\| < \infty$ is the key regularity condition on the estimator. It formalizes the effect of L^2 weight decay regularization during training, which penalizes large weights and keeps the product of operator norms

bounded. Under standard conditions on the regularization parameter and optimization procedure, this bound holds with high probability over the training randomness (Bartlett et al., 2017). The condition is stated as an assumption of the theorem rather than derived from first principles because its verification depends on details of the training procedure that are outside the scope of the present analysis.

Remark S12. Theorem S1 is stated for smooth activations $\sigma \in C^2$. For ReLU networks $\sigma \notin C^2$ and the Hessian calculation in Substep 1c does not apply directly. The first derivative exists almost everywhere with $\sigma' = \mathbf{1}_{(0,\infty)}$ bounded, so Substep 1b goes through unchanged. The distributional second derivative is $\sigma'' = \delta_0$, a Dirac mass at zero, placing $\hat{f}_n$ in a mixed $W^{1,p} \cap BV$ space rather than $W^{2,p}(\Omega)$. A modified version of Lemma S2 applicable in this mixed space, or a restriction to input distributions placing zero mass on the kink set $\{x : a_\ell(x) = 0\}$, would be required to handle this case. We leave this extension for future work.

S25 Proof of Theorem S2 (Gradient Boosting)

S25.1 Setup and Notation

By Proposition 1 of Bühlmann and Yu (2003), the L_2Boost estimator after m iterations can be represented as:

$$\hat{F}_m = (I - (I - S)^{m+1})Y \tag{S25.1}$$

where $S : \mathbb{R}^n \to \mathbb{R}^n$ is the base learner operator. The residual at step m is:

$$u_m = Y - \hat{F}_m = (I - S)^m Y \tag{S25.2}$$

so that each base learner fit is $\hat{f}_m = Su_m$ and the ensemble updates as $\hat{F}_m = \hat{F}_{m-1} + \hat{f}_m$.

Throughout this section $\|\cdot\|_{L^2}$ on $\mathbb{R}^n$ denotes the *normalized* empirical norm $\|v\|_{L^2}^2 = n^{-1}\sum_{i=1}^n v_i^2$, so that it is the $L^2(P_n)$ norm of the function $i \mapsto v_i$ and is directly comparable across sample sizes. This matters because the constant in (S25.5) is $\|Y\|_{L^2}$: under the normalized norm $\|Y\|_{L^2}^2 = n^{-1}\sum_i y_i^2 \to \mathbb{E}[Y^2] < \infty$ almost surely by Assumption 1 and

the law of large numbers, so $\|Y\|_{L^2} = O_p(1)$ uniformly in n. Under the unnormalized Euclidean norm it would instead grow like $n^{1/2}$ and the uniform bounds below would be vacuous.

By Proposition 2 of Bühlmann and Yu (2003), for a symmetric linear learner S with eigenvalues $\{\lambda_k; k = 1, \ldots, n\}$, the eigenvalues of the boosting operator $\mathcal{B}_m = I - (I - S)^{m+1}$ are $\{1 - (1 - \lambda_k)^{m+1}; k = 1, \ldots, n\}$.

Under the eigenvalue condition $0 < \lambda_k \leq 1$ for all k (Bühlmann and Yu, 2003, Theorem 1), let:

$$\rho = \max_k (1 - \lambda_k) \in [0, 1) \tag{S25.3}$$

Since $\lambda_k > 0$ for all k, we have $\rho < 1$. The bias decays exponentially fast: by Theorem 1(a) of Bühlmann and Yu (2003):

$$\|u_m\|_{L^2}^2 = \|(I - S)^m Y\|_{L^2}^2 \leq \rho^{2m} \|Y\|_{L^2}^2 \tag{S25.4}$$

In particular:

$$\|u_m\|_{L^2} \leq C\rho^m, \quad C = \|Y\|_{L^2} = O_p(1) \tag{S25.5}$$

This geometric decay is the engine of the argument below. Note that S is a contraction in the normalized norm exactly when it is one in the unnormalized norm, the normalization being a positive scalar multiple, so the eigenvalue condition and ρ are unaffected.

S25.2 $W^{2,p}$ Tightness and the Conclusion

Step 1: $W^{2,p}$ Bound for a Single Base Learner

Each $\hat{f}_m = Su_m \in W^{2,p}(\Omega)$ by assumption on the base learner class. For shallow neural net base learners this follows from Section S24. For spline base learners it follows from standard spline approximation theory (Wahba, 1990). We require a uniform bound:

$$\sup_m \left\| \hat{f}_m \right\|_{W^{2,p}} \leq C' \|u_m\|_{L^2} < \infty \tag{S25.6}$$

where the bound on $\left\|\hat{f}_m\right\|_{W^{2,p}}$ in terms of $\|u_m\|_{L^2}$ follows from the boundedness of S as an operator from L^2 to $W^{2,p}$.

Step 2: Ensemble Bound via Geometric Decay

By linearity of the $W^{2,p}$ norm and the triangle inequality:

$$\left\|\hat{F}_m\right\|_{W^{2,p}} \leq \sum_{j=1}^{m} \left\|\hat{f}_j\right\|_{W^{2,p}} \leq C' \sum_{j=1}^{m} \|u_j\|_{L^2} \leq C'C \sum_{j=1}^{m} \rho^j \leq \frac{C'C\rho}{1-\rho} < \infty \tag{S25.7}$$

using the geometric decay (S25.5) and the fact that $\sum_{j=1}^{\infty} \rho^j = \rho/(1-\rho) < \infty$ for $\rho < 1$. The bound is uniform in m, and uniform in n as well once the two suprema of (S4.3) are used: $C' \leq \sup_n \|S_n\|_{L^2 \to W^{2,p}}$ and $\rho/(1-\rho) \leq \bar{\rho}/(1-\bar{\rho})$.

What the bound is not is deterministic. Its remaining factor is $C = \|Y\|_{L^2}$, a random quantity, so what Step 2 establishes is

$$\left\|\hat{F}_m\right\|_{W^{2,p}} \leq \frac{\bar{\rho}}{1-\bar{\rho}} \left(\sup_n \|S_n\|_{L^2 \to W^{2,p}}\right) \|Y\|_{L^2} = O_p(1), \tag{S25.8}$$

since $\|Y\|_{L^2} = O_p(1)$ as noted after (S25.5). This is tightness in $W^{2,p}(\Omega)$, which is what Definition S1 requires; a bound by a single deterministic constant would be stronger than the geometric-decay argument can deliver, because nothing caps $\|Y\|_{L^2}$ almost surely under Assumption 1.

Step 3: Apply Lemma S2

L^p consistency is assumed — in L^p and not L^1, matching Definition S1. Tightness in $W^{2,p}(\Omega)$ is (S25.8). Lemma S2 gives $\left\|\nabla \hat{F}_m - \nabla f\right\|_{L^p} \xrightarrow{p} 0$. □

Remark S13 (Why tree base learners are excluded)**.** An earlier form of this theorem carried a second case for tree stump base learners. It is withdrawn, because the hypotheses cannot be met by a tree. The eigenvalue condition $0 < \lambda_k \leq 1$ requires S to have full rank, but a tree is the orthogonal projection onto the piecewise constant functions on its partition, so its spectrum is $\{0, 1\}$ with $n - (\#\text{leaves})$ zeros. Then $\rho = \max_k(1 - \lambda_k) = 1$

and the geometric series driving both steps above diverges — the degenerate situation already noted in Remark S16. Shrinkage does not help, since νS has spectrum $\{0, \nu\}$, and nor does an L^2 leaf penalty; the zero eigenvalues are unavoidable for any finite-rank base learner. A tree whose split is instead chosen from the current residual, as in gradient boosting as it is implemented, is not a fixed linear operator at all, so the representation $\hat{F}_m = (I - (I - S)^{m+1})Y$ does not hold for it and neither does the decay that the representation supplies.

Gradient boosting with greedily grown trees is therefore outside this theorem. It remains inside Theorem 1, which consumes an $L^2(P)$ rate and nothing else, and inside the pairing identity of Theorem 3, a finite sum of piecewise constant trees being of bounded variation. What is not established for it is gradient consistency of the *uncorrected* plug-in, and there is reason to think it does not hold: at the number of rounds minimizing its own risk, such an ensemble can carry substantially more total variation than the function it estimates, with the gap widening in n on targets poorly matched to axis-aligned splits. That is the regularization bias of Section 3.4 appearing in the interpretation theory, and it is an argument for the orthogonal correction rather than an omission from it.

Remark S14. The key step is the geometric decay $\|u_m\|_{L^2} \leq C\rho^m$, which follows from the eigenvalue condition $0 < \lambda_k \leq 1$ via Proposition 1 and Theorem 1 of Bühlmann and Yu (2003). This decay is much faster than any polynomial rate and implies that the geometric series $\sum_j \rho^j$ converges without any condition on step sizes. The eigenvalue condition is the operative regularity assumption of the theorem, replacing the step size conditions common in the boosting literature. It is also what confines the result to full-rank smoothers; see Remarks S16 and S13.

Remark S15. Condition (S4.3) is a genuine restriction, not a technicality. The operator S_n acts on $\mathbb{R}^n$, so its spectrum moves with the sample size, and a gap for each n separately does not bound $\sup_n (1 - \rho_n)^{-1}$, which is what the geometric series in Step 2 produces. The same applies to the constant C' of Step 1, which is the $L^2 \to W^{2,p}$ norm of S_n. Both enter the uniform $W^{2,p}(\Omega)$ bound that Lemma S2 consumes, and that bound is a statement about the sequence in n.

The condition fails whenever the smoother's spectrum accumulates at zero as the sample grows, which includes a smoothing spline at a fixed smoothing parameter: there $\lambda_k^{(n)} = (1 + \gamma d_k)^{-1}$ with the penalty eigenvalues d_k unbounded in n, so $\rho_n \to 1$. It holds for smoothers that are bounded away from singularity by construction, for instance $S_n = \epsilon I + (1 - \epsilon) S_n^0$ with $\epsilon \in (0, 1)$ fixed and S_n^0 symmetric with spectrum in $[0, 1]$, for which $\rho_n \leq 1 - \epsilon$ uniformly. Verifying (S4.3) is therefore part of applying the theorem.

Remark S16. When $\lambda_k \in \{0, 1\}$ for all k — as occurs for linear projection operators — Corollary 1 of Bühlmann and Yu (2003) shows that $\mathcal{B}_m \equiv S$ for all m, meaning boosting does not improve on the base learner. In this degenerate case $\rho = \max_k(1 - \lambda_k)$ takes the value 1 for any k with $\lambda_k = 0$, and the geometric series does not converge. The strict positivity condition $\lambda_k > 0$ excludes this case and corresponds to the requirement that the base learner has positive but incomplete explanatory power — the standard "weak learner" assumption in the boosting literature.

S26 Proof of Theorem S3 (Mondrian Forests)

S26.1 Setup

Let $\hat{f}_n = \frac{1}{B} \sum_{b=1}^{B} T_b$ where T_b are individual Mondrian trees. Each T_b is piecewise constant on a random partition $\mathcal{P}_b$ of $\Omega = [0, 1]^d$ generated by the Mondrian process $MP(\lambda_n, [0, 1]^d)$ independently of the data, with budget $\lambda_n \asymp n^{1/(d+2)}$ (Mourtada et al., 2017). The partition $\mathcal{P}_b$ consists of $K_{\lambda_n} + 1$ hyperrectangular cells, where K_{λ_n} is the number of splits (Roy and Teh, 2009).

S26.2 Step 1: Each Tree is in BV

Each T_b is piecewise constant on a finite partition and therefore in $BV(\Omega)$ by Lemma S3. The forest average $\hat{f}_n = \frac{1}{B} \sum_b T_b$ is a finite convex combination of BV functions, hence $\hat{f}_n \in BV(\Omega)$ with:

$$|D\hat{f}_n|(\Omega) \leq \frac{1}{B} \sum_{b=1}^{B} |DT_b|(\Omega) \tag{S26.1}$$

by convexity of the total variation functional (Ambrosio et al., 2000, Proposition 3.6).

S26.3 Step 2: Reduction to a Single Tree

Step 1 is already the whole of the forest argument. By convexity it suffices to bound one tree:

$$\mathbb{E}\Big[|D\hat{f}_n|(\Omega)\Big] \le \frac{1}{B}\sum_{b=1}^{B}\mathbb{E}[|DT_b|(\Omega)] = \mathbb{E}[|DT|(\Omega)] \tag{S26.2}$$

for a single Mondrian tree T, the trees being identically distributed. No cancellation between trees is needed, and consequently no condition on B: the bound below holds for a forest of any fixed size, including $B = 1$.

Steps 3 to 6 establish $\sup_n \mathbb{E}[|DT|(\Omega)] < \infty$. Write $\bar{y}_l$ for the mean of the responses falling in leaf R_l, with $\bar{y}_l = 0$ when R_l is empty, and N_l for that count.

S26.4 Step 3: The Jump Set is Carried by the Split Faces

Let k index the internal nodes of the tree, let C_k be the cell split at node k along coordinate j_k at threshold t_k, and write

$$F_k = \overline{C}_k \cap \{x : x_{j_k} = t_k\} \tag{S26.3}$$

for the split face. Lemma S3 gives the distributional gradient of a piecewise constant function as a sum over adjacent leaf pairs; the following re-indexes that sum by internal nodes, which is the form the later steps use.

Lemma S6 (Node-face representation)**.** *Let T be piecewise constant on the leaves of a hierarchical axis-aligned partition whose cuts are strictly interior to their cells. Then*

$$|DT|(\Omega) = \sum_k \int_{F_k} |\Delta_k(u)|\; d\mathcal{H}^{d-1}(u), \tag{S26.4}$$

where $\Delta_k(u)$ is the difference of the two leaf values meeting at u.

Proof. Two facts suffice. *First*, each leaf-pair interface lies in the face of that pair's lowest common ancestor. Let R_l and R_m be adjacent leaves with lowest common ancestor k.

They lie in opposite children of k, so after relabelling every $x \in \overline{R}_l$ has $x_{j_k} \le t_k$ and every $x \in \overline{R}_m$ has $x_{j_k} \ge t_k$. A point of $\partial R_l \cap \partial R_m$ lies in both closures, hence satisfies $x_{j_k} = t_k$, and lies in $\overline{C}_k$; so $\partial R_l \cap \partial R_m \subseteq F_k$.

Second, distinct faces meet in an $\mathcal{H}^{d-1}$-null set. Take internal nodes $k \neq k'$. If neither is an ancestor of the other they lie in opposite subtrees of their own lowest common ancestor, so $C_k \cap C_{k'} = \emptyset$. Otherwise let k be a strict ancestor of k', so that $C_{k'}$ lies in one child of k. If $j_{k'} \neq j_k$ the two hyperplanes are transverse and meet in dimension $d-2$, which is $\mathcal{H}^{d-1}$-null. If $j_{k'} = j_k$ they are parallel, and distinct: $t_{k'}$ is strictly interior to the extent of $C_{k'}$ in that coordinate while $C_{k'}$ lies entirely on one side of t_k.

The map sending an adjacent leaf pair to its lowest common ancestor therefore sorts the interfaces into the faces without repetition, and conversely each point in the relative interior of F_k lies between the leaf of $C_k \cap \{x_{j_k} \le t_k\}$ and the leaf of $C_k \cap \{x_{j_k} > t_k\}$ containing it, whose lowest common ancestor is k. Regrouping the sum of Lemma S3 gives (S26.4). □

The argument is deterministic: it uses no probability, no property of the Mondrian process, and holds in every dimension. The hypothesis that cuts are strictly interior holds almost surely here, cut positions being uniform on an open interval, and is what the second fact requires — a cut on a cell boundary would produce an empty child and a face coinciding with an ancestor's.

S26.5 Step 4: Three Facts About the Mondrian Partition

Lemma S7 (Expected interface area)**.** *For $\mathcal{P} \sim MP(\lambda, C)$ on a box C,*

$$\mathbb{E}\left[\sum_k \mathcal{H}^{d-1}(F_k)\right] = d\,\lambda\,\mathrm{vol}(C). \tag{S26.5}$$

Proof. Write $A(C,\lambda)$ for the left side, s for the side lengths of C, $a = \sum_j s_j$ and $V = \mathrm{vol}(C)$. A cut of C in coordinate j has face area $\prod_{m \neq j} s_m = V/s_j$ whatever the cut position, and the two children have volumes summing to V. Conditioning on the first

cut,

$$A(C,\lambda) = \int_0^{\lambda} a e^{-at} \left[\frac{dV}{a} + d(\lambda - t)V \right] dt, \tag{S26.6}$$

where the induction hypothesis $A(\cdot, \lambda') = d\lambda' \,\mathrm{vol}(\cdot)$ has been used on the children and is independent of which coordinate is cut and where. Evaluating,

$$A(C,\lambda) = dV\,\frac{1 - e^{-a\lambda}}{a} + dVa\left[\frac{\lambda}{a} - \frac{1 - e^{-a\lambda}}{a^2}\right] = d\lambda V, \tag{S26.7}$$

the two boundary terms cancelling exactly. □

The cancellation is why the answer is free of the cell's shape. The second fact is the cell law of Mourtada et al. (2017): for $MP(\lambda)$ on $\mathbb{R}^d$ the cell containing a fixed point is $\prod_j [-E_j^-, E_j^+]$ with the $2d$ variables i.i.d. $\mathrm{Exp}(\lambda)$, so each side is $\mathrm{Gamma}(2,\lambda)$ and the sides are independent across coordinates.

Both of the remaining steps need inverse moments of a single side of the cell, and both need them for the *truncated* side, since the cells of $MP(\lambda, [0,1]^d)$ are the whole-space cells intersected with the cube. The following records them once.

Lemma S8 (Inverse moments of a truncated side)**.** *Let* $s = \min(E^-, U) + \min(E^+, 1-U)$ *with* $E^{\pm}$ *i.i.d.* $\mathrm{Exp}(\lambda)$ *and* $U \sim \mathrm{Unif}(0,1)$*, which by projectivity is the law of a side of the cell of* $MP(\lambda, [0,1]^d)$ *containing a uniform point. Then for* $0 < p < 2$*,*

$$\mathbb{E}\big[s^{-p}\big] \le \lambda^p\,\Gamma(2-p)\left(1 + \frac{p}{\lambda}\right) + e^{-\lambda}. \tag{S26.8}$$

Proof. Write $s^{-p} = \Gamma(p)^{-1}\int_0^\infty r^{p-1}e^{-rs}\,dr$ and use $\mathbb{E}[e^{-r\min(E,u)}] = [\lambda + re^{-(\lambda+r)u}]/(\lambda+r)$. Given U the two sides are independent, so integrating $U \sim \mathrm{Unif}(0,1)$ out before r,

$$\mathbb{E}\big[e^{-rs}\big] = \frac{\lambda^2 + 2\lambda r\,(1 - e^{-(\lambda+r)})/(\lambda+r) + r^2 e^{-(\lambda+r)}}{(\lambda+r)^2}. \tag{S26.9}$$

Three elementary integrals follow, using $\int_0^\infty v^{a-1}(1+v)^{-a-b}dv = B(a,b)$ after $r = \lambda v$. The first gives $\Gamma(p)^{-1}\lambda^2\int r^{p-1}(\lambda+r)^{-2}dr = \lambda^p\Gamma(2-p)$. The second, after discarding $1 - e^{-(\lambda+r)} \le 1$, gives $\Gamma(p)^{-1}2\lambda\int r^p(\lambda+r)^{-3}dr = p\,\lambda^{p-1}\Gamma(2-p)$. The third is at most

$e^{-\lambda}$ because $r^2/(\lambda+r)^2 \le 1$ and $\Gamma(p)^{-1}\int r^{p-1}e^{-r}dr = 1$. □

The untruncated value is $\lambda^p\Gamma(2-p)$, so truncation costs the factor $1+p/\lambda$ together with an exponentially small additive term: an $O(1/\lambda)$-thick boundary layer, as one would expect. At $p=1$ this is the bound $\lambda+1+e^{-\lambda}$ used in Lemma S9 below; at $p=1/2$ and $p=3/2$ it is what Lemma S11 needs. It is convenient to write (S26.8) as

$$\mathbb{E}\big[s^{-p}\big] \le c_p(\lambda)\,\lambda^p, \qquad c_p(\lambda) = \Gamma(2-p)\left(1+\frac{p}{\lambda}\right) + \lambda^{-p}e^{-\lambda}, \tag{S26.10}$$

so that $c_p(\lambda)$ decreases to the untruncated constant $\Gamma(2-p)$ and is bounded by $c_p(1)$ for $\lambda \ge 1$. Only boundedness in λ is used below; the rates come from the powers of λ.

Lemma S9 (Aspect ratio)**.** *Let $l(X)$ be the cell of $MP(\lambda,[0,1]^d)$ containing a uniform point X, with side lengths s_j. Then*

$$\mathbb{E}\left[\frac{\operatorname{diam}(l(X))}{s_1(l(X))}\right] \le 2d-1+\frac{4(d-1)}{\lambda}. \tag{S26.11}$$

Proof. The Mondrian is projective, so $MP(\lambda,[0,1]^d)$ is $MP(\lambda,\mathbb{R}^d)$ restricted to the cube and restriction intersects every cell with the sub-box; hence $s_j = \min(E_j^-, X_j) + \min(E_j^+, 1-X_j)$. The coordinates of X are independent and the $E_j^\pm$ are independent across j, so s_m and s_1 are independent and $\mathbb{E}[s_m/s_1] = \mathbb{E}[s_m]\,\mathbb{E}[1/s_1]$. Truncation only shrinks, so $\mathbb{E}[s_m] \le 2/\lambda$.

For the inverse moment apply Lemma S8 with $p=1$: $\mathbb{E}[1/s_1] \le \lambda+1+e^{-\lambda}$, against the value λ on $\mathbb{R}^d$, so $\mathbb{E}[s_m/s_1] \le 2+2(1+e^{-\lambda})/\lambda \le 2+4/\lambda$ for $\lambda \ge 1$. Summing over m with diam $\le \sum_m s_m$ and the $m=1$ term equal to one gives the claim. □

The boundary therefore costs an additive constant rather than a rate: an $O(1/\lambda)$-thick layer contributing $O(1)$ to an integral of size λ. Since $\lambda_n \to \infty$ the constant converges to $2d-1$.

S26.6 Step 5: The Signal Term

Write $\bar{y}_l = \bar{f}_l + \varepsilon_l$, where $\bar{f}_l = N_l^{-1} \sum_{i \in R_l} f(\mathbf{x}_i)$ and $\varepsilon_l = N_l^{-1} \sum_{i \in R_l} (y_i - f(\mathbf{x}_i))$, both zero on empty leaves, and set $T^{\mathrm{s}} = \sum_{N_l \geq 1} \bar{f}_l \mathbf{1}_{R_l}$, $T^{\mathrm{n}} = \sum_{N_l \geq 1} \varepsilon_l \mathbf{1}_{R_l}$, so that $T = T^{\mathrm{s}} + T^{\mathrm{n}}$. By (S26.4) and the triangle inequality,

$$|DT|(\Omega) \leq S_1 + S_0 + N, \tag{S26.12}$$

where N collects the jumps of T^{n}, S_1 the jumps of T^{s} across faces with both sides nonempty, and S_0 those across faces with an empty side.

Fix the coordinate 1; the argument is identical for each. Along the line $\ell(x_{-1}) = \{(t, x_{-1}) : t \in [0,1]\}$ the tree is a step function whose jumps occur at the 1-perpendicular cuts. The following records the only property of such a sum that is used, and in particular how a bound on a jump *between two leaves* becomes a sum over single leaves.

Lemma S10 (Per-leaf charging)**.** *Suppose each jump between consecutive leaves is bounded by $\delta(l') + \delta(l'')$ for the two leaves l', l'' meeting there and some nonnegative δ defined on leaves. Then*

$$\sum_k \int_{F_k} |\Delta_k| \, d\mathcal{H}^{d-1} \leq 2 \sum_l \delta(l) \, \frac{\mathrm{vol}(l)}{s_1(l)} = 2 \, \mathbb{E}_X \left[\frac{\delta(l(X))}{s_1(l(X))} \right], \tag{S26.13}$$

the expectation being over a uniform X with $l(X)$ its leaf.

Proof. Let $l_0, \ldots, l_m$ be the leaves ℓ crosses, in order, with values $v_0, \ldots, v_m$. By hypothesis

$$\sum_{i=1}^{m} |v_i - v_{i-1}| \leq \sum_{i=1}^{m} (\delta(l_{i-1}) + \delta(l_i)) \leq 2 \sum_{i=0}^{m} \delta(l_i), \tag{S26.14}$$

because each index appears in at most two summands — once for the neighbour on its left and once for the neighbour on its right. The neighbouring leaf's contribution is therefore absorbed, and what remains is a sum over single leaves. Integrating over x_{-1} and using that leaf l is crossed by a set of lines of measure $\mathrm{vol}(l)/s_1(l)$ gives the first form; the second is $\sum_l g(l)\mathrm{vol}(l) = \mathbb{E}_X[g(l(X))]$. □

It is (S26.13), with a per-leaf δ, that the next three steps apply: $\delta(l) = L\,\mathrm{diam}(R_l)$ for the signal, $\delta(l) = \|f\|_\infty \mathbf{1}\{N_l = 0\}$ for the empty-leaf term, and $\delta(l) = |\varepsilon_l|$ for the noise. Let $u \in \partial R_l \cap \partial R_m$ be a point of a face at which both leaves are nonempty, and let f be Lipschitz with constant L. Every $\mathbf{x}_i \in R_l$ lies within $\mathrm{diam}(R_l)$ of u, so $|\bar{f}_l - f(u)| \le L\,\mathrm{diam}(R_l)$, and likewise for m; hence

$$\left|\bar{f}_m - \bar{f}_l\right| \le L\left(\mathrm{diam}(R_l) + \mathrm{diam}(R_m)\right). \tag{S26.15}$$

The diameters here are those of the two *leaves* meeting at u, not that of the ancestor cell C_k, which may be larger than either. The bound is pairwise, involving both leaves; it is Lemma S10, applied with the per-leaf quantity $\delta(l) = L\,\mathrm{diam}(R_l)$, that turns it into a sum over single leaves, the neighbour's diameter being absorbed by the factor of two. Then by Lemma S9,

$$\mathbb{E}[S_1] \le 2L\,\mathbb{E}\left[\frac{\mathrm{diam}(l(X))}{s_1(l(X))}\right] \le 2L\left(2d - 1 + \frac{4(d-1)}{\lambda}\right). \tag{S26.16}$$

No rate in n or λ enters: the discrete variation of a Lipschitz function over a refining partition is controlled by the partition's aspect ratio, not by how fine it is.

The bound (S26.15) requires both leaves to be nonempty, since $\bar{f}_l$ is set to zero on an empty leaf and then bears no relation to f near u. Faces with an empty side are collected in S_0 and bounded separately in Step 5a; a jump there is at most $\|f\|_\infty$, which is not small, so the term is controlled by the scarcity of empty leaves rather than by the size of the jump.

S26.7 Step 5a: Empty Leaves

At a face with one side empty the jump of T^{s} is at most $\|f\|_\infty$, and at a face with both sides empty it is zero. By (S26.13),

$$S_0 \le 2\,\|f\|_\infty \sum_{l\,:\,N_l = 0} \frac{\mathrm{vol}(l)}{s_1(l)}. \tag{S26.17}$$

Conditionally on the partition $\mathbb{P}(N_l = 0) = (1-\mathrm{vol}(l))^n \le e^{-n\,\mathrm{vol}(l)}$, so with $V = \mathrm{vol}(l(X))$ as in Lemma S11,

$$\mathbb{E}[S_0] \le 2\,\|f\|_\infty\,\mathbb{E}\left[\frac{e^{-nV}}{s_1}\right] \le 2\,\|f\|_\infty\,n^{-1/2}\,\mathbb{E}\left[s_1^{-3/2}\right]\prod_{j\ge 2}\mathbb{E}\left[s_j^{-1/2}\right], \tag{S26.18}$$

using $e^{-x} \le x^{-1/2}$, valid for $x > 0$ since $\sup_{x>0} x^{1/2}e^{-x} = (2e)^{-1/2} < 1$. The moments are those of Lemma S8, so by (S26.10)

$$\mathbb{E}[S_0] \le 2\,\|f\|_\infty\;c_{3/2}\,c_{1/2}^{d-1}\,\lambda^{1+d/2}n^{-1/2}, \tag{S26.19}$$

the same order as the noise term, and so $O(1)$ at $\lambda_n \asymp n^{1/(d+2)}$. Empty leaves are rare at this budget because the typical cell holds $n\,\mathrm{vol}(l) \asymp n^{2/(d+2)}$ observations, and that is what the bound exploits.

S26.8 Step 6: The Noise Term

Lemma S11 (Noise)**.** *Under Assumption 1,*

$$\mathbb{E}[N] \le 2\bar{\sigma}C_d(\lambda)\,\lambda^{1+d/2}n^{-1/2}, \qquad C_d(\lambda) = \sqrt{2}\,c_{3/2}\,c_{1/2}^{d-1} + c_{3/2}^{(2+d)/3}, \tag{S26.20}$$

with $c_p = c_p(\lambda)$ as in (S26.10). For $\lambda \ge 1$, $C_d(\lambda) \le C_d(1) < \infty$, and $C_d(\lambda)$ decreases to $\sqrt{2\pi}(\sqrt{\pi}/2)^{d-1} + \pi^{(2+d)/6}$.

Proof. Given the partition, $N_l \sim \mathrm{Bin}(n, \mathrm{vol}(l))$ and $\mathbb{E}[|\varepsilon_l| \mid N_l] \le \bar{\sigma}N_l^{-1/2}$ on $\{N_l \ge 1\}$. With $\mu = n\,\mathrm{vol}(l)$ and $\psi = \mathbb{E}[N^{-1/2}\mathbf{1}\{N \ge 1\}]$ we have $\psi \le \mu$, since $N^{-1/2} \le 1$ there, and also $\psi \le \sqrt{2}\,\mu^{-1/2}$: indeed $N^{-1/2} \le \sqrt{2}(N+1)^{-1/2}$ on $\{N \ge 1\}$, Jensen gives $\mathbb{E}[(N+1)^{-1/2}] \le (\mathbb{E}[(N+1)^{-1}])^{1/2}$, and $\mathbb{E}[(N+1)^{-1}] = (1-(1-p)^{n+1})/((n+1)p) \le 1/\mu$. So $\psi \le \min(\sqrt{2}\mu^{-1/2}, \mu)$.

Write $V = \mathrm{vol}(l(X)) = \prod_j s_j$, so that $\mu = nV$. The sides s_j of the cell containing a uniform point are independent across j and each has the truncated law of Lemma S8; we use that law throughout rather than the whole-space $\mathrm{Gamma}(2,\lambda)$, since both truncation effects — shorter sides and hence larger inverse moments — go against the bound. By

(S26.13) and $\sum_l g(l)\,\mathrm{vol}(l) = \mathbb{E}_X[g(l(X))]$ it suffices to show $\mathbb{E}[\psi/s_1] \le C_d(\lambda)\,\lambda^{1+d/2} n^{-1/2}$.

On $\{\mu \ge 1\}$ use $\psi \le \sqrt{2}\mu^{-1/2} = \sqrt{2}\,n^{-1/2} V^{-1/2}$ and independence across coordinates:

$$\mathbb{E}\left[\frac{\psi}{s_1}; \mu \ge 1\right] \le \sqrt{2}\,n^{-1/2}\,\mathbb{E}\left[s_1^{-3/2}\right] \prod_{j\ge 2} \mathbb{E}\left[s_j^{-1/2}\right] \le \sqrt{2}\,c_{3/2}\,c_{1/2}^{d-1}\,\lambda^{1+d/2} n^{-1/2}, \tag{S26.21}$$

by (S26.10) twice, the exponents combining as $3/2 + (d-1)/2 = 1 + d/2$. The moment $\mathbb{E}[s_1^{-3/2}]$ is finite because the law of a side has density vanishing linearly at the origin, so that $t^{-3/2} \cdot t$ is integrable; $p = 2$ is the threshold in Lemma S8 for exactly this reason. It is the last inverse moment that exists, and $3/2$ is precisely what the factor $1/s_1$ from (S26.13) and the factor $\mu^{-1/2}$ from the binomial jointly require.

On $\{\mu < 1\}$ use $\psi \le \mu \le 1$, then Hölder with exponents $3/2$ and 3, then the Chernoff bound $\mathbb{P}(V < w) \le w^s\,\mathbb{E}[V^{-s}] = w^s \prod_j \mathbb{E}[s_j^{-s}] \le w^s (c_s \lambda^s)^d$ at $s = 3/2$:

$$\mathbb{E}\left[\frac{\psi}{s_1}; \mu < 1\right] \le \left(\mathbb{E}\left[s_1^{-3/2}\right]\right)^{2/3} \mathbb{P}(V < 1/n)^{1/3} \le c_{3/2}^{(2+d)/3}\,\lambda^{1+d/2} n^{-1/2}. \tag{S26.22}$$

Adding the two gives the claim. □

The small-cell regime, where ψ behaves like μ rather than $\mu^{-1/2}$, needs no separate treatment: there the bound $\psi \le \mu$ is the stronger of the two, and all that is required is that $\{V < 1/n\}$ carry little mass.

S26.9 Step 7: Uniform BV Boundedness

Summing Steps 5, 5a and 6 over the d coordinates and inserting $\lambda_n \asymp n^{1/(d+2)}$, for which $\lambda^{1+d/2} n^{-1/2} = O(1)$ because $(1 + d/2)/(d+2) = 1/2$ — the constants in those three steps being bounded in λ by (S26.10) —

$$\sup_n \mathbb{E}[|DT|(\Omega)] < \infty, \tag{S26.23}$$

and by (S26.2) the same bound holds for $\hat{f}_n$ for every $B \ge 1$.

S26.10 Step 8: Apply Lemma S4

The L^1 consistency condition follows from the L^2 consistency result of Mourtada et al. (2017) under the budget $\lambda_n \asymp n^{1/(d+2)}$ by the Cauchy–Schwarz inequality on the bounded domain Ω:

$$\left\| \hat{f}_n - f \right\|_{L^1} \leq |\Omega|^{1/2} \left\| \hat{f}_n - f \right\|_{L^2} \xrightarrow{p} 0 \tag{S26.24}$$

Uniform BV boundedness is Step 7, for every $B \geq 1$. Lemma S4 then gives $D\hat{f}_n \xrightarrow{w*} Df$ in $\mathcal{M}(\Omega)^d$. □

Remark S17. Theorem S3 is stated for Mondrian forests, whose partitions are generated independently of the data. Independence enters in exactly two places, both in Step 4: the interface identity of Lemma S7 is a statement about the Mondrian process alone, and the cell law behind Lemma S9 likewise. Neither survives data-dependent splitting. For Breiman forests the partition geometry and the response variation are coupled — splits are chosen to maximize response differences across boundaries, which is the worst case for a bound on total variation — so both lemmas would need replacing, not merely adapting. That is left as future work.

What the argument no longer needs is any cancellation between trees. An earlier version bounded the second moment of $|D\hat{f}_n|(\Omega)$ and relied on cross terms vanishing, which required B to grow with n. The per-tree bound of Steps 3 to 7 removes both the cross-term argument and the condition on B, so the theorem now covers forests of the fixed size that is used in practice.

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